UNIVERSITY OF CHINESE ACADEMY OF SCIENCES

DOCTORAL THESIS

The study of Higgs properties in the four-lepton final state at CMS

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Dedicated to my family who endured missed birthdays, missed festivals and countless days alone so that I may make history

Chapter 1

Introduction

The Standard Model (SM) of the particle physics describes the nature of the constituents of the matter and the fundamental forces through which they interact. The three fundamental forces (electromagnetic, strong and weak) are described by three dedicated theoretical descriptions Quantum Electrodynamics (QED), Quantum Chromodynamics (QCD) and Quantum Flavourdynamics (QFD) which are encoded in the SM. Later, electromagnetic and weak interactions were unified into electroweak (EWK) interaction at the unification energy scale of the order of 100 GeV with significant contributions by Sheldon Glashow, Abdus Salam, and Steven Weinberg. The experimental verifications of the existence of the electroweak interaction were established by the existence of neutral currents by the Gargamelle collaboration in 1973 and the discovery of the two massive (W and Z) gauge bosons by UA1 and UA2 collaborations in 1981. These boson should be massless (like photon in QED) according to Goldstone's theorem. To overcome this issue, spontaneous symmetry breaking was introduced that generates the masses of the particles by the Higgs mechanism and separates weak interaction from electromagnetic (EM) interaction [**Reference7**, **Reference8**, **Reference9**, **Reference10**, **Reference11**, **Reference12**]. A new scalar particle-Higgs boson was introduced by the theory as a last missing patch of the SM. This triggered the search for that last missing patch of the SM which continued for many years. One of the main motivations of the construction of the Large Hadron Collider (LHC) at CERN was also to look for this new particle within full mass range allowed given unitary and above the Large Electron-Positron (LEP) constraints, i.e from 114 GeV to 1 TeV. After a long search, the CMS and the ATLAS collaborations, the two general purpose experiments at LHC, announced the existence of the heavy scalar boson on 4 July 2012 [**Reference20**, **Reference21**]. Further tests to measure the properties of this scalar boson immediately started. The mass of newly discovered particle, the only free parameter in Higgs sector, is measured to be $m_H = 125.09 \pm 0.21(stat.) \pm 0.11(syst.)$ GeV [**Reference22**]. The spin and parity [**Reference21a**] and couplings to other particles [**Reference21b**, **Reference21c**] have also been measured and found to be compatible with SM predictions. Therefore, the new particle very probable appear appears as the SM Higgs boson, affiliated with EWK spontaneous symmetry breaking. SM has passed enormous experimental tests in last 50 years and no significant deviation has been seen yet. The biggest achievement of the SM is the experimental evidence of the Higgs boson. Other success includes the predictions of the W and Z boson, the gluon, top and charm quark before they have been verified experimentally. The ratio of the masses of W and Z bosons was also found to be consistent with SM predictions.

Although SM has passed numerous experimental tests with great precision, it is not considered to be theory of everything. There are many phenomenons which are not explained by SM i.e Hierarchy problem, existence of dark matter and dark energy, neutrinos masses etc. There are several theoretical extension posed in SM such as Minimal Super Symmetrical Model(MSSM) and Next-to-Minimal Super Symmetric Standard Model (NMSSM) or completely new theoretical description, such as string theory, extra dimensions. More precise Higgs boson properties measurements are needed to capture any possible deviation from SM predictions which can be a doorway to new physics.

One of the important properties of the Higgs boson is its model independent production cross section for a particular process, where the cross section illustrates the possibility of the occurrence of that process. In this dissertation, measurements of the integrated and differential fiducial cross sections for the Higgs boson production in the $H \to 4\ell$ decay channel ($\ell = e, \mu$) in proton-proton collisions at $\sqrt{s} = 7, 8, 13$ TeV are presented. These measurements provide a direct test of perturbative QCD calculations in the Higgs sector of Standard Model at the TeV energy scale. Any deviation of the integrated or the differential cross sections from SM expectations would provide a direct hint to BSM contributions. Nearly similar measurements have already been reported by the ATLAS experiment in $H \to \gamma\gamma$ [**ATLAS:Hgg_8TeV**, **ATLAS:Hgg_8TeV_13TeV**], $H \to 4\ell$ [**ATLAS:H4l_8TeV**, **ATLAS:H4l_8TeV_13TeV**] decay channels, combination of both channels [**ATLAS:H4l_Hgg_8TeV**, **ATLAS:H4l_Hgg_8TeV_13TeV**] and by the CMS experiment in $H \to \gamma\gamma$ decay channel [**CMS:Hgg_8TeV**].

The Higgs cross section measurement helps to probe the structure, nature of the interaction of Higgs boson with decaying particles. Differential fiducial cross sections are produced using the 13 TeV data as a function several kinematic observables which are sensitive to the Higgs-boson production mechanisms: rapidity and transverse momentum of the four-lepton system, associated jet multiplicityand transverse momentum of the leading jet. These differential distributions are sensitive to gluon fusion production mechanism, PDFs of colliding protons, theoretical modeling of hard quarks radiations and relative contributions of different Higgs boson production mechanisms.

The fragmentation of the dissertation is systematized as follows. An overview of the SM, Higgs boson production and decays are described in Chapter 2. The overview of LHC and CMS detector are presented in Chapter 3. The calibrations and reconstructions of the physics objects in the CMS detector are described in Chapter **??**. Chapter 4 presents the measurements of differential and integrated fiducial cross sections for Higgs boson production at $\sqrt{s} = 13$ TeV. Finally, the conclusions and outlook of the dissertation are summarized in Chapters **??** and **??**.

Chapter 2

The Standard Model of Particle Physics

The aim of particle physics is to understand the nature of the constituents of the matter and the fundamental forces through which they interact. There are many theories purposed to understand the laws of particle physics but the Standard Model is the theory which stood firm against high precision tests [**Reference1**]. The Standard Model is a collaborative theory of the fundamental world, developed in 20th century. The Standard Model expresses the fundamental particles, boson, fermions and interaction between them by using mediating gauge bosons. The last unverified part of the Standard Model, the Higgs boson, has also been discovered in 2012 by CMS and ATLAS experiments at LHC at European Organization for Nuclear Research(CERN) [**Reference20**, **Reference21**]. Despite of incredibly successful theory of particle physics, the Standard Model fail to fully describe the known universe.

This chapter is structured as follows. Details of the fundamental particles and the interactions would be described in Section 2.1. The Higgs mechanism, Higgs boson production and decays at LHC would be described in Sections 2.2 and 2.3. Experimental Observations of the Standard Model would be described in Section 2.4. Physics beyond the Standard Model will be described in Section 2.5.

2.1 Fundamental Particles and Interactions

2.1.1 Fundamental Particles

Particle with unknown structure is called fundamental particle. Fundamental particles can be divided into two classes, fermions and bosons. Particles which are characterized by Fermi-Dirac statistics and Bose-Einstein statistics are called fermions and bosons respectively. According the spin-statistics theorem, particles with integral spin are called bosons, while particles with half-integral spin are called fermions. Fermions includes all leptons, quarks and baryons which are composite particles made up of three quarks. Matter is made up of leptons and quarks which are divided into three "generations" as shown in Table 2.1. All quarks generations include of an "up" type quark with electric charge $+\frac{2}{3}$ and a "down" type with electric charge $-\frac{1}{3}$. Quarks also carry color charge - red, green, or blue, which is responsible for strong interaction. The types of the quarks in order of increasing mass are u, d, s, c, b, t. The whole visible matter is only made of lightest type quarks,"up" and "down" . The heavier quarks are only produced in high energy collisions (cosmic rays, particle accelerators) and quickly decays into "up" and "down" quarks.

Quarks interact through all three Standard Model forces, weak, strong, and electromagnetic. As showed by the theory of the strong force, Quantum Chromodynamics (QCD), a quark antiquark pair or three quarks can combine as mesons ($\mathrm{q_1\bar{q}_2}$) or baryons ($\mathrm{q_1q_2q_3}$) respectively to make color-neutral hadrons. Out of hadrons, protons and neutrons are most widely recognized making nuclei of matter which are formed of uud and udd quarks respectively.

Fermions	1st Gen.	2nd Gen.	3rd Gen.	Charge	Interactions
Quarks	u	c	t	$+\frac{2}{3}$	all
	d	s	b	$-\frac{1}{3}$	
Leptons	e	μ	τ	-1	weak, electromagnetic
	ν_e	ν_μ	ν_τ	0	weak

TABLE 2.1: Generations, charge and interaction types of quarks and leptons.

Leptons also have three generations like quarks. Leptons only interact via electromagnetic and weak interaction, since they have no color charge. In fact, quarks and gluons are only particles carry color charge. Each generation of leptons have a charged lepton and a neutrino associated. Neutrinos are electrically neutral and carry no color charges, so interact through weak force only. Charged leptons masses are listed in order of increasing mass $m_e = 0.511$ MeV, $m_\mu = 105.7$ MeV and $\mathrm{m}_\tau = 1.78$ GeV. Charged leptons carry -1 and their antiparticle carry $+1$ electric charge. In addition, all leptons lepton flavor quantum number: all leptons have assigned $+1$ while antileptons have assigned -1.

Bosons are the carriers of the fundamental forces, more details would be described in Section 2.1.2. Electromagnetic force is mediated by massless, neutral, spin-1 photon. Weak force is mediated by $W^\pm$ and Z bosons having electric charges ±1 and 0 respectively. The masses of $W^\pm$ and Z bosons are 80.4 GeV and 91.2 GeV respectively. At last, strong forces is carried by massless, neutral spin-1 gluons. Gluon is the only gauge boson which carries color and anti-color charge, thus it can interact with other gluons.

2.1.2 Fundamental Interactions

The three fundamental forces (electromagnetic, strong and weak) are described by three dedicated theoretical descriptions Quantum Electrodynamics (QED), Quantum Chromodynamics (QCD) and Quantum Flavourdynamics (QFD) which are encoded in the Standard Model. Later, electromagnetic and weak interactions are

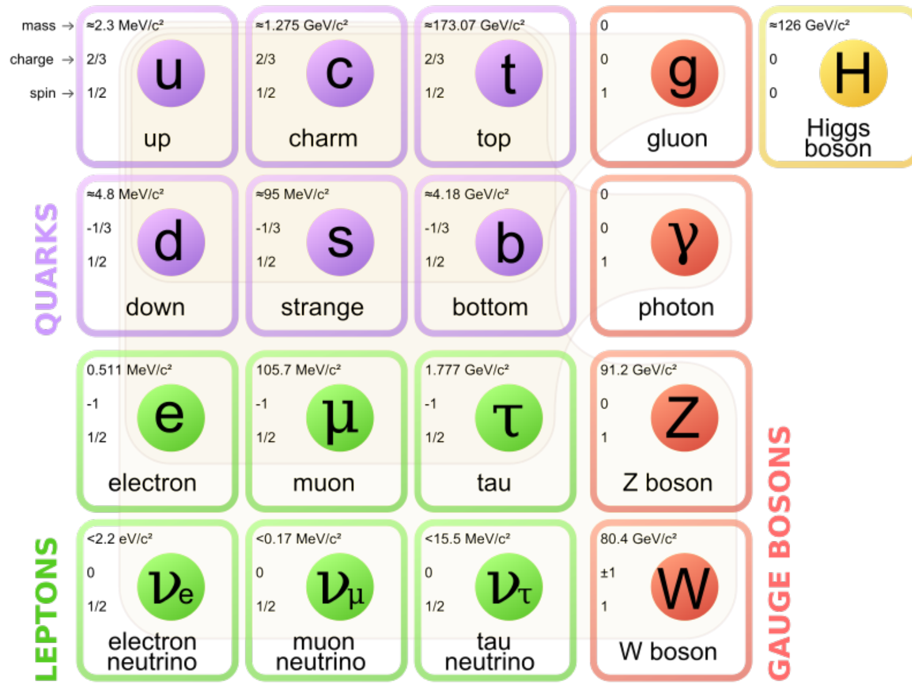


FIGURE 2.1: The Standard Model of elementary particles, with the three generations of matter, gauge bosons in the fourth column, and the Higgs boson in the fifth.

TABLE 2.2: Gauge bosons and their masses, coupling constants and interaction ranges for the three forces of the Standard Model.

Force	Weak	Electromagnetic	Strong
Exchange Boson	$W^{\pm}, Z$	photon (γ)	Gluon (g)
Mass(GeV)	80.4, 91.2 GeV	0	0
Coupling Constant	$G_F \approx 1.2 \times 10^{-5}$ GeV^{-2}	$\alpha = \frac{1}{137}$	$\alpha_s \approx 0.1$
Interaction Range	10^{-16} cm	∞	10^{-13} cm

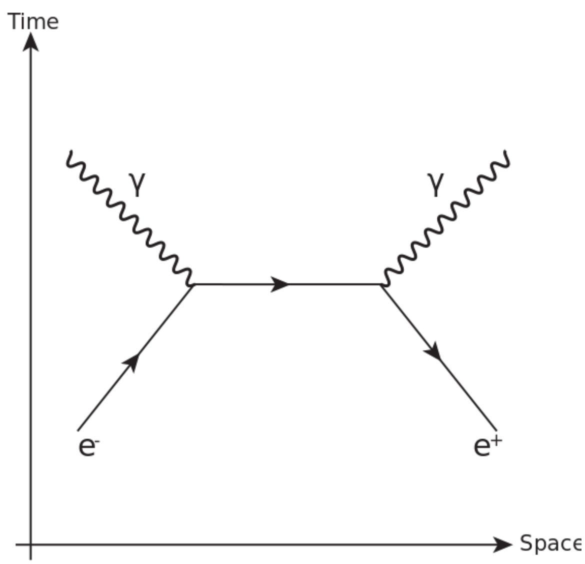


FIGURE 2.2: Feynman diagram of electron positron annihilation.

unified into electroweak (EWK) interaction at the unification energy scale of the order of 100 GeV with significant contributions by Sheldon Glashow, Abdus Salam, and Steven Weinberg. The mathematical description of interaction of charged particles are described by QED. QED photon is the force carrier of electromagnetic force. QED is the first theory which explains the interaction between matter and light that patch up special relativity and quantum mechanics. Figure 2.2 shows an example of QED, electron-positron annihilation. Annihilation is the process of destruction of particle and anti particle pair to any set of particles obeying conservation of quantum numbers, energy and momentum. QED is perturbative theory based on fundamentals of quantum mechanics established by P. A. M. Dirac, W. Heisenberg, and W. Pauli during the 1920s. Later, Freeman J. Dyson, Richard P. Feynman and Julian S. Schwinger showed that charged particle can possibly interact through a sequence of processes with growing complexity, and all of these processes could be described by Feynman developed graphical technique. QED is an abelian gauge theory with the symmetry group $U(1)$.

Weak interaction, one of the four fundamental forces, was first revealed by the understanding of beta decay in 1933 by Enrico Fermi called Fermi's interaction [**Referencea**]. Weak force is due to exchange of $W^{\pm}$ and Z bosons. In terms of field theory, it represented by gauge theory with symmetry group $SU(2)$, having three gauge bosons (W^{+}, W^{-}, Z). Weak interaction rarely called quantum flavordynamics (QFD), it is the only force which is proficient to change the flavor of quarks. This is very short range force because exchange bosons ($W^{\pm}$, Z) are massive. All fermions and the Higgs boson can interact via weak force. In 1970s, weak and electromagnetic forces are unified into electroweak force above unification energy(on the order of 100 GeV), the Nobel Prize in Physics was awarded to Sheldon Glashow, Abdus Salam, and Steven Weinberg for their role. In the field theory representation, Electroweak unification is settled under an $\mathrm{SU(2)_L \otimes U(1)_\gamma}$ gauge group. Experimental observation of existence of electroweak interaction was accomplished in two phases, the first was disclosure of existence of neutral current in scattering of neutrino by Gargamelle collaboration in 1973 and the second discovery of W and Z gauge bosons in proton-antiproton collisions by both the UA1 and the UA2 collaborations in 1983 [**Reference2**, **Reference3**, **Reference4**, **Reference5**].

The SM is represented by $\mathrm{SU(3)_C \otimes SU(2)_L \otimes U(1)_\gamma}$ gauge symmetry. Where $\mathrm{U(1)_\gamma}$ is representation of QED, $\mathrm{SU(2)_L}$ of weak theory and $\mathrm{SU(3)_C}$ of QCD. By the unification of electromagnetic and weak force represented by $\mathrm{SU(2)_L \otimes U(1)_\gamma}$ gauge symmetry [**Reference6**], Glashow, Salam and Weinberg was able to formulate the fields to account the $W^{\pm}$, Z and γ (A) boson in footings of the original basis fields A' and B as expressed in equations 2.1, 2.2 and 2.3. The spontaneous symmetry breaking of $\mathrm{SU(2)_L \otimes U(1)_\gamma}$ gauge symmetry is the primary reason that generated the masses of the particles and separate electroweak (EW) interaction from electromagnetic (EM) and weak interactions. At higher energies($> 100\ GeV$), both weak and EM forces acts as similar style. More details would be described in Section 2.2.

$$W^{\pm}_{\mu} = \sqrt{\frac{1}{2}}(A^1_\mu \pm iA^2_\mu) \qquad M_W = \frac{1}{2} g \mathrm{v} \tag{2.1}$$

$$Z^{\pm}_{\mu} = \sqrt{\frac{1}{g^2 + g'^2}}(gA^3_\mu - g'B_\mu) \qquad M_Z = \frac{1}{2} \mathrm{v} \sqrt{g^2 + g'^2} \tag{2.2}$$

$$A_{\gamma} = \sqrt{\frac{1}{g^2 + g'^2}}(gA^3_\mu - g'B_\mu) \qquad M_A = 0. \tag{2.3}$$

The strong interaction, one of four fundamental forces, is responsible for strong nuclear force which bounds the nucleus of an atom. As the name speaks for itself, strong force is strongest among four fundamental forces, although it act at a distance of 10^{-15} meter (femtometer) or less. Strong force is described by QCD and only exist between quarks and gluons which carry non-vanishing color charges. In the field theory representation of the SM, the strong interaction is represented by the $\mathrm{SU(3)_C}$ symmetry group. Here, C represents QCD color, which must be conserved in all strong interactions. The strong force is mediated by gluons. Contrary to the other fundamental forces, strong force does not diminish with gain in distance between quarks know as color confinement. As an implication of this property, no signature of elementary quark or gluon particle are observed in High Energy Physics (HEP) colliders, instead they bound with each other to form streams of hadrons by phenomenon known as hadronization called jets.

2.2 The Brout-Englert-Higgs Mechanism (BEH)

Gauge fields must acquire mass in order correctly describe weak interaction and to be consistent experimental findings. Before electroweak symmetry breaking (EWSB), all particles supposed to be massless, however a mass term in electroweak

Lagrangian for gauge bosons must be introduced to guarantee the renormalization of theory that would violate gauge invariance. Which hints that some basic thing is missing in the theory apart from fermions and gauge boson fundamental fields, which can maintain the local gauge invariance. The Higgs mechanism was postulated as a part of SM, which introduces the existence of a new scalar field, so called Higgs field. The Higgs field transformation ensures the renormalizability of the theory and gauge invariance. By this formulation the concept of breaking of symmetry and origin of mass of fermions and boson ($W^{\pm}, Z$ boson) are introduced [**Reference7**, **Reference8**, **Reference9**, **Reference10**, **Reference11**, **Reference12**]. The Higgs mechanism introduces a doublet (under isospin) of complex scalar field

$$\phi = \begin{pmatrix} \phi^{\dagger} \\ \phi^{0} \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} \phi^{1} + i\phi^{2} \\ \phi^{3} + i\phi^{4} \end{pmatrix} \tag{2.4}$$

which can be substituted in electroweak lagrangian given below

$$\mathscr{L}_{EWSB} = (D_{\mu}\phi)^{\dagger}(D_{\mu}\phi) + \mathrm{V}(\phi^{\dagger}\phi) \tag{2.5}$$

here $D_{\mu} = \delta_{\mu} - igt_{a}W_{\mu}^{a} + \frac{i}{2}g'YB_{\mu}$ represents covariant derivative. The lagrangian described in Equation 2.5 is invariant under $\mathrm{SU}(2)_{\mathrm{L}} \otimes \mathrm{U}(1)_{\gamma}$ transformation. The kinetic energy term is written in terms of covariant derivatives, and the potential only depends upon on $\phi^{\dagger}\phi$. The scalar field ϕ can be codify in terms of isospin (I_3) , hypercharge (Y) and electric charge (Q) quantum numbers:

	I_3	Y	Q
$\begin{pmatrix} \phi^{\dagger} \\ \phi^{0} \end{pmatrix}$	$\begin{pmatrix} \frac{1}{2} \\ \frac{1}{-2} \end{pmatrix}$	$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$	$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$

while the potential term in Equation 2.5 can also be expressed as

$$\mathrm{V}(\phi^{\dagger}\phi) = -\mu^{2}\phi^{\dagger}\phi - \lambda(\phi^{\dagger}\phi)^{2} \tag{2.6}$$

here μ is mass term and the parameter λ represents the strength of the interaction. In the scenario, $\mu > 0$ and $\lambda > 0$ and the potential shapes becomes very interesting, which is essential for Higgs mechanism shown in Figure 2.3, has a minimum for

$$\phi^{\dagger}\phi = \frac{1}{2}(\phi_{1}^{\dagger} + \phi_{2}^{\dagger} + \phi_{3}^{\dagger} + \phi_{4}^{\dagger}) = \frac{-\mu^{2}}{2\lambda} \equiv \frac{\mathrm{V}^{2}}{2} \tag{2.7}$$

but on the circle of points where the magnitude of z is ϕ. This minimum does not exist at a single value of ϕ, but at circular non-zero values.The ($\phi^{\dagger}\phi^{0}$) selection that affiliated with the ground state (i.e. the lowest energy state, or vacuum) is arbitrary and the selected point is not invariant under rotations in the ($\phi^{\dagger}\phi^{0}$) plane: this is called spontaneous symmetry breaking. If one select to settle the ground state on

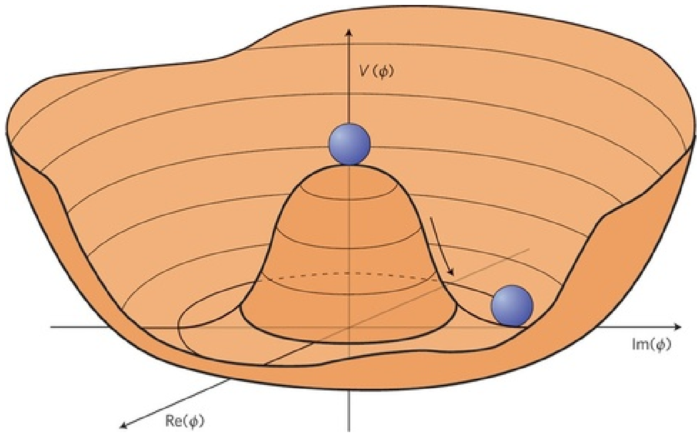


FIGURE 2.3: Shape of the Higgs potential in Eq.2.6

the ϕ^0 axis, the vacuum expectation value of the ϕ field is

$$\langle\phi\rangle = \frac{1}{\sqrt{2}} = \begin{pmatrix} 0 \\ \mathrm{V} \end{pmatrix}, \qquad \mathrm{V}^2 = -\frac{\mu^2}{\lambda}. \tag{2.8}$$

The field ϕ given below, can be derived by rewriting it in a generic gauge, on footings of its vacuum expectation value:

$$\phi = \frac{1}{\sqrt{2}} e^{\frac{i}{\nu}\phi^a t_a \phi} = \begin{pmatrix} 0 \\ H+\nu \end{pmatrix}, a = 1,2,3 \tag{2.9}$$

here ϕ_a and $\phi_4 = H + \nu$ represents massless fields called Goldstone fields. New degrees of freedom is introduced along with six degrees because of polarizations of the massless vector boson $W^{\pm}$ and Z, as they are massless and scalar. A unitary gauge transformation is fixed by

$$\phi' = e^{-\frac{i}{\nu}\phi^a t_a}\phi = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ H+\nu \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ \phi^4 \end{pmatrix}$$

so here we have zero expectation value for the Higgs field.
Now if we rewrite the lagrangian in Eq. 2.5 with field ϕ in unitary gauge, $\mathscr{L}_{EWSB}$ can be disintegrated into three components

$$\mathscr{L}_{EWSB} = \mathscr{L}_H + \mathscr{L}_{HW} + \mathscr{L}_{HZ} \tag{2.10}$$

where three terms is given, assuming approximation $\mathrm{V} = \mu^2 H^2 + \cos t$ and ignoring higher order terms:

$$\mathscr{L}_H = \frac{1}{2}\delta_a H \delta^a H + \mu^2 H^2$$

$$\begin{aligned} \mathscr{L}_{HW} &= \frac{1}{4}\nu^2 g^2 W_a W^{\dagger a} + \frac{1}{2} H W_a W^{\dagger a} \\ &= m_W^2 W_a W^{\dagger a} + g_{HW} W_a W^{\dagger a} \end{aligned} \tag{2.11}$$

$$\begin{aligned} \mathscr{L}_{HZ} &= \frac{1}{8}\nu^2 (g^2 + g'^2) Z_a Z^a + \frac{1}{4}\nu (g^2 + g'^2) H Z_a Z^a \\ &= \frac{1}{2} m_Z^2 Z_a Z^a + \frac{1}{2} g_{HZ} H Z_a Z^a. \end{aligned} \tag{2.12}$$

Mass terms for $W^{\pm}$ and Z bosons are included in Eqs. 2.11 and 2.12: $W^{\pm}$ and Z bosons have acquired mass and three additional degree of freedom, related to the longitudinal polarization. Also, three of the four Goldstone bosons have vanished from the Lagrangian $\mathscr{L}_{EWSB}$, thus sustaining the total number of degrees of freedom. A new scalar field has been introduced referred as Higgs field, introduced in the theory. The boson associated with Higgs field is called Higgs boson. The mass of the Higgs boson is linked with a free parameter in the theory λ that depends upon the self-interaction of Higgs field.

Briefly, the BEH mechanism ensures the renormalizability of theory and explain existence of the massive weak gauge bosons, despite of the fact that the fundamental gauge boson in theory still stays to be massless. In fact, the symmetry still exist in equation so using “hidden“ is preferred over broken. According to BEH mechanism, fermions also acquire masses via their interaction with Higgs field. The invariance of the minimum for the Higgs field for $\mathrm{U}(1)_\gamma$ group shows the electromagnetic symmetry is not spontaneously broken that keeps photon massless.

2.2.1 Vector Boson Masses and Couplings

Eqs. 2.11 and 2.12 shows the masses of $W^{\pm}$ and Z bosons depends upon a parameter ν which is vacuum expectation value that defines a mass scale property of EWSB and to electroweak coupling constants:

$$\begin{cases} m_W = \frac{1}{2}\nu g \\ m_Z = \frac{1}{2}\sqrt{g^2 + g'^2} \end{cases} \rightarrow \frac{m_W}{m_Z} = \frac{g}{\sqrt{g^2 + g'^2}} = \cos(\theta_W). \tag{2.13}$$

Also, Higgs boson couplings to vector bosons can also be obtained by Eqs. 2.11 and 2.12

$$g_{HW} = \frac{1}{2}\nu g^2 = \frac{2}{\nu} m_W^2 \tag{2.14}$$

$$g_{HZ} = \frac{1}{2}\nu (g^2 + g'^2) = \frac{2}{\nu} m_Z^2. \tag{2.15}$$

Using above Eqs. 2.14 and 2.14 helps to get the ratio of Higgs decay to $W^{\pm}$ and Z bosons

$$\frac{\mathcal{B}(H \to W^+W^-)}{\mathcal{B}(H \to ZZ)} = \left(\frac{g_{HW}}{\frac{1}{2}g_{HZ}}^2\right) = 4\left(\frac{m_W^2}{m_Z^2}\right)^4 \approx 2.4. \tag{2.16}$$

At last, the characteristic mass scale for EWSB may be obtained by the relation between the Fermi constant G_F and the ν parameter:

$$\nu = \left(\frac{1}{\sqrt{2}G_F}\right) \approx 246GeV. \tag{2.17}$$

2.2.2 Fermions Masses and Couplings

Fermions couples to Higgs field to acquire mass by Higgs mechanics. This can be achieved by introducing a $\mathrm{SU(2)_L} \times \mathrm{U(1)}_\gamma$ invariant term (called yukawa term) in SM lagrangian that describes the interactions between the fermion fields and the Higgs. As ϕ is an iso-doublet and fermions can be disintegrated up into right-handed and left-handed singlet, the yukawa term for each fermion will have expression given below for leptons:

$$\mathscr{L}_\ell = -G_\ell.\bar{l}_\ell\phi\ell R + \bar{\ell_R}\phi^\dagger l_\ell. \tag{2.18}$$

As the first component of ϕ is zero, so a mass term will only emerge for the second component of l_ℓ: this exactly repeat the fact that neutrino is massless.

$$\mathscr{L}_\ell = -\frac{G_{H\ell}}{\sqrt{2}}\nu\bar{\ell}\ell - \frac{G_{H\ell}}{\sqrt{2}}H\ell\bar{\ell}. \tag{2.19}$$

For quarks, the down type quarks (d, s, b) behave in similar way as leptons while up type quarks (u , c, t) must couple to charge conjugate of ϕ

$$\phi_c = -i\tau_2\phi^* = \frac{1}{\sqrt{2}}\begin{pmatrix} \phi^3 + i\phi^4 \\ -\phi^1 + i\phi^4 \end{pmatrix} \tag{2.20}$$

in unitary gauge that becomes

$$\phi_c = \frac{1}{\sqrt{2}}\begin{pmatrix} \eta + \nu \\ 0 \end{pmatrix}. \tag{2.21}$$

Therefore, the yukawa Lagrangian can be expressed as

$$\mathscr{L} = -G_{H\ell}\bar{L_L}\phi\ell_R - G_{Hd}\bar{Q_L}\phi d_R - G_{Hu}\bar{Q_L}\phi^c u_R + h.c.. \tag{2.22}$$

The couplings constant and mass of a fermion can be derived from Eq. 2.19, written as

$$m_f = \frac{G_{Hf}}{\sqrt{2}}\nu \tag{2.23}$$

$$g_{Hf} = \sqrt{G_{Hf}}\sqrt{2} = \frac{m_f}{\nu}. \tag{2.24}$$

The fermions mass can not be predicted by the theory as G_{Hf} is free parameters. The interaction of Higgs field with fermions transforms right-handed chirality in a left-handed chirality and vice versa in a way that object acquire mass. The strength of the interaction of Higgs fields with particles is reflected by its mass.

This evaluation is rather interpretative, as it not accumulating the fact that there are three families of quarks and leptons in reality. The different between weak mass eigenstates and the physical mass eigenstates makes real analysis complicated. Transformation of the weak mass basis to physical basis in essential to deal with the physical particles. This is achieved by the usual unitary transformation matrices, Cabibbo-Kobayashi-Maskawa (CKM) and Pontecorvo-Maki-Nakagawa-Sakata (PMNS) matrices, for quark and lepton sectors respectively.

2.3 Production and Decay of the Higgs Boson

After electroweak unification, spontaneous symmetry breaking and Higgs mechanism solves all the puzzles with the assumption of existence of the Higgs boson. Higgs boson was considered as the last missing patch of the Standard Model. Higgs boson has been searched at LHC within full mass range allowed given unitary and above LEP constraints, i.e from 114 GeV to 1 TeV. In this Section, Higgs boson production and decay modes would be discussed. This will allow us to identify the most promising channels in the perspective of Higgs boson discovery and its properties measurements. Theory does not predict the Higgs mass and coupling of the Higgs with bosons are directly proportional to the square of their masses as shown in Eqs. 2.14 and 2.15. In case of fermions, their couplings with the Higgs is directly proportional to the mass of the fermion as in Eq. 2.24. For this reason, the most promising channels for Higgs searches are its decays to W^+W^-, ZZ and third generation of leptons. Also, photon being massless particle does not couple to Higgs directly but through loop contributions dominated by top quark and $W^{\pm}$ boson.

2.3.1 Higgs Production

The dominant production modes of the Higgs boson in pp collisions at LHC are shown in Figure 2.4, corresponding production cross sections as a function of the Higgs mass at center-of-mass energies $\sqrt{s}$ = 7, 8 and 13 TeV are shown in Figure 2.5. A considerable increase in the Higgs production cross section is found as moves towards higher center-of-mass energies shown in Figure 2.6 **[Referenceb]** for Higgs mass 125 GeV.

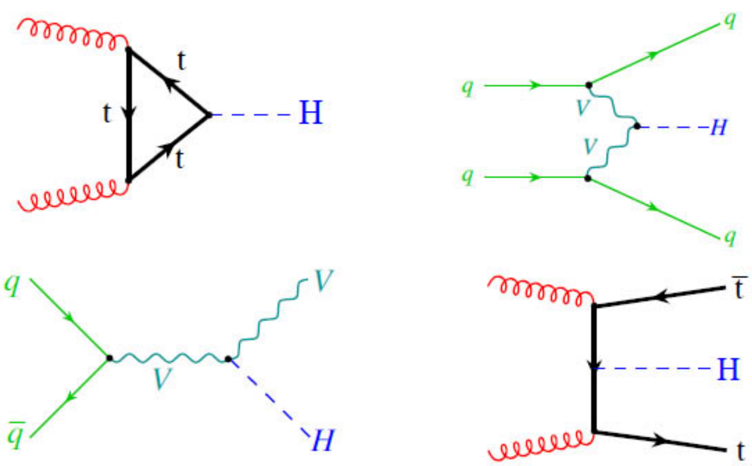


FIGURE 2.4: Higgs production mechanisms at tree level in proton-proton collisions: gluon-gluon fusion (top left);VV fusion (top right); W and Z associated production (bottom left); $t\bar{t}$ associated production (bottom right).

Gluon-gluon Fusion

This is most dominant Higgs production mode at LHC in whole mass range and center-of-mass energies range is gluon-gluon fusion ($\mathrm{gg} \rightarrow \mathrm{H}$), illustrated in Figure 2.4 (top left) [**Referenceb**]. As LHC and tevatron are hadron colliders, it highly probable that two gluons responsible for binding of hadron collide. The gluons couples to Higgs through the loop of virtual quarks, as coupling to Higgs is directly proportional to the mass of fermions, top quark loop is most dominant. The ggH cross section for Standard Model Higgs at $m_H = 125$ GeV at center-of-mass energies of 7 TeV, 8 TeV and 13 TeV are approximately 15, 19, 44 pb respectively. As we move towards higher energies the production cross section increases significantly, i.e 27% from 7 TeV to 8 TeV and 227% from 8 TeV to 13 TeV at $m_H = 125$ GeV. The cross section numbers in Table 2.3 are computed at NNLO QCD and NLO EW accuracies.

Vector Boson Fusion

Vector boson fusion (VBF) is the second most important Higgs production mode at LHC and LEP as shown in Figure 2.4 (top right). VBF Higgs production cross section is roughly 10 times lower than ggH at $m_H = 125$ GeV, and becomes comparable at $m_H = 1$ TeV. The VBF cross section for Standard Model Higgs at $m_H = 125$ GeV at center-of-mass energies of 7 TeV, 8 TeV and 13 TeV are approximately 1.2, 1.6, 3.8 pb respectively. The cross section numbers in the Table 2.3 are computed at NNLL QCD and NLO EW precision [**Referenceb**]. In VBF production mode, two $W^{\pm}$ or Z boson fusion create Higgs that have been emitted by a (anti)fermion

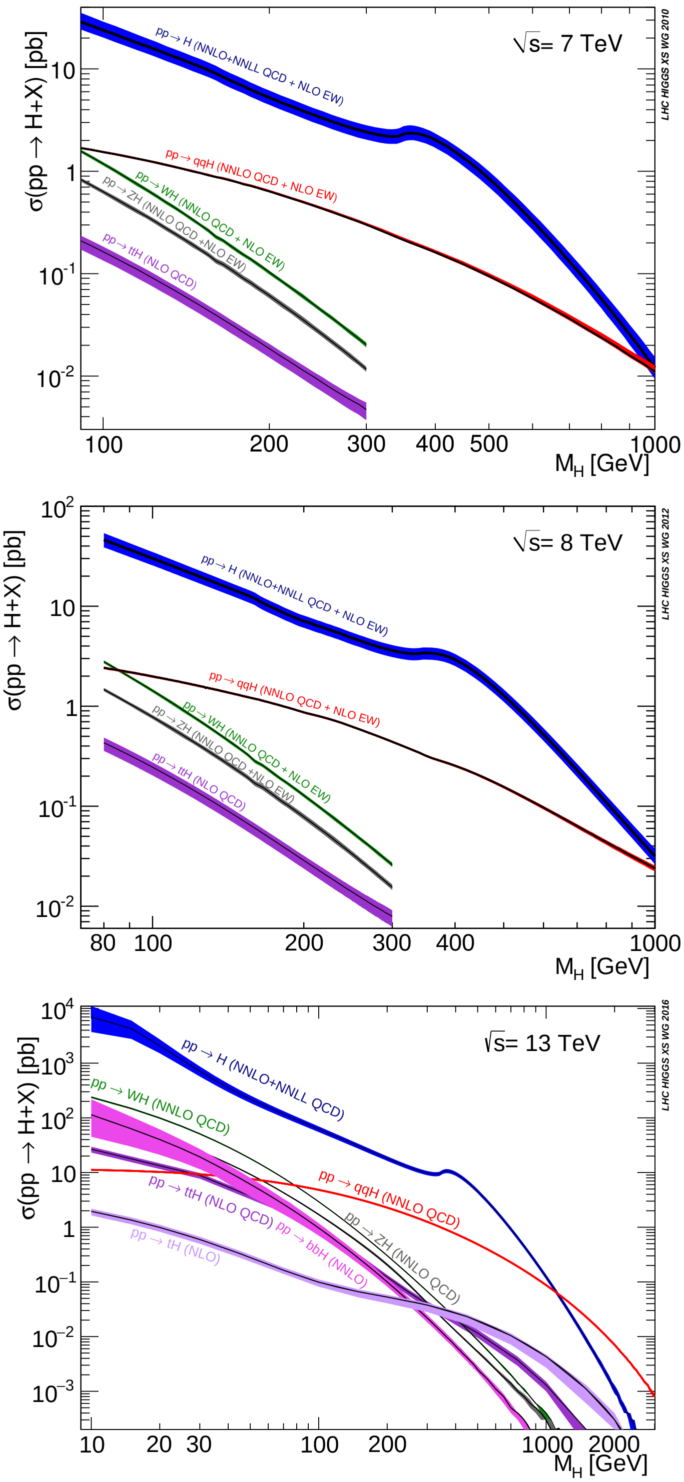


FIGURE 2.5: Standard Model Higgs boson production cross sections at 7, 8 and 13 TeV as function of mass of Higgs boson.

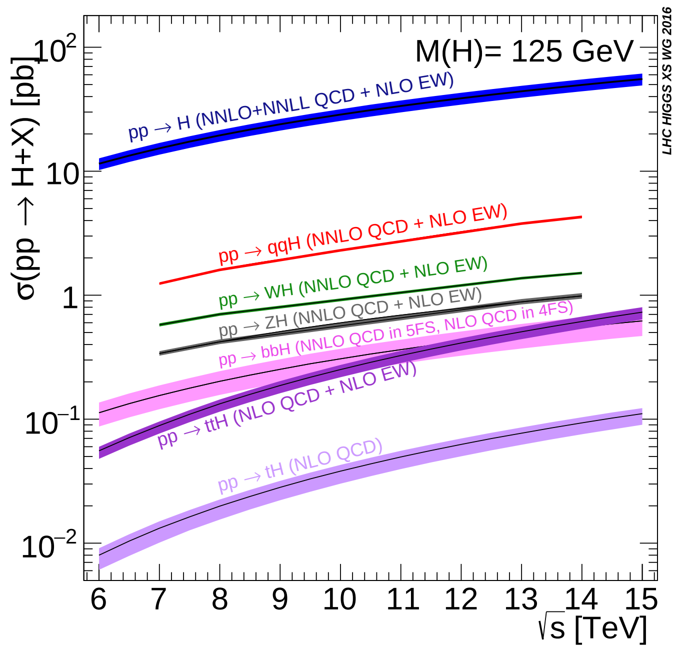


FIGURE 2.6: Higgs boson production cross sections as a function of center-of-mass energies at $m_H = 125$ GeV

TABLE 2.3: Higgs boson production cross sections in pb for different Higgs masses and center-of-mass energies by ggH and VBF productions mechanism at LHC.

$m_H(GeV)$	ggH			VBF		
	7 TeV	8 TeV	13 TeV	7 TeV	8 TeV	13 TeV
120	16.61	21.07	47.38	1.30	1.68	3.9
125	15.31	19.47	44.14	1.24	1.60	3.78
130	14.15	18.03	41.23	1.19	1.53	3.64

pair collision. The VBF experimental signal is very clean and can be well distinguished from ggH with two high energy jets in forward region of the detector. These jets signature helps to increase signal to background ratio.

W and Z Associated Production

W and Z associated production also called Higgs Strahlung is defined by Higgs production in association with $W^{\pm}$ or Z boson as shown in Figure 2.4 (bottom left). In this production mechanism, Higgs is radiated by high energetic $W^{\pm}$ or Z bosons that is produced by fusion of two fermions. This production mechanism was leading production mode at LEP and sub-leading at Tevatron. At LEP it becomes very probable to produce $W^{\pm}$ or Z boson with electron positron collisions. In case of hadron collider, tevatron makes quark antiquark collisions more likely

TABLE 2.4: Higgs boson production cross section in pb for different Higgs masses and center-of-mass energies by associative productions mechanism at LHC.

$m_H(GeV)$	WH			ZH			ttH		
	7 TeV	8 TeV	13 TeV	7 TeV	8 TeV	13 TeV	7 TeV	8 TeV	13 TeV
120	0.66	0.81	1.57	0.39	0.48	0.99	0.11	0.15	0.57
125	0.58	0.70	1.37	0.34	0.42	0.88	0.09	0.13	0.51
130	0.51	0.62	1.21	0.30	0.37	0.79	0.08	0.12	0.45

as compared to LHC. AT LHC this production mechanism is at third number. The WH and ZH cross section for Standard Model Higgs at center-of-mass energies of 7 TeV, 8 TeV and 13 TeV are listed in Table 2.4 which are computed at NNLL QCD and NLO EW precision [**Referenceb**].

Top Fusion

The least dominant process of Higgs production is illustrated in Figure 2.4 (bottom right). In this process two gluons collide producing pair of top antitop which produces Higgs by fusion in association $t\bar{t}$. The ttH cross section for Standard Model Higgs at center of mass energies of 7 TeV, 8 TeV and 13 TeV are listed in Table 2.4 which are computed at NLO QCD and NLO EW precision [**Referenceb**].

2.3.2 Higgs Decay

As Higgs boson is electroweak particle, so it wants to decay quickly. The Higgs boson decay likelihood depends on a variety of aspects, i.e mass of the Higgs boson, interaction strength etc, most of these factors are controlled by SM except the Higgs boson mass itself. The life time of SM Higgs boson with mass of 126 GeV is about 1.6×10^{-22} s. Higgs boson can decay to all SM particles with mass but it likes to decay most heavier particles available. Higgs boson decay branching ratio to a variety of particles are illustrated in Figure 2.7 [**Referenceb**].

The most dominant decay channel of the Higgs boson $H \to b\bar{b}$ at $m_H \approx 125$ GeV. But the QCD background in this channel is of many order higher than signal which makes it less sensitive channel. However, the $b\bar{b}$ channel has been exploited in the boosted regime, in association with a vector boson decaying leptonically, by CMS [**Reference13**], by ATLAS [**Reference14**] and by CDF and D0 [**Reference15**]. B quarks are also difficult to reconstruct in a detector. When they decay in the detector, they fragment into jets of many hadrons. These jets are difficult to reconstruct perfectly. Because of this, it is necessary to exploit other channels in this region. $H \to \gamma\gamma$ is one most favorable decay channel due to signature with two high energy photons with excellent mass resolution.

The main backgrounds in this channel are $q\bar{q} \to \gamma\gamma$, $\mathrm{gg} \to \gamma\gamma$ and instrumental background in which one or two jets are identified as photons. As the Higgs mass at 125 GeV, Higgs boson decay to diboson(WW, ZZ) boson are also available, where one or both boson can be off-shell. In case of WW decay, its impossible to

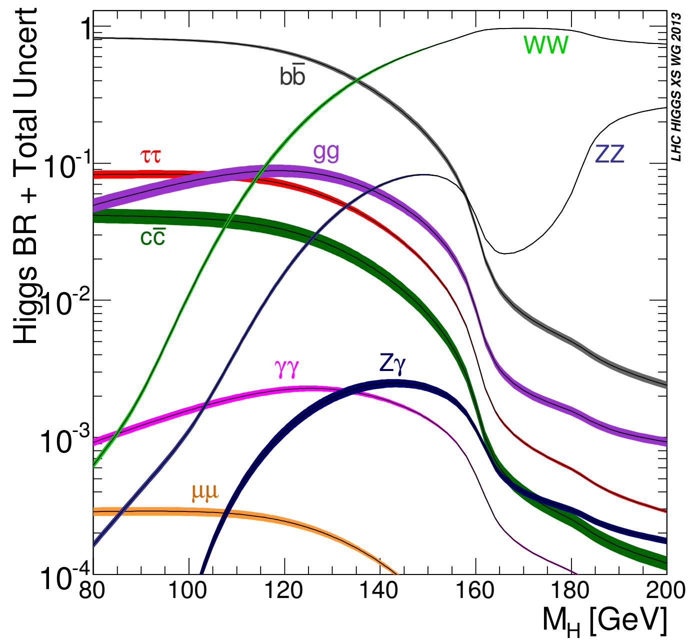


FIGURE 2.7: Branching ratios for different Higgs boson decay channels as a function of the Higgs mass.

completely reconstruct WW boson in detector as it decays $H \to WW \to \ell\nu\ell\nu$ have two neutrinos and $H \to WW \to \ell\nu jj$ have one neutrino and two jets. These objects pulls mass distribution to be wide and signal to background ratio to be low, which makes WW decay channel less sensitive. While $H \to ZZ \to \ell\ell\ell\ell$ decay is labeled as golden channel in spite of low branching ratio. The clean signature of at least four leptons makes this channel almost background free. These leptons are high energetic leptons which can reconstructed with excellence in the detector. The branching ratio of the Higgs boson in intermediate range is illustrated in Figure 2.8 [**Referenceb**].

2.4 Experimental Observations

The electroweak unification theory lead to series of impressive experimental discoveries. In 1973, the evidence of neutral current through inelastic scattering was found by Gargemelle buddle chamber collaboration at CERN [**Reference16**]. This evidence provide starting platform to the Standard Model of electroweak interaction.

Super Proton Synchrotron proton-atniproton collider at CERN played its role to next breakthrough that came a decade later. The existence of W, Z boson announced by UA1 and UA2 collaboration through the channels $Z \to e^+e^-$, $Z \to \mu^+\mu^-$, $W \to e\nu$ [**Reference2**, **Reference3**, **Reference4**, **Reference5**]. This was a first

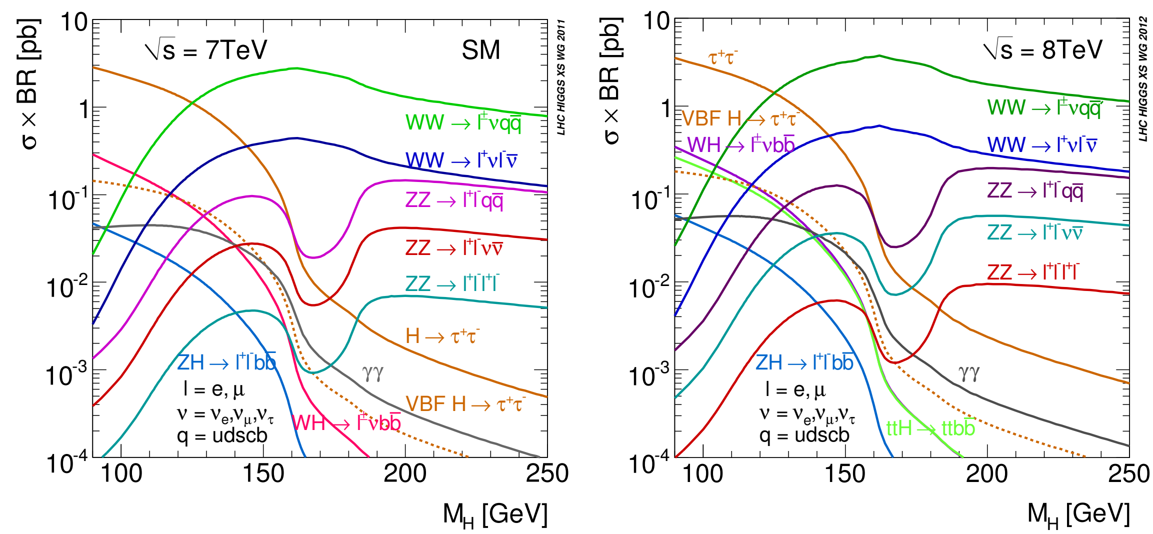


FIGURE 2.8: Higgs cross sections times branching ratios 7 TeV (left) and 8 TeV (right).

observation of W, Z boson that mediates weak force. Carlo Rubbia and Simon van dar Meer was awarded with Nobel Prize for their contribution in 1984.

These important experimental discoveries and growing knowledge on weak interaction convinced to construct the Large Electron Positron collider. at CERN, for deeper look into W, Z further decays, their precise mass measurement [**Reference17**, **Reference18**]. LEP was equipped with four detectors around the ring; ALEPH(Apparatus for LEP Physics), DELPHI (Detector with Lepton and Hadron Identification), OPAL (Omni-Purpose Apparatus for LEP) and L3 (Letter of Intent 3). These experiments reported many precise results of the Standard Model, specially measuring the mass of W and Z boson and confirming three neutrino species consistent with Standard Model [**Reference18**].

Around 2000, Large Hadron Collider [**Reference19**] at CERN construction started to look for last missing patch of electroweak unification, the Higgs boson. LHC is equipped with four major detectors; ATLAS (A Toroidal LHC Apparatus), CMS (Compact Muon Solenoid), ALICE (A Large Ion Collider Experiment), LHCb (Large Hadron Collider beauty). LHC recorded data of proton-proton collisions at center-of-mass energies 7 and 8 TeV in 2011 and 2012. After a long search , the CMS and ATLAS collaboration, two general purpose experiment at LHC, announced the exist of the heavy scalar boson on 4 July 2012 [**Reference20**, **Reference21**]. Further test to measure of the properties of this scalar boson immediately begins, in March 2013, CERN announced the tentative confirmation of consistency of its properties with SM [**Reference22**, **Reference23**, **Reference24**, **Reference25**].

2.5 Beyond Standard Model

Although SM has passed numerous experimental test with great precision, it is not considered to be theory of everything. There are many phenomenons which are not explained by SM i.e Hierarchy problem, existence of dark matter and dark energy, neutrinos masses etc. There are several theoretical extension posed in Standard Model such as Minimal Super Symmetrical Model(MSSM) and Next-to-Minimal Super Symmetric Standard Model (NMSSM) or completely new theoretical description, such as string theory, extra dimension. The simplest broadening of the SM Higgs sector is the the electroweak singlet (EWS) model [**Reference26**]. In this model, a heavy singlet is defined along with SM scalar Higgs field and then two CP-even Higgs bosons comes out as result of spontaneous symmetry breaking. Anther generic extension to the SM Higgs is the 2 Higgs Doublet Model (2HDM), which introduces two scalar, complex doublet ϕ_1 and ϕ_2 [**Reference27**]. Spontaneous symmetry breaking results five Higgs boson, a neutral CP-odd one (A), two CP-even particles (h and H) and two charged ($H^{\pm}$). Experiments are looking for any direct or indirect hints of new physics. Some of main aspects where SM is unable to explain are discussed below.

2.5.1 Hierarchy Problem

The larger discrepancy between the features of the gravity and the weak force is known as "Hierarchy Problem". There is no convincing explanation of the problem that why the gravity force is 10^{32} times weaker than weak force. The field theory breaks down because of existence of quantum gravity at plank scale ($M_p \approx 10^{19}$ GeV). Writing the Standard Model Lagrangian for the Higgs mass will lead to following expression below

$$m_H^2 = m_{oH}^2 + \delta m_H^2 \tag{2.25}$$

here m_{oH}^2 represents mass parameter exist in the Standard Model Lagrangian, and Δm_H^2 represents the quantum radiative corrections to the mass that are predicted to be heavier. These corrections can be found as

$$\Delta m_H^2 \approx \frac{\lambda^2}{16\pi^2}\Lambda^2 \tag{2.26}$$

here λ represents the Yukawa coupling constant, and Λ represents the energy at which the Standard Model is invalid, or about 1 TeV. This was also one of the reason to expect Higgs boson mass at order of few hundred GeV. The Super symmetry (SUSY) provide the solution of Hierarchy problem by predicting the existence of heavier super partners of all SM particles (bosons and fermions). Several direct and indirect searches has been conducted at LHC to hunt for SUSY, but no evidence has been found so far.

2.5.2 Dark Matter and Dark Energy

Latest observation from cosmological experiments shows that universe is composed of 4.9% ordinary matter, 26% dark matter and 70% dark energy. Standard Model does not provide any clue about existence of dark matter and dark energy.

2.5.3 Matter Antimatter Asymmetry

Today, if we look around in the universe, we see matter dominance over antimatter. At the time of Big-Bang matter and antimatter are supposed to be produced in equal quantities. This give rise to one of the greatest mysteries, where all antimatter has gone. Standard Model is unable to explain this imbalance between matter and antimatter.

2.5.4 Grand Unification and Beyond

One of main goal of the particle physics, is to unify all fundamental force at some energy scale to describe the theory of everything as Standard Model unify electromagnetic and weak force in to electroweak. But Standard Model does not provide any clue about unification of gravity along with other fundamental force generally referred as Grand Unification Theory (GUT). This implies that Standard Model itself is a patch of Grand Unification. The search for any new physics (beyond Standard Model) is important and justified that could improve our understanding about these unsolved mysteries.

2.6 Higgs Boson as a Probe to BSM

After the discovery of the Higgs boson, it became very natural and important to measure Higgs boson properties as precise as possible. Measurements of the Higgs boson properties provide a direct probe into SM Higgs sector perturbative QCD calculations at the TeV energy scale.

The couplings of the Higgs boson with fermions are predicted by SM through Yukawa interactions. Quarks and leptons acquire masses through Yukawa interactions with Higgs boson. While in SM the coupling strength of Higgs boson with any other particle is directly proportional to the mass of that particle. The masses of the leptons and quarks are well know by experiments, so it possible to estimate the decay rates of Higgs to fermions precisely. Experimental measurements can test these prediction which will provide a direct hint to BSM if any deviation is observed.

The cross sections measurement of the Higgs boson illustrate the possibility of the occurrence of that process. Any deviation from SM prediction in integrated or differential cross sections will guide to BSM effects. The observation and studies of the Higgs boson decays like gluon gluon fusion ($\mathrm{gg} \to \mathrm{H}$), vector boson fusion (VBF), associated production with a vector boson (WH, ZH) and top quark pair ($t\bar{t}\mathrm{H}$) also shed light on the coupling of the Higgs with vector boson, top quark etc.

Chapter 3

Experiment

3.1 The Large Hadron Collider

The Large Hadron Collider, the most powerful particle accelerator of the world that has tendency to smash two protons beams at the energy up to 7.0 TeV per beam. It is the most advanced, complex and largest experiment in the history of mankind, designed and constructed by a collaborative team of more than 10,000 technician, engineers and scientists from over 100 countries. The LHC was constructed at CERN from 1998 to 2008, in a tunnel of 27 Kilometers (17 mi) circumference below the France-Switzerland border close to Geneva, Switzerland. The LHC is accommodated in a 3.8 meters wide circular tunnel at a depth varies from 75-150 meters underground. Previously, this tunnel also hosted the Large Electron-Positron collider constructed from 1983 to 1988.

The first proton beam was injected into LHC on September 10, 2008. The LHC was able to fire the protons around the tunnel in the segments of three kilometers each time. The initial test to circulate the beams in clockwise and counterclockwise throughout the collider was also successfully completed by the same day. On 19 September 2008, an electric defect triggered the magnet quench in about 100 bending magnets. The consequences of the incident led to an explosive loss of six tonnes of liquid helium that damaged 50 superconducting magnets and their mounts. Three months later, CERN unfolded the details of the incident and published technical report. Particle accelerator was repaired after this incident that took about an year. Finally, on November 20, 2009, LHC became operational again and a low energy beam was successfully rotated within tunnel. On November 30, shortly after, LHC achieved 1.18 TeV per beam and became world's strongest particle accelerator, defeating the Tevatron's record of 0.98 TeV. In 2010, LHC kept boasting the beam energies and reached up to 7 TeV center-of-mass energy (3.5 TeV per beam). In 2011 and 2012, LHC consistently delivered data at 7 and 8 TeV center-of-mass energies respectively. During 2013 and 2014 , the machine was shutdown to do major parts upgrade on the LHC machine and detectors installed on it. In 2015, upgraded LHC started again at 13 TeV center-of-mass energy close to full stipulated operating energies (14 TeV). After successfully delivering proton-proton collisions data during Run 2 till the end of 2018, LHC machine is shutdown for Long Shutdown 2 (LS2) upgrades and is expected be operational again for Run3 in 2021 after several upgrades.

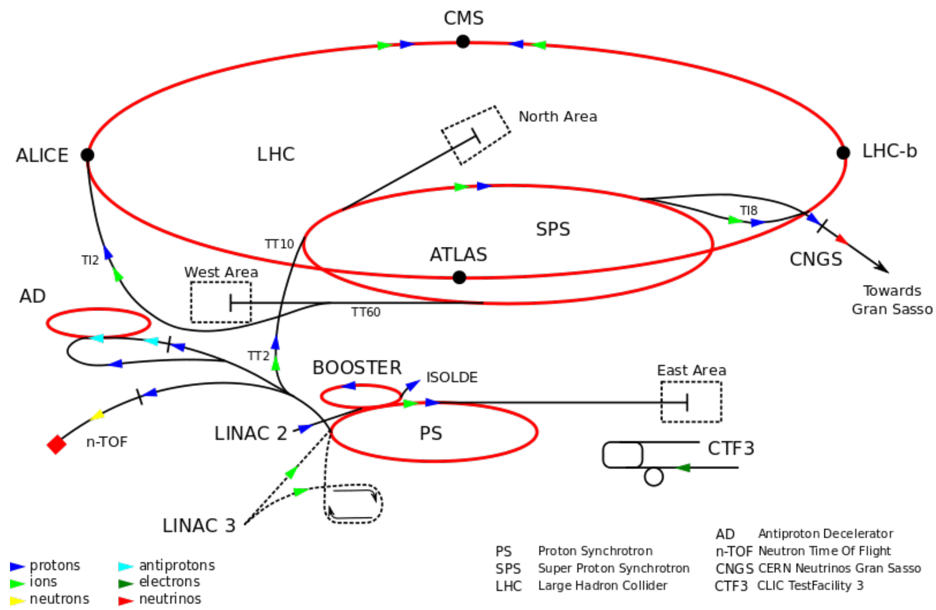


FIGURE 3.1: LHC experiment at CERN

3.1.1 Machine Layout and Performance

Machine Layout

The basic layout of LHC and LEP machines are similar, having eight linear and eight arcs sections shown in Figure 3.2 [**Reference28**]. The LHC is designed to have two separate rings containing clockwise and counterclockwise proton beams, equipped with separate magnetic fields and vacuum chambers. These two rings only intersect each other at the points where detectors are installed. Point 1 and point 5, opposite to each other, where two general purpose detectors ATLAS and CMS are installed respectively. The other two important detectors are installed at point 2 (ALICE) and point 8 (LHCb) designed to study heavy-ion and B Physics respectively. The two protons beams are injected in the vertical plane in both rings close to point 2 and point 8. Among 8 sections, there are four with no beam crossing at all. Point 3 and point 7 are equipped with two independent collimation systems for both rings. At point 4, two RF system are installed one for each beam. While at point 6, kicker magnets are installed to dump the beams.

Performance Goals

The aim of the LHC to test Standard Model Physics to ultimate precision and hunt for any kind of new physics up to 14 TeV. The total total number of events (N) of a certain process produced at LHC can be given as

$$N = \mathscr{L}\sigma_{process} \tag{3.1}$$

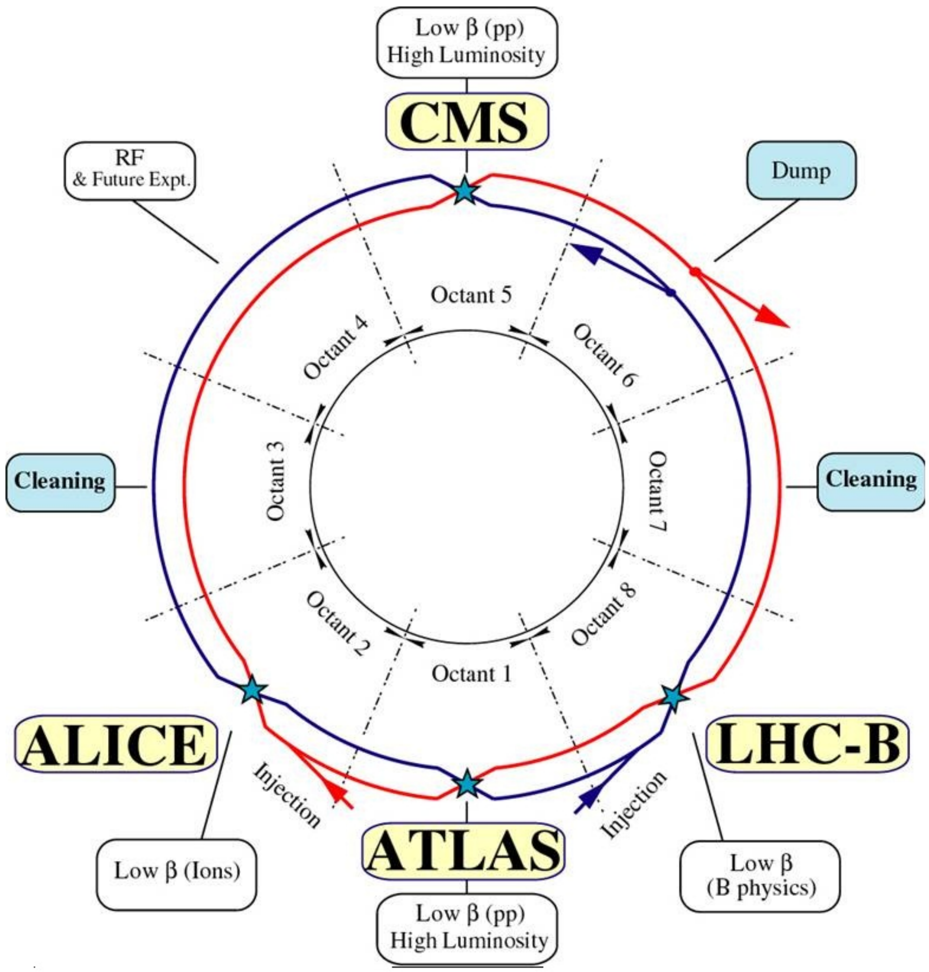


FIGURE 3.2: Schematic layout of the LHC

Here $\mathscr{L}$ and $\sigma_{process}$ represent machine luminosity and the cross section for a certain process respectively. Cross sections for all processes produced by proton-(anti) proton collisions at machine luminosity $\mathscr{L} = 10^{33}\ cm^2\ s^{-1}$ are shown in Figure 3.3. The Higgs boson production at LHC is rare as compared to other QCD and weak boson (W, Z) productions which act as backgrounds in the Higgs boson searches. In order to reconstruct Higgs boson at LHC, one need to exploit specific final states providing enough signal to background discrimination. From Figure 3.3 its clear that Higgs boson production rises with increase in center-of-mass energies. So increase in machine luminosity and center-of-mass energies enables us to get more statistics of Higgs boson. The Nominal luminosity for CMS and ATLAS experiment is $\mathscr{L}$ = $10^{34}\text{cm}^2\text{s}^{-1}$, and LHC nominal center-of-mass energy is $\sqrt{s} = 14$ TeV. The cross section in Figure 3.3 [**Reference29**] shows that LHC has possibility to produce a single Higgs boson after every 100 s at mass 500 GeV. To get ultimate number of Higgs bosons produced at certain luminosity, one need to multiply this with specific branching ratio, selection and reconstruction efficiencies.

Nominal Luminosity and Beam Parameters

The machine luminosity is function of beam parameters and can be expressed as below

$$\mathscr{L} = \frac{\gamma_r N_b^2 f_{rev} n_b}{4\pi\epsilon_n\beta^*}F \tag{3.2}$$

here γ_r, N_b, n_b represents relativistic factor, number of particles per bunch and number of bunches per beam. While β^* represents the beta function at the collisions point and F is the factor to account for geometrical depletion in luminosity due to cross angle at interaction point (IP). This factor can be written as below

$$F = \left(1 + (\frac{\theta_c\theta_X}{2\sigma^*})^2\right) \tag{3.3}$$

where θ_c, θ_z and σ^* are crossing angle at interaction point, root mean square (RMS) bunch length and σ^* the transverse RMB beam size at at IP. The expression above is valid for two circular beams with same parameters.

The two major experiment at LHC, the CMS [**Reference30**] and the ATLAS [**reference31**] are two general purpose detectors designed to operate at high luminosity of $\mathscr{L} = 10^{34} cm^2 s^{-1}$. The summary of the parameters to obtain this nominal luminosity is summarized in Table 3.1. Such high luminosity can not be attained by proton-anti-proton collisions, leaving only possibility of proton proton collisions for LHC machine. To align two opposite moving proton beams in both rings for collisions needs dipole magnetic with opposite fields direction. The LHCb [**Reference32**] detector is specially designed to study B Physics at the peak luminosity of $\mathscr{L} = 10^{32} cm^2 s^{-1}$. At last, the ALICE [**Reference33**] is the detector to study heavy-ion collisions targeting peak luminosity of $\mathscr{L} = 10^{27} cm^2 s^{-1}$.

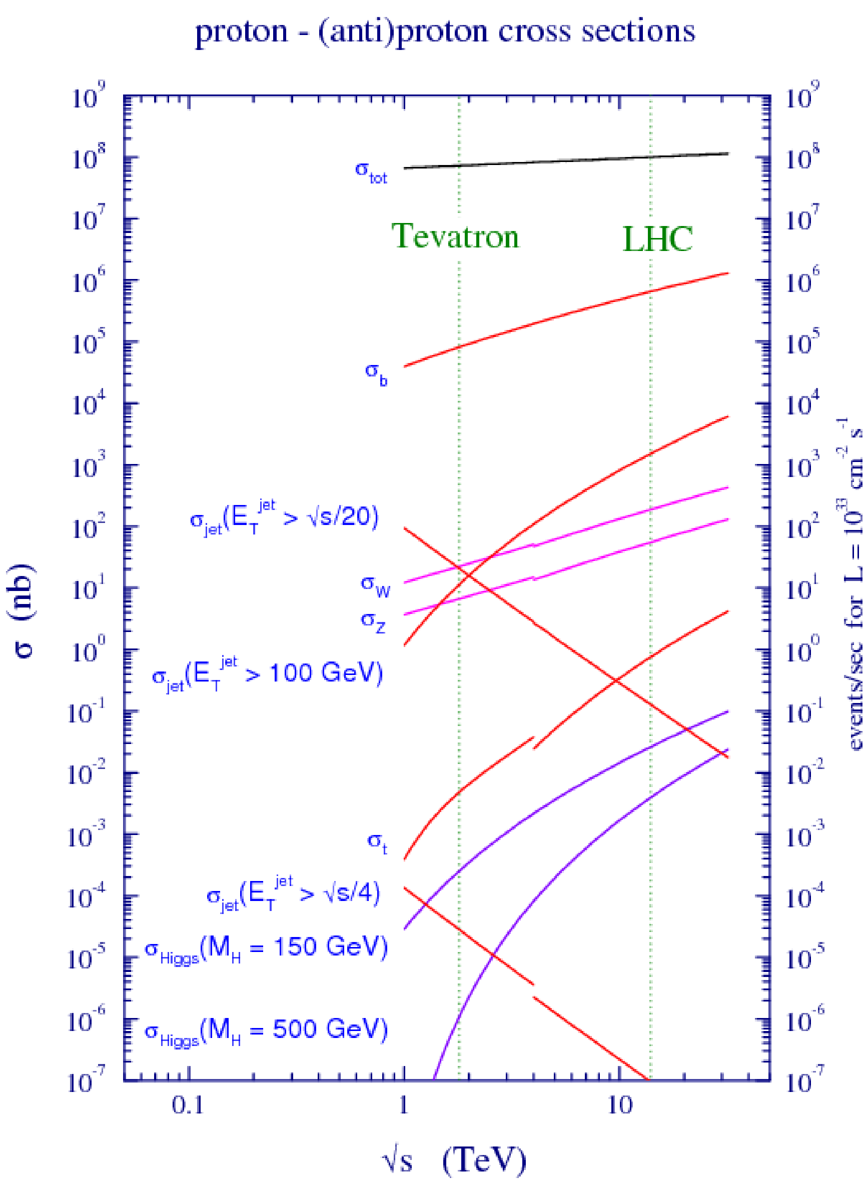


FIGURE 3.3: Cross section values for different process corresponding to machine luminosity $\mathscr{L} = 10^{33}\,cm^2\,s^{-1}$.

TABLE 3.1: Parameters values to obtain LHC nominal luminosity for detectors.

Parameter	Value
circumference	26.7 km
center-of-mass energy ($\sqrt{s}$)	14 TeV
Nominal luminosity	$10^{34}cm^2s^{-1}$
Bunch spacing	25 ns
distance between bunches	7.5 m
Number of bunches in ring	2808
Number of protons per bunch	1.15×10^{11}
beta function at the collision point(β^*)	0.55 m
Transverse RMS beam size at impact point(σ^*)	16.3 μm
Dipole field at 7 TeV(B)	8.33 T
Dipole temperature (T)	1.9 K

3.1.2 Data Taking from 2015 to 2018 (Full Run 2)

After successful data taking during Run 1, LHC machine was shutdown for two years in 2013 and 2014 for major upgrades in LHC machine and detectors. LHC started again in 2015 at center-of-mass energies to 13 TeV(close to its design value i.e 14 TeV). LHC delivered 4.2 fb^{-1}during 2015 at 13 TeV center-of-mass energy as shown in Figure 3.4(a). Subsequently, it delivered 41.0 fb^{-1}, 49.8 $\mathrm{fb}^{-1}$and 67.9 fb^{-1}during the years 2016, 2017 and 2018 as shown in the figures Figure 3.4(b), Figure 3.4(c) and Figure 3.4(d) respectively.

3.2 The Compact Muon Solenoid Detector

The Compact Muon Solenoid is 28 m long, 15 m high and 14,000 tonnes heavy general purpose particle detector installed on LHC ring. CMS detector is capable to study a large range of physics from Standard Model Physics (including Higgs Physics) to search for extra dimensions and the dark matter. CMS is efficient enough to cover a large range of physics with energy ranges from a few 100 GeV to TeV scale [**Reference34**]. CMS and ATLAS experiment shares the same physics goal though both independent experiment uses different technical solution and detectors design. Figure 3.3 shows that the total proton-proton cross section at $\sqrt{s} = 14$ TeV is approximately 100 mb. Keeping in mind the nominal luminosity will lead to have approximately 10^9 inelastic events/s. To deal with such high rate collisions is a experimental challenge. CMS trigger system helps to select online events and reducing the huge rate to approximately 100 events/s to be stored and used for physics analysis later. The read-out and trigger system designed to meet the short bunch crossing time, 25 ns.
Being CMS as general purpose detector, followings are the main requirements to fulfill physics goal of LHC:

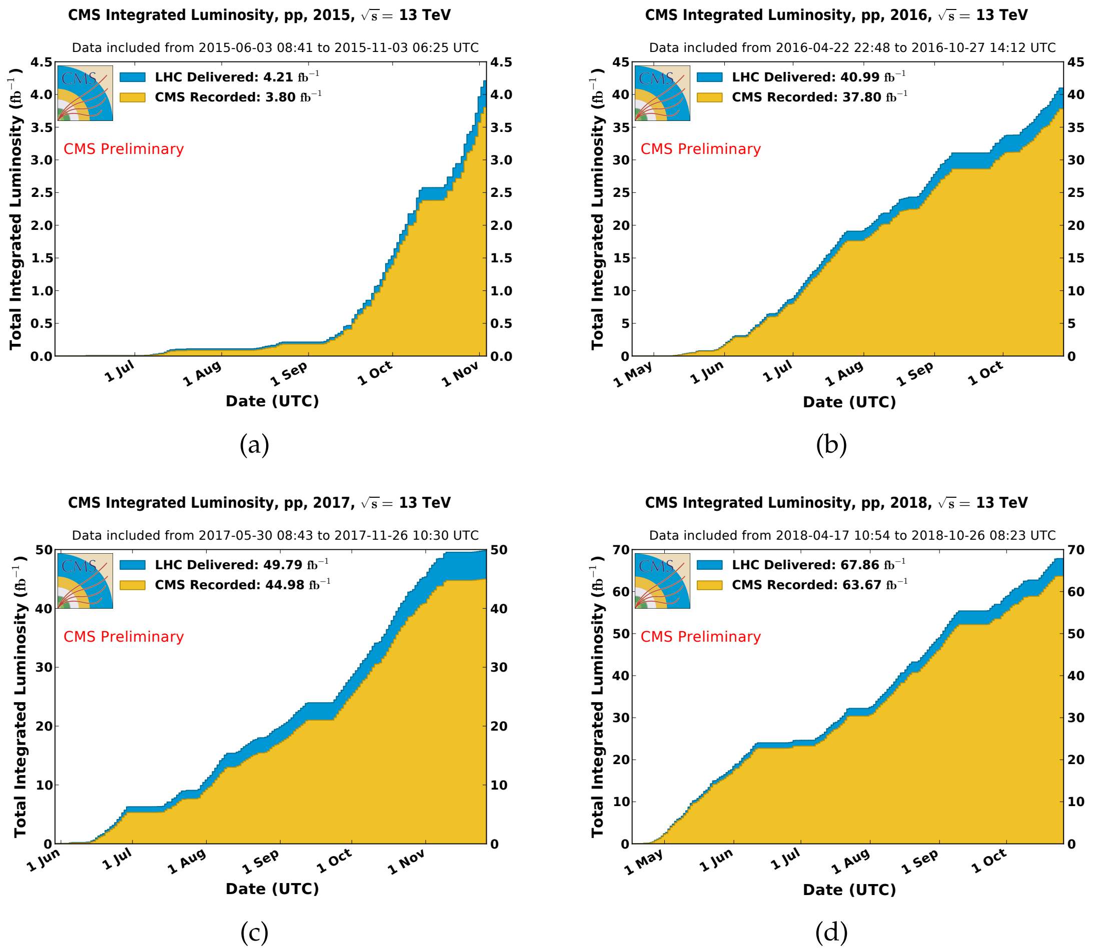


FIGURE 3.4: Integrated luminosities of LHC delivered (blue) and CMS recorded (yellow) data for 2015(a), 2016(b), 2017(c) and 2018(d).

- Good identification and momentum resolution of muons covering wide range of momenta and angles and determination of the charge on muons with ultimate precision with momentum up to 1 TeV.
- Healthy inner tacker reconstruction efficiency and resolution of charged particle momentum. Efficient triggering and discrimination of τ's and b-jets.
- Good dielectron and diphoton mass and energy resolutions.
- Good dijet-mass and missing-transverse-energy (MET) resolutions .

One of the main features of a particle detector is the choice of its magnetic field configuration. Stronger magnetic field is needed to measure high energy charged particle momentum with precision. In CMS, this is achieved by selecting the superconducting magnets.

CMS overall design is shown in Figure 3.5. A 13-m long and 6-m inner-diameter superconducting solenoid producing a magnetic field of 3.8-T is placed at the heart

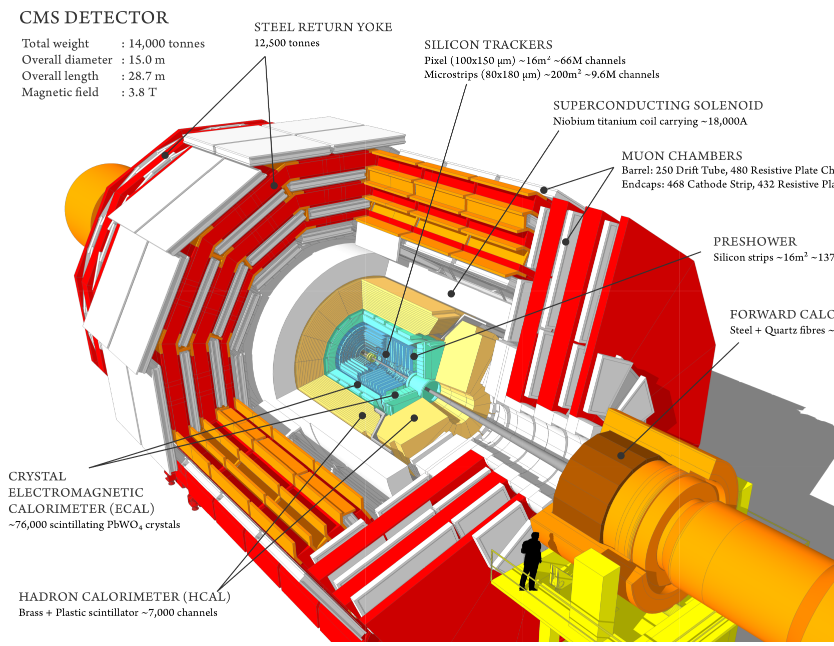


FIGURE 3.5: The CMS detector sectional view.

of CMS. The inner tracker, electromagnetic and hadronic calorimeters both are fitted inside the large bore of the magnet. Inner tracker of the CMS have large track multiplicity as being closest to beam pipe where collisions happens. The electromagnetic calorimeter (ECAL) is made of lead tungstate crystals that covers pseudorapidity up to $|\eta| < 3.0$. After ECAL, hadronic calorimeter is installed which is made of brass. Finally, Muon system is installed at most outer layer of CMS detector. Details of sub-detectors, trigger and data acquisition systems would be described in next Sections.

3.2.1 CMS Detector Geometry

The coordinate system would be described in this Section and will be used throughout. The CMS detector is like a closed cylinder around beam axis (Z axis) as illustrated in Figure 3.5. The center of the experiment lies inside beam pipe where two beams collides; the y axis is directed vertically upwards; and the x axis points horizontally towards the center of the LHC ring. In the transverse (xy) plane, the radial component is represented by r and the azimuthal angle measured from the x axis is denoted by ϕ. The polar angle θ is measured from the z axis as shown in Figure 3.6. The direction of any particle in CMS detector is described by (η, ϕ), where η is called pseudorapidity and defined as $\eta = -\ln \tan(\theta/2)$. As the LHC is protons collider, which contains partons (quarks and gluons) that carry different fractions of proton's total momentum. As a results, events in CMS detector will have boosted center-of-mass as compared to lab frame along z axis. New defined quantity pseudorapidity is invariant under these boosts. In short, pseudorapidity

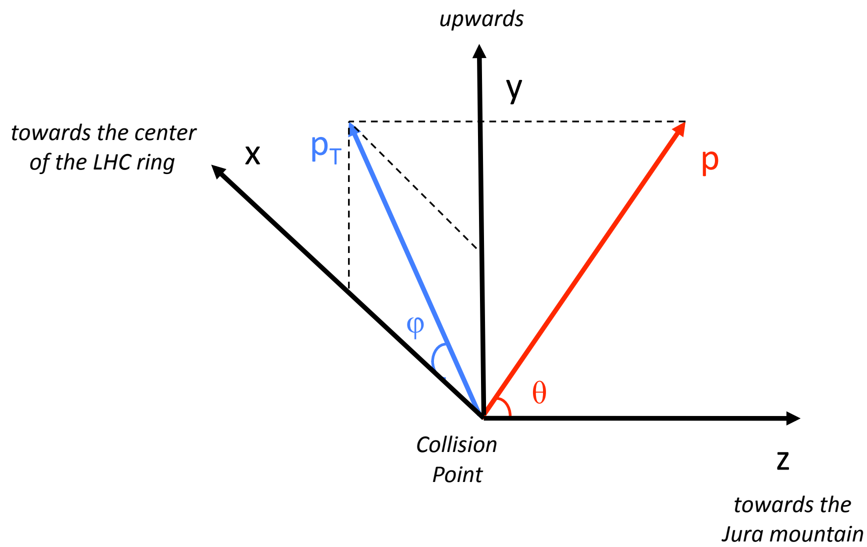


FIGURE 3.6: The CMS Experiment coordinate system.

is approximation of rapidity in case of high energetic particles which are assumed to be massless and is defined below

$$y = \ln(\frac{E + p_z}{E - p_z}) \tag{3.4}$$

$$\approx \ln(\frac{1 + \cos\theta}{1 - \cos\theta)} \tag{3.5}$$

$$= \ln(\cot^2\theta). \tag{3.6}$$

Pseudorapidity also helps to distinguish the barrel (central) and endcaps regions of sub detectors. The barrel and endcap regions are defined by $|\eta| < 1.5$ and $1.5 < |\eta| < 3.0$ respectively. Formally, electromagnetic calorimeter and tracking system provides a coverage a region up to $|\eta| < 2.5$, while muon station covers up to $|\eta| < 2.4$. So in CMS muons and electrons are selected with $|\eta| < 2.4$ and $|\eta| < 2.5$ respectively. As there is no transverse momentum (along y axis) before collisions leads to have transverse momentum approximately zero. The transverse energy of any particle is orthogonal to z axis, defined as $E_T = E\sin\theta$ and similarly the transverse momentum $p_T = p\sin\theta$. For a massless particle approximation, electrons and muons, transverse momentum is equal to transverse energy i.e. $E_T = p_T$.
As the CMS detector is hermetic, the missing transverse energy is understood as the energy of those particle which escaped from detector. These particles can be new physics particles, neutrinos and neutralinos which have no or very little interaction with detector material.

TABLE 3.2: Types of detectors installed in CMS tracking system.

Section	Radial distance (cm)	Detector type	Cell size
Pixel	$r < 10$	pixel	$100\mu m \times 150\mu m$
TIB+TID	$20 < r < 55$	microstrips	$10cm \times 150\mu m$
TOB+TEC	$55 < r < 110$	strips	up to $25cm \times 180\mu m$

3.2.2 Inner Tracking System

The CMS inner tracking system is most inner sub-detector surrounding the beam pipe and interaction region as shown in Figure 3.5. It has diameter of 2.5 m and a length of 5 m feeding $|\eta|$ coverage up to 2.5. The purpose of inner tracking system is to measure the charged particles trajectories needed to calculate their momentum and reconstruction of secondary vertices. The whole tracker is placed in a uniform magnetic field of 3.8-T provided by CMS solenoid. At LHC nominal luminosity of $10^{34} cm^2 s^{-1}$, there is an average of more than 20 proton-proton interactions (for 2018 it was above 30) leading to average about 1000 particles emerging after every bunch crossing, 25 ns. To distinguish the trajectories and assign correct bunch crossing, fast response and high granularity detector technology is needed. To protect against radiation damage caused by intense particle flux and priority to minimize the material to avoid from multiple scattering, nuclear reaction, bremsstrahlung and photon conversion, a compromise was found to choose silicon detector technology. CMS inner tracker is the largest silicon tracker ever made with about 200 m^2 of active silicon.

CMS tracker detector should be robust which can reconstruct tracks of charged particle in whole pseudorapidity range with momentum above 1 GeV to achieve LHC physics goals. Tracker also helps to identify other particles like electron, muons along with electromagnetic calorimeter. CMS inner operate under intense particle flux with hit density varies from 1 MHz/mm^2 at radial distance of 4cm decrease to 60 KHz/mm^2 at radial distance of 22cm and 3 KHz/mm^2 at distance of 115cm. Keeping in view of hit densities, different types of detectors are installed at various radial distances of inner tracker listed in Table 3.2. Silicon detectors are chosen due its unique properties of fast response and good spatial resolution which enables it to work efficiently even in high radiation environment. To limit the damage mainly by radiation the whole tracker system is kept at about -20° C.

The CMS inner tracker schematic drawing (as it stood before phase 1 upgrades) is shown in Figure 3.7 [**Reference34**]. Tracker most inner region is the pixel detector that is nearest to the interaction point consists of disk modules on each side and barrel layers. It provides accurate traces in $r - \phi$ and z with a pixel cell size of $100 \times 150 \mu m^2$ covers the $|\eta|$ range up to 2.5. Pixel detector also plays its role for impact parameter resolution that helps to reconstruct secondary vertex [**Reference34**]. Pixel detector delivers three spatial point with excellent precision throughout the coverage of pixel.

During Run 1, some dynamic inefficiencies were observed in the pixel subdetector, of the order of about 5% at an event pile-up of 40 in the first barrel layer,

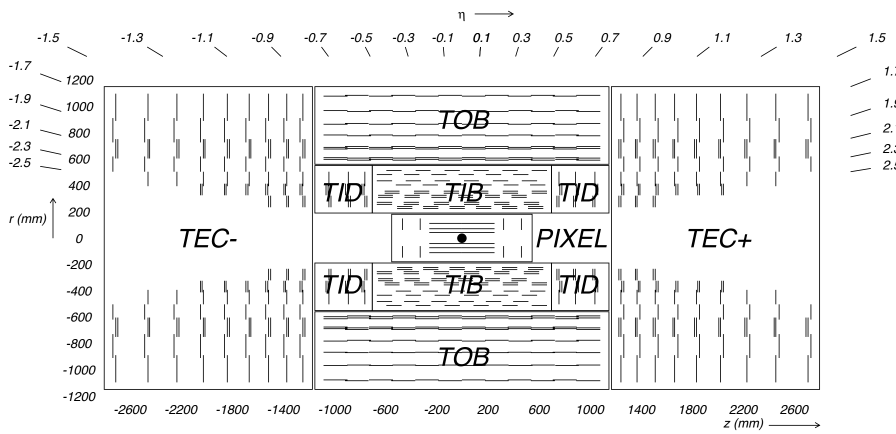


FIGURE 3.7: Schematic diagram of CMS tracker system. Lines represent detector modules (before phase 1 pixel upgrades).

due to the limited size of the readout bandwidth. A new pixel detector was installed in March 2017 Ref. see comment, equipped with one additional barrel layer and one additional forward disk. In the new pixel detector, the innermost layer is closer to the interaction point and the outermost one is further away from it as it is shown in Fig. 3.8 It also uses new Readout Chips (ROCs) with reduced data loss at higher collision rates with respect to Phase 1 (160 Mbit/s), and the bandwidth of the readout electronics was increased approximately by a factor of eight (320 Mbit/s). In addition, DC-DC power converters were installed to allow reusing the existing fibers and cables. Hence, the upgraded pixel detector has a higher rate capability and number of channels, providing increased tracking efficiency, reduced fake rate, and improved impact parameter resolution. In the figure 3.9 clear improvments in the dynamic efficiency of 2017 (with pixel upgrades) with respect to the 2016 (without pixel upgrades) can be seen.

Second layer detector after pixel is silicon strip detector installed at radial distance from 20cm to 116cm. This detector surrounds both sides of pixel with its tacker inner barrel (TIB) and inner tracker disk (TID) regions. There are 3 disks on each endcap and 4 barrel layers where silicon microstrips of thickness 320μm are installed. Microstrips are aligned along with beam axis in barrel region and in radial direction on the disks. The strip pitch in barrel region varies from 80μm to 120μm from inner to outer barrel layer. In the endcap region strip pitch changes from 100μm to 141μm. Finally, the most outer part of CMS tracker system is tracker outer barrel (TOB) installed at radial distance from 55cm to 116cm containing 500μm thick 6 barrel layers with strip pitch of 183μm and 122μm. TOB provides z coverage up to 118 cm, beyond this point 9 disks of tracker endcaps (TEC+, TEC-) are located that covers 124cm$< |z| <$282cm and 22.5cm$< |r| < 113.5$cm.

To measure the momentum in CMS tracker, we simply exploit the property of bending of the charged particles in the magnetic field. High momentum particles will bend less in detector and vice versa, making resolution worse due to less curvature.

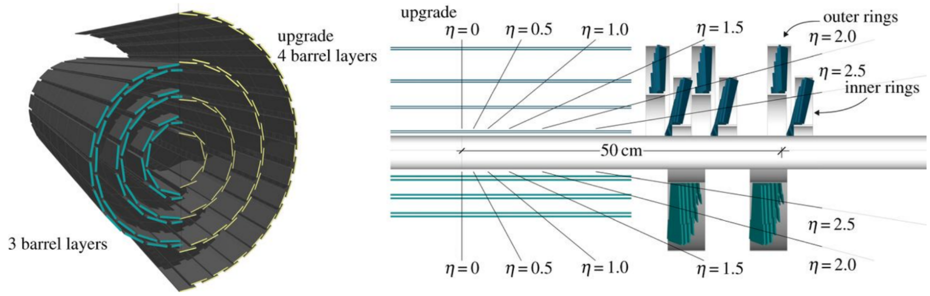


FIGURE 3.8: Schematic drawing of the upgrade of the pixel detector. The lower half shows the old pixel detector, and the upper half represents the newly-installed detector.

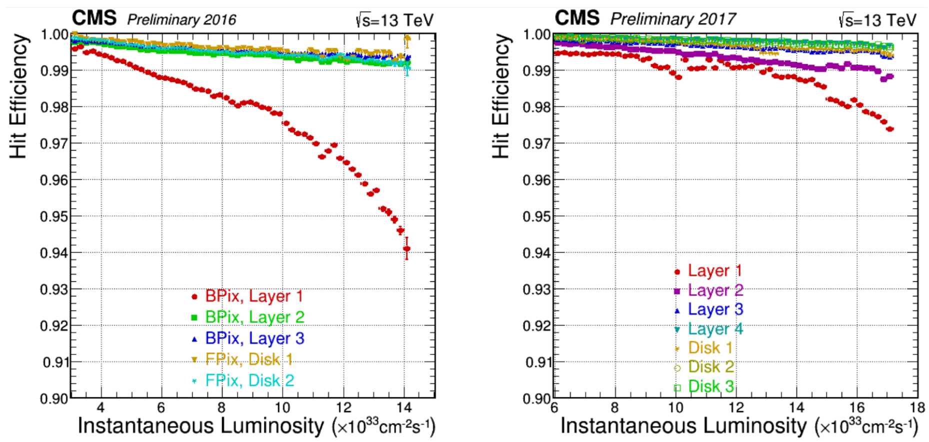


FIGURE 3.9: . Hit efficiency of the pixel detector as a function of the instantaneous luminosity measured with 2016 data (left) and measured with 2017 data (right) Ref.in comments

The upgraded pixel detector improved $\mathrm{H} \rightarrow 4\ell$ analysis as there was significant increase in number of leptons in an event improving the reconstruction of Z candidates and hence an increase in the selection efficiency upto much extent. (Ref. CMS-TDR-011)

3.2.3 Electromagnetic Calorimeter

Electromagnetic calorimeter (ECAL) is the cylindrical shape hermetic homogeneous calorimeter made of lead tungstate ($PbWO_4$) crystals. Barrel region of the ECAL (EB) is providing pseudorapidity coverage of $|\eta| < 1.479$ and contains 61200 $PbWO_4$ crystals. Endcap region of the ECAL (EE) is providing pseudorapidity coverage up to $\eta < 3$ containing 7324 crystals in each of the two endcaps. The crack region between barrel and endcaps ($1.4442 < |\eta| < 1.56$) as shown in Figure 3.10 **[Reference34]** is excluded for physics analysis. A preshower detector is installed in endcap regions to distinguish between single and diphoton close to each other e.g neutral pion decay ($\pi^\circ \rightarrow \gamma\gamma$). Specially designed Avalanche photodiodes (APDs) and Vacuum phototriodes (VPTs) are used in barrel and endcaps region respectively ensure efficient working under high magnetic field. One of the ECAL key feature in the design is to reconstruct electrons and photons with high resolution enabled us to discover Higgs boson in both $H \rightarrow \gamma\gamma$ and $H \rightarrow ZZ \rightarrow 4\ell$ favorable channels.

ECAL Crystals and Geometry

To make a compact and high granularity calorimeter which could fit inside the CMS solenoid, lead tungstate was the best option which is denser than steel (8.28 g/cm^3) with short radiation length (X○) of 0.89 cm [**cms:ECALTDR**]. The scintillator ($PbWO_4$) emit light directly proportional to the energy of the incident particle through ionization. Fast response (same order of LHC bunch cross crossing i.e. 25ns) of $PbWO_4$ makes it ideal for ECAL. In barrel region, there are 8.14 m^3 active crystals installed weighting up to 67.4 tonne. The each crystals cross section is measured to be 22 × 22 mm^2 in barrel and 28.62 × 28.62 mm^2 at endcaps. The length of the crystals in barrel and endcaps is 230mm and 220 mm which is 25.8 and 24.7 times of the radiation length.

The amount of emitted light from scintillator amplification of photodiodes both are temperature dependent, the nominal temperature of ECAL operation is 18°C. For this reason, cooling system has been installed in ECAL to maintain the temperature at nominal value. The cooling system rotates water at 18°C through parallel grid of aluminium pipes a to maintain the temperature in the detector. In the barrel region, each supermodule has its independent supply of water. The cooling system of the ECAL work efficiently and maintain the temperature of $18 \pm 0.05°C$.

TABLE 3.3: ECAL crystal details at endcaps and barrel regions.

	Barrel	Endcaps
Number of crystal	61200	14648
Crystal cross section in (η, ϕ)	0.0174×0.0174	variable
Cross section of front face of crystal	$22 \times 22mm^2$	$28.62 \times 28.62mm^2$
Cross section of rear face of crystal	$26 \times 26mm^2$	$30 \times 30mm^2$
Crystal length	230 mm	220 mm
Crystal length(in terms of $X_\circ$)	25.8$X_\circ$	24.7$X_\circ$

Photodetectors

The requirements on photodetectors are to be fast, robust and able to work efficiently in a region that contains 3.8-T magnetic field. Considering these requirement, different types of photodetectors are installed in the barrel and the endcaps regions. In the barrel region, a pair of avalanche photodiodes are mounted with active area of 5 × 5 mm^2 on each crystal. APD are working at gain 50 and in parallel read out. In the endcap region, a specially designed vacuum phototriodes at

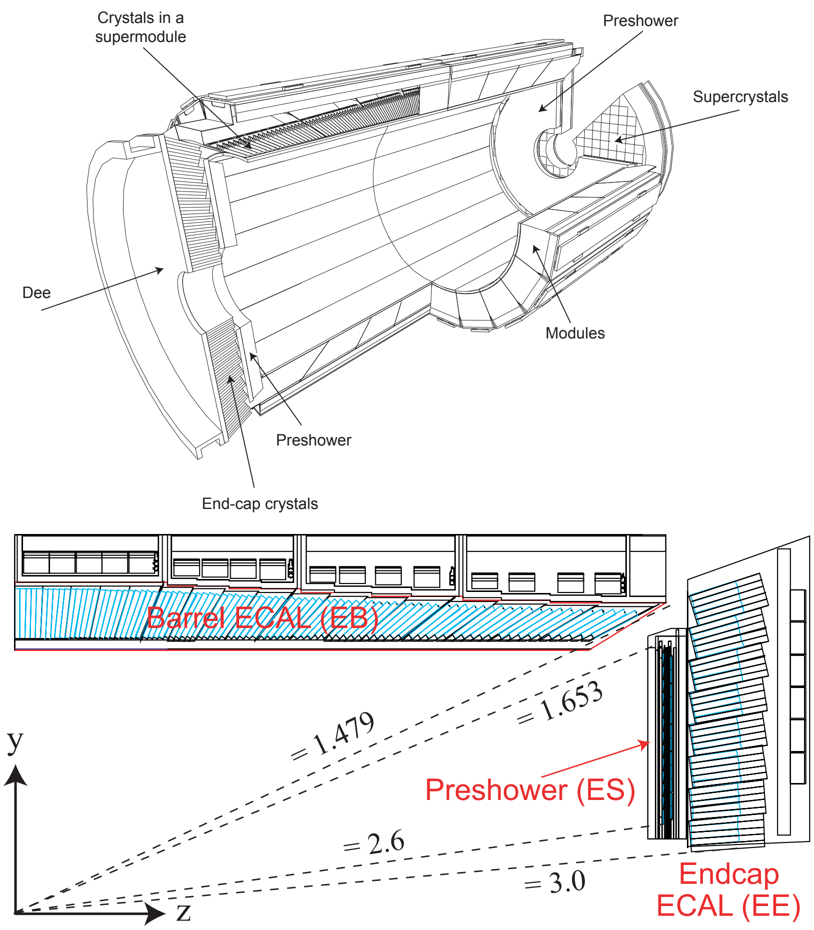


FIGURE 3.10: CMS ECAL layout and pseudorapidity coverage.

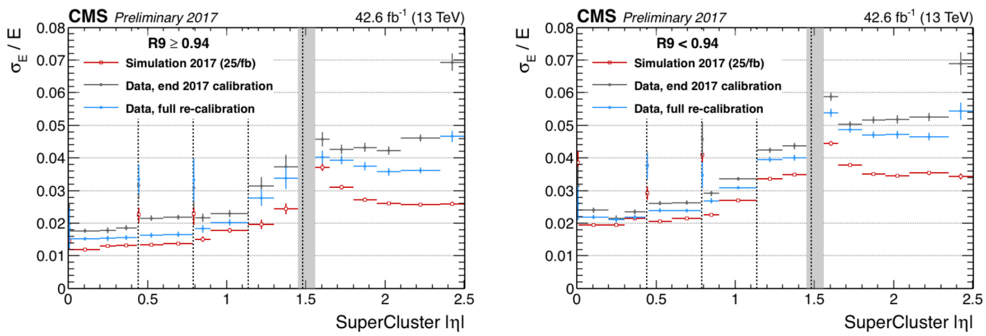


FIGURE 3.11: CMS ECAL energy resolution.

National Research Institute Electron in St. Petersburg for CMS ECAL was used. These are single gain photomultipliers with mean gain of 10.2 at zero field. The diameter of each VPT is 22mm. The total active area is almost 280mm^2 where one VPT is attached at the back of every crystal. Their fine coper anode makes them capable to work under intense magnetic field. As compared to APDs, VPTs has lower quantum efficiency and internal gain which is balanced out with their larger back face crystal coverage. The summary of number of the crystals with their dimensions are summarized in Table 3.3.

Preshower

The purpose of the installation of preshower detector is to to identify neutral pions in the endcaps regions. The preshower provides the pseudorapidity coverage of $1.653 < |\eta| < 2.6$. It also helps to distinguish between electron and minimum ionizing particles (MIPs) and the position determination of electrons and photon due to its high granularity.

Energy Resolution

The energy resolution can be defined as

$$\left(\frac{\sigma}{E}\right)^2 = \left(\frac{S}{\sqrt{E}}\right)^2 + \left(\frac{N}{E}\right)^2 + C^2 \tag{3.7}$$

where C is the constant term extracted by fitting of measured points to the function, N is noise term and S is stochastic term. Recently relatively energy reslution study of ECAL using 2017 data and MC has been performed using Z events decaying to a pair of electrons. The resolution plots for bremsstrahlung electrons (R_9>0.94 and R_9<0.94) is separately shown for data and MC in the Figure 3.11 **[comments]**. For H $\rightarrow 4\ell$ analyses, excellent energy resolution for leptons is important for precise measurements for which ECAL of CMS plays an important role.

3.2.4 Hadron Calorimeter

CMS being a general purpose detector is capable to study several final state signatures involved in a large range of high energy processes. The hadron calorimeter (HCAL) provides the measurement of energies and positions of hadronic jets (for example neutrons, protons, kaons and pions)[**cms:HCALTDR**]. HCAL being hermetic is also helpful to calculate MET that can be assigned to neutrinos or other invisible particles. There are no gaps or cracks in HCAL to make sure to capture all known particles, leaving MET with only possibility of escaped particles. HCAL of CMS is fitted between ECAL and CMS magnet solenoid at a radial distance from 1.77 m to 2.95 m. HCAL provided a pseudorapidity coverage up to $|\eta| < 3.0$, beyond this point a forward hadron calorimeter (HF) is installed at a distance of 11.2 m from the collision point on both ends extending pseudorapidity coverage to $|\eta| < 5.2$. As HCAL is placed inside 3.8-T of magnetic field, so absorber must be a non-magnetic material. Also, due to requirement on maximum number of interaction length, cost and mechanical property brass has been chosen for CMS HCAL.

As usual HCAL can also be splitted into barrel (HB) and endcap (HE) regions with barrel and endcaps pseudorapidity coverage of $|\eta| < 1.33$ and $1.3 < |\eta| < 3$ respectively. HB consists of 36 similar wedges each weighing 26 tonnes that makeup two half-barrels (HB^+ and HB^-) illustrated in Figure 3.12. The wedges are made of plane brass absorber plates kept along beam axis. All plates are made of brass with physical properties described in Table 3.4 except inner and outer most made of steel for structural strength. Table 3.5 shows the thickness of each layer. HCAL uses the alternate layers of absorber brass and plastic scintillator to measure the particle position, energy and arrival time. The thickness of the brass plates are 79 mm with 99 mm gaps for scintillators installation. Optical fibers collect all the light emitted when a particle passes through scintillator and passed it to readout box. Total energy is calculated by summation of the light over many layers in depths called tower. The scintillator in HCAL are arranged in the from of 70,000 tiny tiles. individual tiles makes scintillator tray by grouping them in ϕ layers to limit the number of readout boxes. These trays works as individual units, so it can be replaced or tested easily. The focus of the design of HCAL endcap was to minimize the cracks between HB and HE rather than single-particle energy resolution.

Forward Calorimeter

The two HF are placed on both sides of CMS to catch up the particle emerging from proton- proton collisions at very small angle relative to beam axis as shown in Figure 3.12 [**Reference34**]. HF is most forward region of the detector where enormous particle fluxes are expected. On average, the energy deposited in HF only is 8 times more as compared to the whole CMS detector. The design of HF calorimeter for such harsh conditions was a considerable challenge. HF will absorb a radiation dose of about 10 MGy after an integrated luminosity of $5 \times 10^5\ \mathrm{pb}^{-1}$at

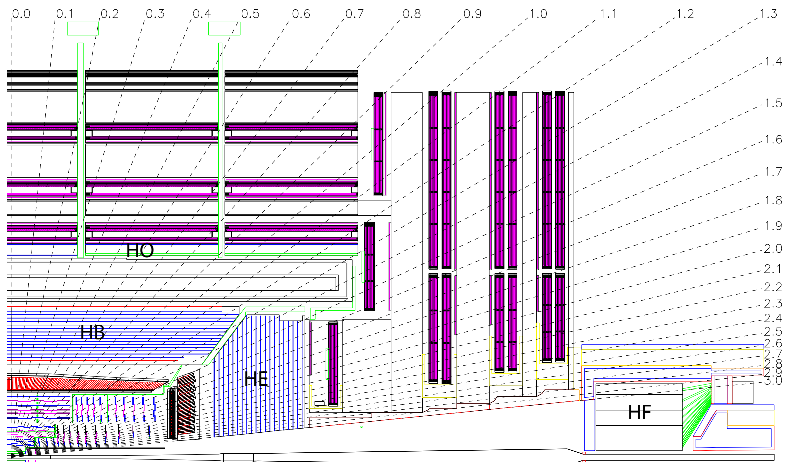


FIGURE 3.12: CMS HCAL layout.

TABLE 3.4: Properties of brass of HCAL.

chemical composition	70% Cu, 30% Zn
density	8.53 g/cm^3
radiation length	1.49 cm
interaction length	16.42 cm

$|\eta| = 5$. The rate of charged hadrons is also extremely high. Because of hard conditions HF material and design is different from rest of the HCAL.

The energy resolution of the HCAL is given as

$$\left(\frac{\sigma(E)}{E}\right)^2 = \left(\frac{90\%}{\sqrt{E}}\right)^2 + (4.5\%)^2 \tag{3.8}$$

TABLE 3.5: HCAL absorber thickness.

layer	material	thickness
front plate	steel	40mm
1-8	brass	50.5mm
9-14	brass	56.5mm
back plate	steel	75mm

and for HF is given as

$$\left(\frac{\sigma(E)}{E}\right)^2 = \left(\frac{170\%}{\sqrt{E}}\right)^2 + (9.0\%)^2. \tag{3.9}$$

There were subsequent upgrades in HCAL during Run 2 era. In 2017, PMTs were replaced with new multi-anode PMTs and additionally electronics replaced for better differentiation of real hit from the muons which hit the PMT window. In 2017, HE of HCAL was upgraded and photo detectors were replaced with latest silicon photo multipliers (SiPMs) for noise elimination and better photo detection efficiency. (reference in comments)

3.2.5 Superconducting Magnet

The CMS central is superconducting solenoid [**cms:MFTDR**, **Reference36**, **Reference37**, **Reference38**] designed in 6 m diameter and 12.5 m long bore producing magnetic field of 3.8-T which is 10,000 time stronger than earth magnet. The CMS solenoid is the largest ever built magnet weighs 12,000 tonnes, powerful enough to provide a strong magnetic field throughout the detector. The CMS inner tracker, ECAL and HCAL are fitted inside the bore of the magnetic field. The task of the magnetic field is to bend the charged particles when they pass through. More the momentum of the charged particles less they bend, so to measure the momentum of high energetic charged particles a stronger magnetic field is required. CMS aim to have excellent momentum resolution with such strong magnetic field specially for muons. The CMS magnetic field is a solenoid made of coil of NbTi wire which supplies the uniform magnetic field when current pass through it. The NbTi becomes superconductor at $-268.5°$ C (4 K) which allows a huge amount of current (10,000 A) to pass through it. To maintain this temperature liquid helium is used. A large bending is achieved in tacker by stronger magnetic field providing precise and efficiencies momentum measurement. The magnetic field direction inside (tracker) and outside the solenoid (muon system) allows to make second momentum measurement for muons.

3.2.6 Muon System

Muon identification in CMS is provided by its outer most layer of sub-detector called muon system [**cms:muonsystem_TDR**]. The detection of the muons is very important for many physics process. In particular, for SM Higgs boson decay to ZZ/ZZ^* which is labeled a golden channel if each Z boson decay to a pair of opposite charged muons. The best mass resolution can be obtained with four muons in final state as muons are less effected by detector material as compared to electrons because they interact weakly with detector material. Along with this example, SUSY and other model for new physics searches also encourages the necessity for their reconstruction of muons with excellent resolution.

To have a robust and precise muon system was one of the core motive of CMS as indicated by detector name. Identification, triggering and momentum measurement of the muons are the tasks assigned to the muon system. The flux-return yoke and strong magnetic field are playing crucial role for excellent muons momentum resolution and triggering. The flux-return yoke by absorbing the hadrons helps for muons identification.

Muon system was essentially designed as cylindrical shape to provide full pseudorapidity coverage with detection area of 25000 m^2, having barrel and endcaps regions. Muons detection is based on gasses detectors of three types. In the barrel region, the magnetic field is uniform with less neutron-induced background drift tube are installed providing pseudorapidity coverage up to $|\eta| < 1.2$, as shown in Figure 3.13. The barrel region is arranged to have four muons station, measuring $r - \phi$ bending plane and distance along beam side. The arrangement of the drift cells are offset by the half of the width of the cell providing dead spot in efficiency, determination of standalone bunch crossing and excellent time resolution by utilizing mean timer circuits. The direction and number of chambers are optimized to get maximum efficiency of combination of the hits from different stations to get single track which is helpful to reject the background. In the endcap region, where non-uniform magnetic field and more background is expected cathode strip chambers (CSCs) are installed that providing coverage $0.9 < |\eta| < 2.4$. Like DT, there 4 CSCs stations on each endcap. Each station is composed of chambers, which are capable to work independently, aligned perpendicularly to z-axis and interspersed between the flux-return plates. To overcome the uncertainties in background rates and assignment of correct bunch crossing number, resistive plate chambers (RPCs) are installed both in barrel and endcap regions. There is a dedicated trigger system for each type of muons gasses detector including RPCs. Good time resolution, fast response and coarser spatial resolution makes RPCs distinct from CSCs and DT. The RPCs also plays vital role for the combination of tracks from multiple hits a chamber. The RPCs are installed in between of DT and CSCs region the barrel and endcap region.

Drift Tubes

The DTs are installed in the barrel region of muon system that provides pseudorapidity coverage up to $|\eta| < 1.2$ illustrated in Figure 3.13. Whole DT system consists of 5 wheels along z-axis, and 4 stations along radial direction (MB1, MB2, MB3, MB4) each made of 12 chambers except fourth which contains 14 chambers. Each drift chamber of an average size of 2m$\times$2.5m is made of 12 aluminium layers, making 3 groups of four. The central group provides the coordinates measurement along z-axis and other two groups measures $r - \phi$. There are about 60 drift tubes in each chamber, each 4 cm wide made of drift cells each of size 42 mm $\times$ 13 mm. The 50μm wire made of stainless steel serves as anode while two "I-shaped" aluminum tape each of thickness 50μm are cathodes of the drift cell as shown in Figure 3.14. As any charges particle pass by the drift cell, it ionizes the gas by removing electrons from its atom. This establish a electric field finishing of at positively charged

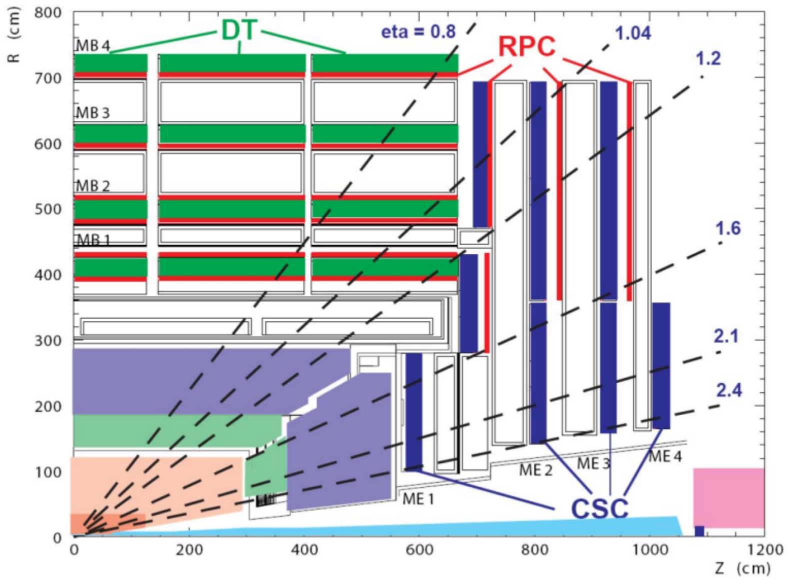


FIGURE 3.13: CMS muon system layout.

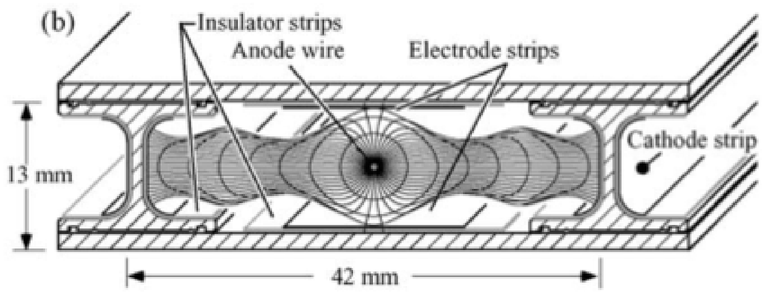


FIGURE 3.14: DT electric field created by cathode, anode and insulating strips .

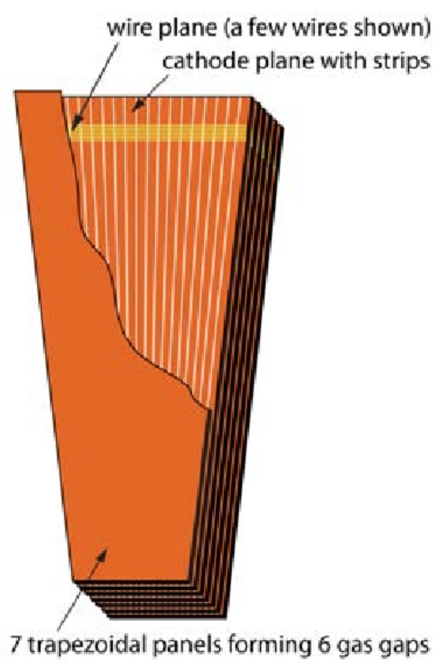
wire. Distance between the wire and the muon track is estimated by drift time of these electrons. The cell is filled with gas mixture of CO_2 (15%) Ar (85%) which is inexpensive and safe. Other qualities of this mixture are good saturation drift velocity (5.4 cm/μs) and quenching. To keep drift velocity steady, the filled gas is kept at a constant pressure inside the chambers. The pressure in each wheel is continuously monitored by sensors. It is expected to inject 10% new gas everyday.

Cathode Strip Chambers

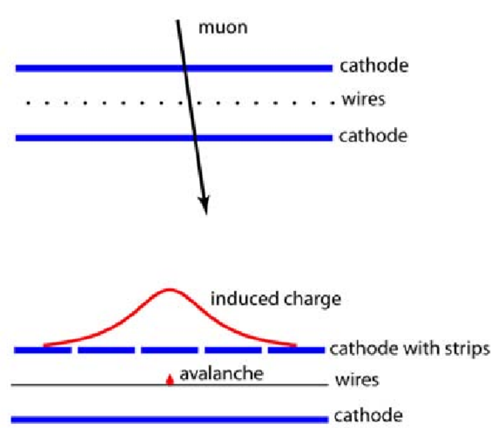
The cathode strip chambers are installed at endcap region of CMS muon system providing pseudorapidity coverage of $|0.9| < |\eta| < 2.4$, illustrated in Figure 3.13. The magnetic field in the endcap is non-uniform which forced to use different technology for barrel and endcap regions of the muon system. Another challenge for

CSCs is higher expected neutron-induced background as compared to the barrel region. The endcaps of the muon systems are made up of 468 CSCs arranged in four station like the barrel region (ME1, ME2, ME3 and ME4). The pseudorapidity range from $|0.9| < |\eta| < 1.2$ is covered by both CSCs and DTs. The chambers are aligned trapezoidal, each chamber is covering 10° or 20° in ϕ. RPCs are also installed between CSCs stations.

A single chamber is the basic unit which capable to work independently which is shown in Figure 3.15(a). The CSC is multiple wire detector made of 6 layers of positively charged anode interleaved with 7 layers of negatively charged cathode wires made of cooper filled within gas volume as shown in Figure 3.15(a). The cathode and anode wires are perpendicular to each other and provides two position coordinates of each particle. As the charged particle pass through chamber, it will ionize the gas atoms producing avalanche of electrons to anode wires, illustrated in Figure 3.15(b) . As the wires are nearby, CSCs not only provides excellent momentum resolution but helps to trigger the event as well. It feeds $r - \phi$ resolution of 2 mm to first level trigger, 75μm off-line spatial resolution in $r - \phi$ for ME1/1 and ME1/2 chambers and 150μm for all other stations. The CSCs also help to assign bunch crossing with an efficiencies of 92% which can be enhanced up to 99% by simple majority out of all chambers. The tendency to operate at high rates and non uniform magnetic field makes CSCs robust which does not require precise gas pressure and temperature. Multilayer CSCs is useful for the combination of hits in CSCs and tracker detectors.



(a)



(b)

FIGURE 3.15: CSC layout (left) and induced current (right) when a charged particle pass thorugh it .

Resistive Plate Chambers

Resistive plate chambers are installed in CMS muon system in both barrel and endcap region. The motivation of RPCs are to get excellent time resolution along with good spatial resolution. The RPCs are very fast responsive detector providing another muon trigger along with CSCs and DTs. RPCs are capable to provide a time resolution of just one nanosecond.

RPCs are made of two high resistive plastic positive and negative charged plates, kept parallel to each other, with gas filled inn between them. The gas mixture consists of 95% of $C_2H_2F_4$ and 5% of C_4H_{10}. As the muon pass through the RPCs, primary and secondary ionization of the gas create avalanche of electrons which is picked up by external metallic strips after a small but precise time delay. Quick momentum measurement is feeded to trigger the event. The RPCs excellent time resolution assign bunch crossing with ultimate precision.

Momentum Resolution

CMS detector is capable to measure muons transverse momentum independently by using tracker or muon system. The expected transverse momentum resolution of muons system has been studied for the using muons of $p_T = 40$ GeV from Z and W boson decays and found to be 10%(20%) in barrel and endcap regions. Global muons can also be reconstructed at CMS by the combination of the information of the tracker and muons system. For muons with p_T below 100 GeV tracker provide better resolution than muon system and vice versa. The transverse momentum resolution is found to be improved sufficiently by using global muons i.e 1%(2%) for $p_T < 100$ GeV in barrel(endcap) regions. For muons with $p_T > 100$ GeV, the transverse momentum resolution of muon system improves to 8% (10%) in the barrel and endcap region. The momentum resolution of CMS muon system alone, tracker alone and combination of both are shown in Figure 3.16 [**cms:muonsystem_TDR_2**].

3.2.7 Trigger and Data Acquisition System

The LHC running at it nominal parameters provides collisions at the rate of 40 MHz. Most of these events are soft collisions and uninteresting for new physics as they are like minimum bias collisions. Though CMS detector is capable to reconstruct all of these events but storing all these events will fill the CMS storage in days. So trigger system of CMS helps of select interesting events out of all for further steps. The events which does not pass the first step trigger are deleted permanently. Trigger system performs the first step for physics event selection which reduces the rate from 40 MHz to 1kHz into two layers Level-1 (L1) trigger and high-level trigger (HLT).

The L1 trigger is made of programmable electronics, dropping the rate drastically to 100 kHz.To be fast, L1 trigger events by their energy and momentum

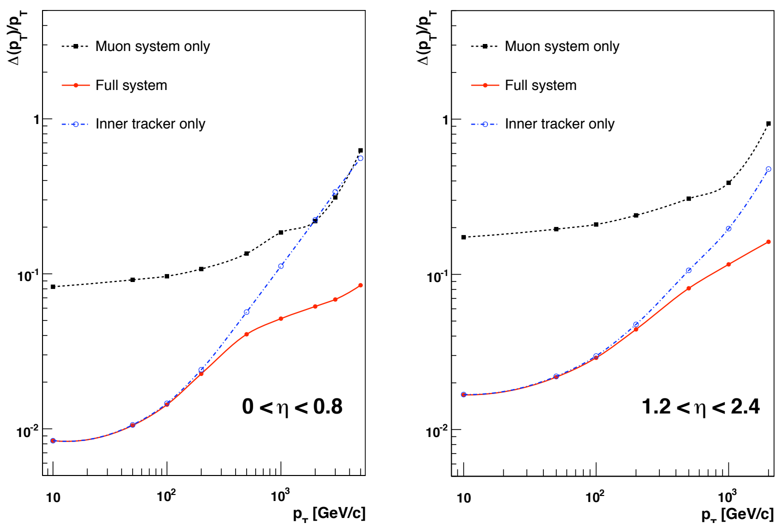


FIGURE 3.16: Muon momentum resolution as a function of p_T of muons in barrel and endcap regions.

calculations estimated by only using coarse in muon system and calorimeter instead of complete high resolution information. The L1 trigger decides to select or reject event in 3.2μm.

All events with positive L1 decision, with their complete information, are transferred to HLT which is a farm of one thousand commercial processors. It uses the algorithms with growing complexity that is based on fine granularity of the event. HLT decision time is different for each event with mean time of about 50 ms. The HLT has excess of complete information of the event that demands a greater bandwidth of the order of 1Tb/s.

For example, to look for Higgs boson decay into 4-leptons final state, the trigger would look for energetic leptons in the detector. For L1 trigger, electron and muons points to a narrow energy deposit in ECAL and hit pattern in tracker and muon system respectively. While for HLT, the total energy and momentum of the leptons are reconstructed by ECAL and combination of muon system and inner tracker.

Level-1 Trigger Architecture

The configuration of L1 trigger is shown in Figure 3.17. L1 trigger can be divided into calorimeter and muon triggers. Each ECAL and HCAL has independent regional trigger feeding to global calorimeter trigger. Similarly, muon trigger has local triggers for each sub-detector merging into global muon trigger. Finally, global calorimeter and muon triggers merge into global trigger which makes L1 final decision.

Each local trigger feeds to global/regional trigger by providing trigger primitives groups (TPGs) based on coarse of sub-detector. Regional calorimeter trigger (RCT) builds objects which deposit their energy in ECAL and HCAL such as

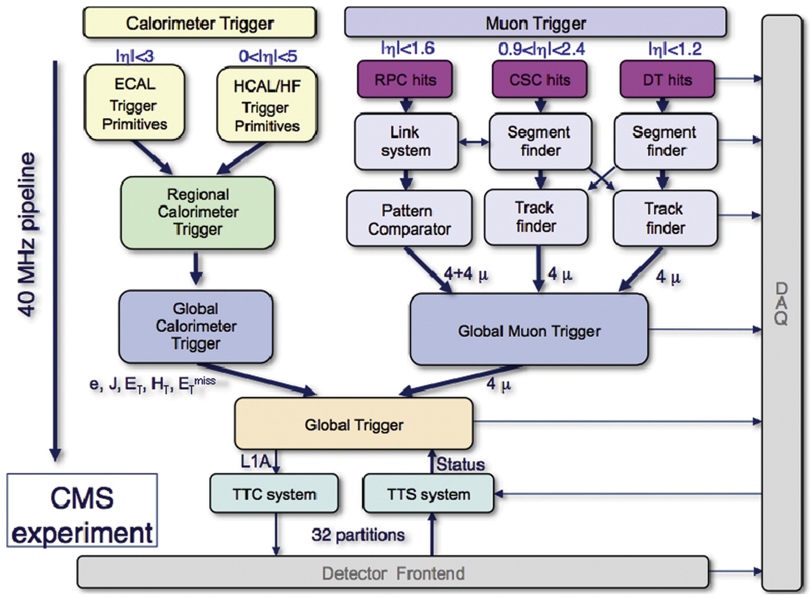


FIGURE 3.17: The configuration of L1 trigger.

jets, γ, electrons, photons and tau's. Similarly, in muon triggering, DT/CSC/RPC track finders feeds muons tracks to muon global trigger (MGT). At most four best candidates are sent to global muon and calorimeter trigger which further sent to global trigger (GT). GT compares these candidates with L1 trigger menu to decide weather to keep to reject the event. Some triggers in the menu evolved with increasing center of mass energy of LHC by tightening the requirement on candidate or pre-scaled (selecting only the nth event) to control the output rate. Candidates able to fire at least one trigger out of whole trigger menu is kept for further investigation by HLT. For example, the trigger/bit "L1 SingleEG8Iso" in the L1 trigger menu requires at least one isolated (candidate with no or little hadronic activity around it) electron or photon with transverse momentum greater than 8 GeV.

High-Level Trigger Architecture

HLT is three level (2, 2.5 and 3) trigger with growing complexity starting at level 2 which usually works on top of the information provided by L1 trigger. Goal of the HLT is to reduce the event rate 100kHz upto 1 kHZ. Level 2 HLT is to build fine granularity objects by combining the information from sub-detectors e.g from preshower and ECAL energy clusters are combined for electrons. Bremsstrahlung of the electrons causes its energy to spread into several neighbor cells. To compensate this effect, clusters of ECAL are combined to form superclusters.

At HLT level 2.5, the inner tracker information is combined and matched with calorimeters and muon system. In case of electron, energy supercluster is matched with the hits in the tracker, if matches it form a seed. For muon, hits in the tracker are matched with the hits in muon system to reconstruct global muon. HLT level 3 reconstructs the complete track, if its found electron seed or matching hits of

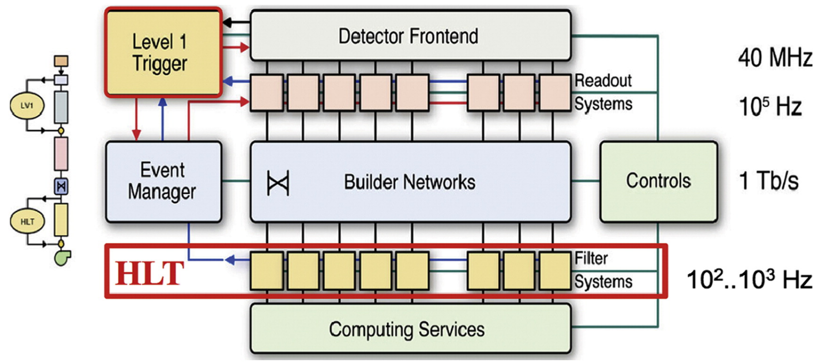


FIGURE 3.18: The CMS DAQ system structure.

muon. After these three stages, precise energy and momentum are calculated of the objects that decides to accept or reject the event.

There are more than 200 different HLT paths each derived by at least one L1 bit. The data which pass the HLT trigger is organized into different datasets according to the objects reconstructed in the event and the passed selection e.g an event with two muons reconstructed with transverse momentum greater than 8 and 17 GeV will be placed in "DoubleMu" dataset.

Data Acquisition System

The schematic skeleton of CMS Data Acquisition (DAQ) system is shown in Figure 3.18. CMS DAQ duty is to provide data from 650 different readout channels to HLT along with enough computing resources to limit the output data rate **[cms:DAQ_TDR]**. An event fragments with a size of 2KB is provided by each module. A network with a bandwidth of 100 Gb/s is utilized by Front end drivers (FEDs) to read the data from detector sensors. These FEDs are located underground at the level of the detector called counting room. Events are transferred to the event filter systems at a maximum rate of 100 kHz. The complete reconstruction of the objects of the event and high machine luminosity is the reason for this high rate.

3.2.8 Computing Grid

An important and last step after data collection is to process it with full detailed reconstruction algorithms and made available for physics analysis.Although, trigger system of CMS drastically reduces the amount of data but even then CMS delivers data at the order of petabytes every year. To analyze and store this huge amount of the data, LHC has connected huge number of computing resources from collaborative institutes/universities, computer centers and middleware providers through the Worldwide LHC Computing Grid (WLCG).

There are three layers of Tier (Tier-0, Tier-1 and Tier-2) for data flow between different computing sites. The types of dataset is described below briefly

- RAW that contains all the recorded information from detector along with triggers decisions and other metadata. A copy of RAW data is achieved and placed at permanent storage element. A occupancy of around 1.5 (2.0) MB/event is needed to store data (simulated data) in RAW version.
- Many algorithms and pattern recognition like cluster and track-finding; primary and secondary vertex reconstruction etc are applied to RAW to obtain RECO data format. RECO events occupancy is roughly three times less than RAW (0.5 MB/event).
- The most compact format of data AOD (analysis object data) is obtained by dropping unnecessary information and reducing per event size up to 100 KB.

Tier-0 is located at CERN with primary duties to copy recorded data to permanent storage, promptly reconstruct data from RAW version to have first version of prompt-RECO data and exporting a copy of RAW data to Tier-1 centers. Tier-1 are mainly hosted by national computing centers and labs from collaborative countries. At each Tier-1, a fraction of detailed the reconstruction of RAW data simulated data has been performed to get AOD. Tier-1 also responsible to store and export AOD format data to Tier-2. The last layer Tier-3 is hosted by CMS collaborative institutes where main analysis activity, alignment, detector related studies and offline calibration is done. Tier-3 are used for Monte Carlo (MC) data production which can be exported to Tier-1 for long term storage.

3.3 Physics Objects

3.3.1 Electrons

Electron Reconstruction

More details on electron reconstruction can be found in Ref. [**ElectronLegacy**].

Electron candidates are preselected using loose cuts on track-cluster matching observables, so as to preserve the highest possible efficiency while rejecting part of the QCD background. To be considered for the analysis, electrons are required to have a transverse momentum $p_T^e > 7$ GeV, a reconstructed $|\eta^e| < 2.5$, and to satisfy a loose primary vertex constraint defined as $d_{xy} < 0.5$ cm and $d_z < 1$ cm. Such electrons are called **loose electrons**.

The data-MC discrepancy is corrected using scale factors as is done for the electron selection with data efficiencies measured using the same tag-and-probe technique outlined later (see Section 3.3.1). These studies for reconstructions are carried out by the EGM POG and the results are only summarised here.

The electron reconstruction scale factors are shown Fig. 3.19 and are applied as a function of the super cluster η and electron p_T.

Electron Identification and Isolation

One of the main improvements brought in the analysis is the usage of a new multivariate discriminant for electron selection in all data taking periods.

Reconstructed electrons are now identified and isolated by means of a Gradient Boosted Decision Tree (GBDT) multivariate classifier algorithm, which exploits observables from the electromagnetic cluster, the matching between the cluster and the electron track, observables based exclusively on tracking measurements as well as particle flow isolation sums. The full list of observables used can be found in the Table 3.6.

observable type	observable name
6*cluster shape	RMS of the energy-crystal number spectrum along η and φ; $\sigma_{i\eta i\eta}$, $\sigma_{i\varphi i\varphi}$
	super cluster width along η and ϕ
	'ratio of the hadronic energy behind the electron'ă supercluster to the supercluster energy, H/E
	circularity $(E_{5\times5} - E_{5\times1})/E_{5\times5}$
	sum of the seed and adjacent crystal over the super cluster energy R_9
	for endcap traing bins: energy fraction in pre-shower E_{PS}/E_{raw}
2*track-cluster matching	energy-momentum agreement E_{tot}/p_{in}, E_{ele}/p_{out}, $1/E_{tot} - 1/p_{in}$
	position matching $\Delta\eta_{in}$, $\Delta\varphi_{in}$, $\Delta\eta_{seed}$
5*tracking	fractional momentum loss $f_{brem} = 1 - p_{out}/p_{in}$
	number of hits of the KF and GSF track N_{KF}, N_{GSF} $(\cdot)$
	reduced χ^2 of the KF and GSF track χ^2_{KF}, χ^2_{GSF}
	number of expected but missing inner hits $(\cdot)$
	probability transform of conversion vertex fit χ^2 $(\cdot)$
3*isolation	Particle Flow photon isolation sum $(\cdot)$
	Particle Flow charged hadrons isolation sum $(\cdot)$
	Particle Flow neutral hadrons isolation sum $(\cdot)$
1*For PU-resilience	Mean energy density in the event: ρ $(\cdot)$

TABLE 3.6: Overview of input variables to the identification classifier. Variables not used in the run I MVA are marked with $(\cdot)$.

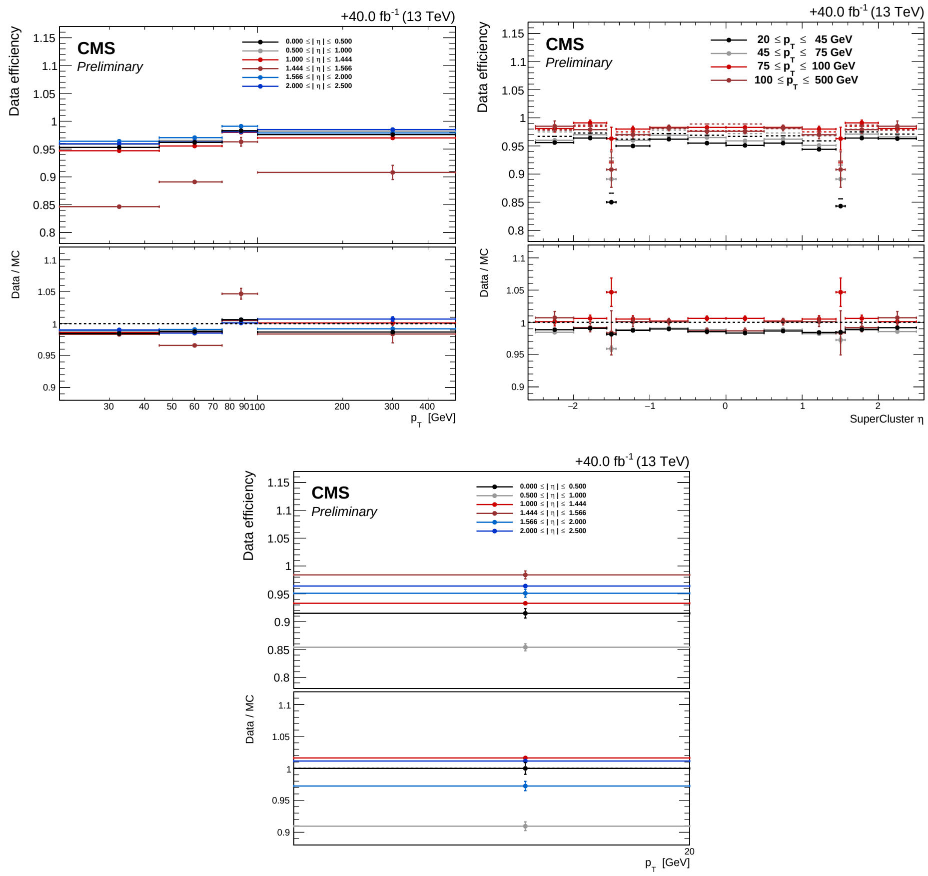


FIGURE 3.19: Electron reconstruction efficiencies efficiency in data versus p_T (left) and η (right) for electrons with $p_T > 20$ GeV (top) and $p_T < 20$ GeV (bottom) with corresponding data/MC scale factors as‘ăprovided by the EGM POG.

The classifier is trained on 2017 Drell-Yan plus jets MC sample for both signal and background. Fig. 3.20 shows the output of the multiclassifier discriminant for prompt electrons from DY events and fakes from jets in DY events.

The impact of the transition from the TMVA(v1) to the xgboost(v2) training library is shown in Fig. 3.21. The working points shown are chosen so as to get the same signal efficiency as a cut on MVA ID and a cut on the PF isolation, in each p_T and η bin.

Table 3.7 lists the cut values applied to the BDT score for the chosen working point. For the analysis, loose electrons have to pass this MVA identification and isolation working point.

Electron Impact Parameter Selection

In order to ensure that the leptons are consistent with a common primary vertex we require that they have an associated track with a small impact parameter with

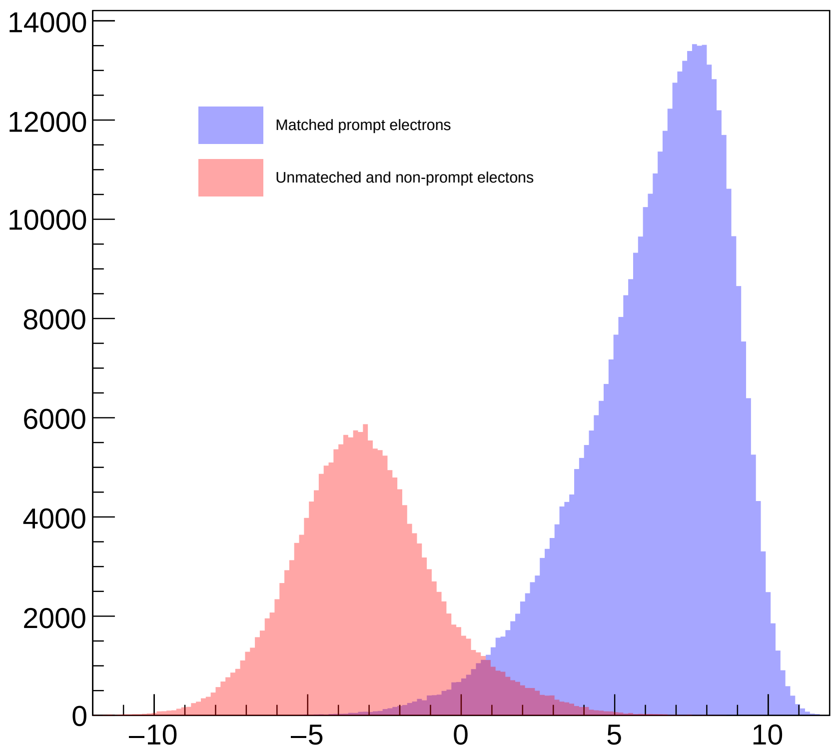


FIGURE 3.20: Output of the multiclassifier discriminant for prompt electrons matched to truth electrons from Z decay (blue) and for fakes (red). Events are all taken from DY events MC sample.

	Cut on BDT score	Signal eff.	Background eff.
	$\lvert\eta\rvert < 0.8$		
$5 < p_T < 10$ GeV	1.264	81.04%	4.4%
$p_T > 10$ GeV	2.364	97.1%	2.9%
	$0.8 < \lvert\eta\rvert < 1.479$		
	Cut on BDT score	Signal eff.	Background eff.
$5 < p_T < 10$ GeV	1.178	79.3%	4.6%
$p_T > 10$ GeV	2.078	96.3%	3.6%
	$\lvert\eta\rvert > 1.479$		
	Cut on BDT score	Signal eff.	Background eff.
$5 < p_T < 10$ GeV	1.330	72.97%	3.6%
$p_T > 10$ GeV	1.080	95.7%	6.7%

TABLE 3.7: Minimum BDT score required for passing the electron identification, together with the corresponding signal and background efficiencies.

respect to the event primary vertex. We use the significance of the impact parameter to the event vertex, $|\mathrm{SIP}_{3\mathrm{D}} = \frac{\mathrm{IP}}{\sigma_{\mathrm{IP}}}|$, where IP is the lepton impact parameter in three dimensions at the point of closest approach with respect to the primary interaction vertex, and σ_{IP} the associated uncertainty. Hereafter, a "primary lepton"

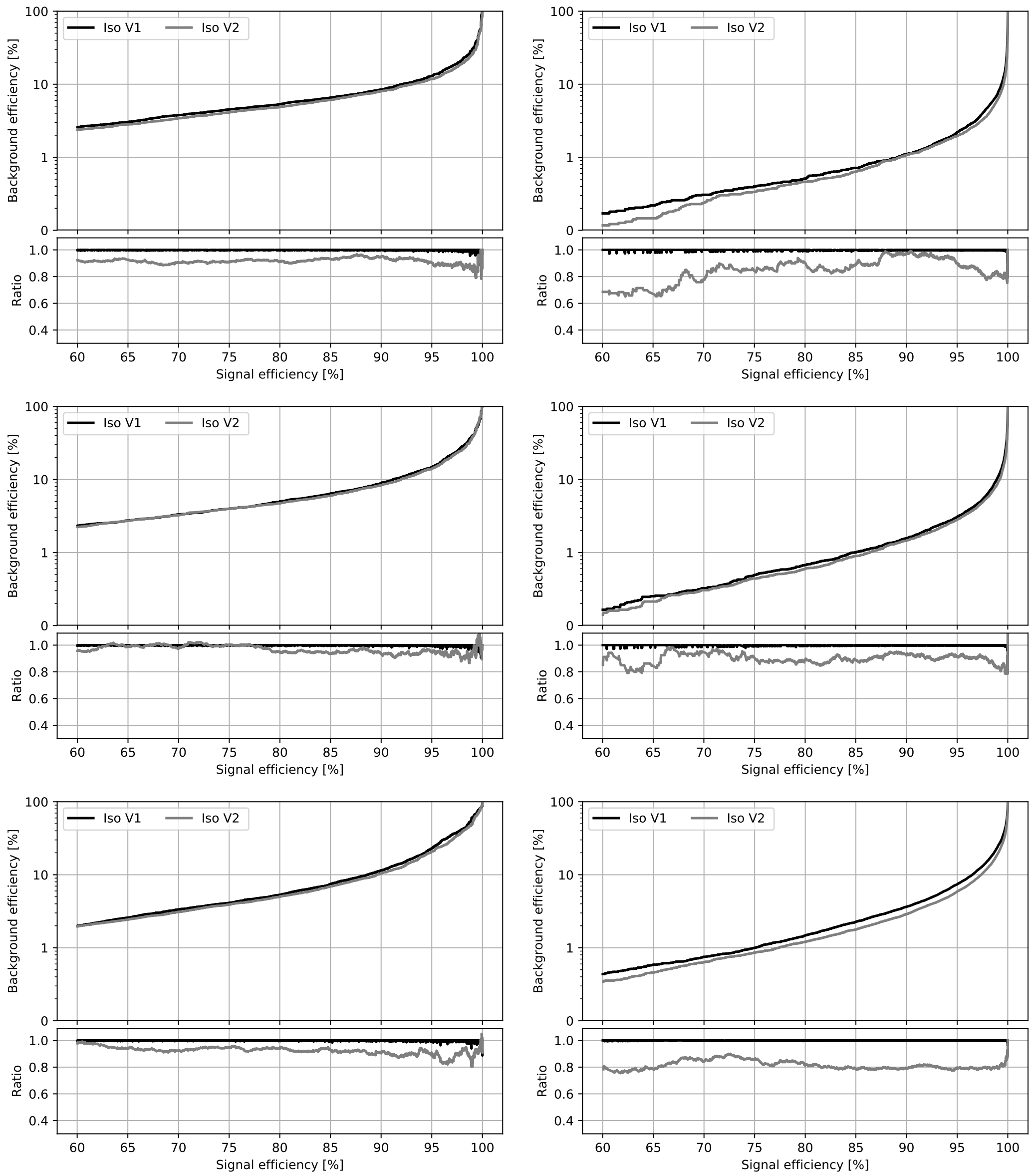


FIGURE 3.21: The receiver operating characteristic curves (background efficiency vs signal efficiency) of the MVA's trained on the 2017 simulation and applied on the 2018 simulation. The training combines identification and isolation variables. Performance are shown for electrons with $5 < p_T < 10$ GeV (left), $p_T > 10GeV$ (right) and $|\eta| < 0.8$ (top), $0.8 < |eta| < 1.479$ (middle) and $|eta| > 1.479$ (bottom). V1 and V2 versions of training are compared, exploiting TMVA and xgboost training libraries respectively.

is a lepton satisfying $|\mathrm{SIP}_{3\mathrm{D}}| < 4$.

Electron Energy Calibrations

Electrons in data are corrected for features in ECAL energy scale in bins of p_T and $|\eta|$. Corrections are calculated on a $\rightarrow$ ee sample to align the dielectron'ă mass spectrum in the data to that in the MC, and to minimize its width.

The $\rightarrow$ ee mass resolution in Monte Carlo is made to match data by applying a pseudorandom Gaussian smearing to electron energies, with Gaussian parameters varying in bins of p_T and $|\eta|$. This has the effect of convoluting the electron energy spectrum with a Gaussian.

The electron energy scale is measured in data by fitting a Crystall-ball function to the di-electron mass spectrum around the Z peak in the $Z \rightarrow$ ee control region. The energy scale for the full 2018 dataset is shown in Fig. 3.22(a) and agrees decently with the MC with the preliminary corrections released by EGAMMA POG for Moriond.

Electron Efficiency Measurements

The Tag-and-Probe study was performed on the EGM dataset using the golden JSON of 58.83 fb^{-1}. More details on the Tag-and-Probe method can be found in Ref. [**AN-15-277**].

Tag electrons need to satisfy the following quality requirements:

- trigger matched to HLT_Ele32_WPTight_Gsf_L1DoubleEG_v*
- $p_T > 30$ GeV, super cluster (SC) $\eta < 2.17$

For the bin between 7 and 20 GeV, additional criteria are required: the tag has to pass a cut on the MVA ID> 0.92, $\sqrt{(}2*PFMET*p_T(tag)*(1-cos(\phi_{MET}-\phi_{tag}))) < 45GeV$.

Probe electrons only need to be reconstructed as GsfElectron while the FSR recovery algorithm is not applied in efficiency measurement.

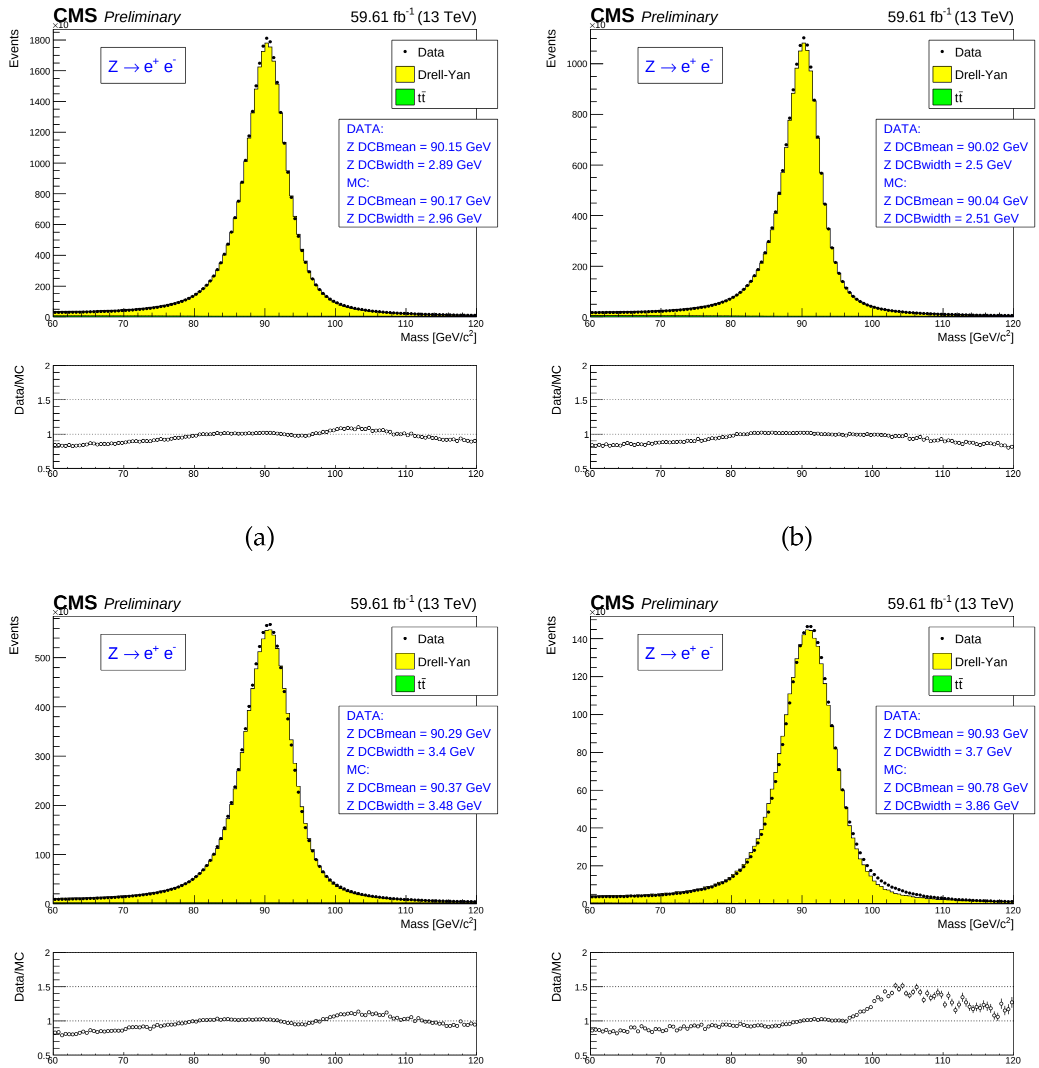
The nominal MC efficiencies are evaluated from the LO MadGraph Drell-Yan, while the NLO systematics use the MadGraph_AMCatNLO sample.

For the efficiency measurements a template fit is used. The m_{ee} signal shape of the passing and failing probes is taken from MC and convoluted with a Gaussian. The data is then fitted with the convoluted MC template and a CMSShape (an Error-function with a one-sided exponential tail). This change follows from the usage of the T&P tool developed by the EGM POG.

The electron selection efficiency is measured as a function of the probe electron p_T and its SC η, and separately for electrons falling in the ECAL gaps. Figure 3.23 and **??** shows the p_T and η turn-on curves measured in data.

The EGM recommendations on the evaluation of Tag-and-Probe uncertainties for efficiency measurements are followed. Specifically, we consider

- Variation of the signal shape from a MC shape to an analytic shape (Crystal Ball) fitted to the MC



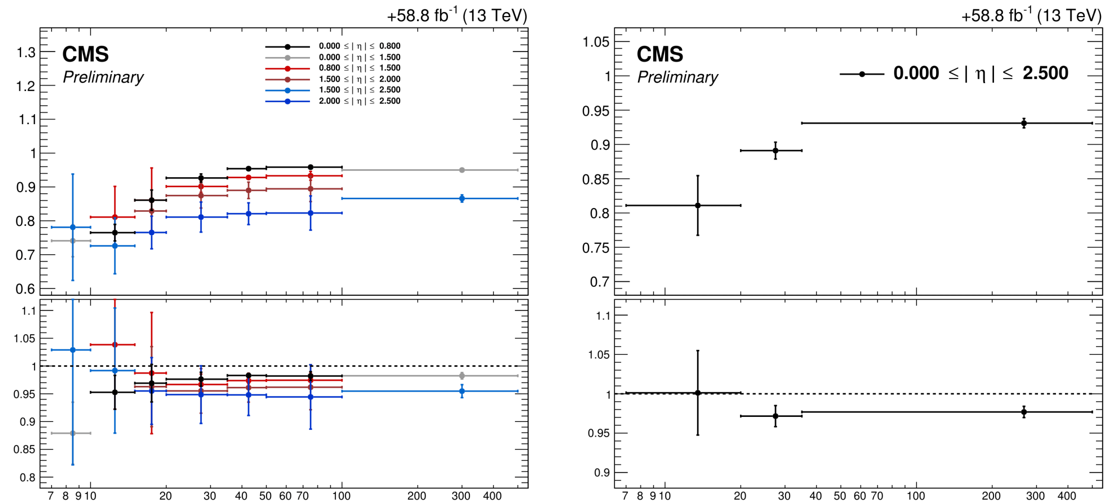
(a) (b)

(c) (d)

FIGURE 3.22: (a): electron energy scale measured in the $Z \rightarrow ee$ control region for all electrons, for both electrons in the barrel (b), for one electron in the barrel, one in the endcaps (c) and for both electrons in the endcaps (d). The results of the Crystall-ball fit are reported in the figures.

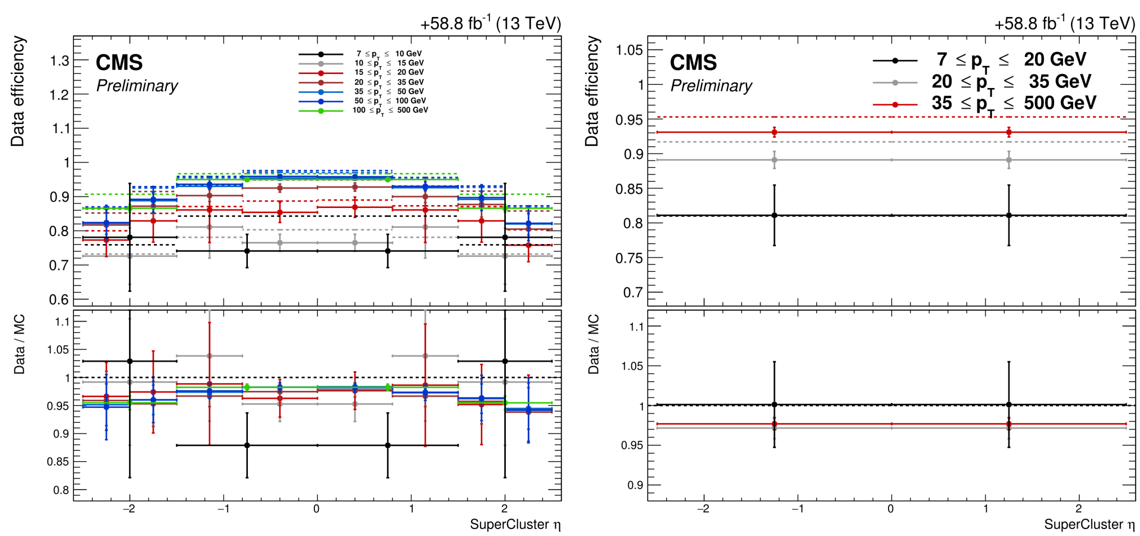
- Variation of the background shape from a CMS-shape to a simple exponential in fits to data
- Using an NLO MC sample for the signal templates

The total uncertainty for the measurement of the scale factors is the quadratic sum of the statistical uncertainties returned from the fit and the aforementioned systematic uncertainties.



(a) (b)

FIGURE 3.23: Electron selection efficiencies vs p_T measured using the Tag-and-Probe technique described in the text, non-gap electrons (left) and gap electrons (right), together with the corresponding data/MC ratio (bottom).



(a) (b)

FIGURE 3.24: Electron selection efficiencies vs η measured using the Tag-and-Probe technique described in the text, non-gap electrons (left) and gap electrons (right), together wit the corresponding data/MC ratio at the bottom of each plot.

3.3.2 Muons

Muon Reconstruction and Identification

More details on muon reconstruction can be found in Ref. [**AN-15-277**]. We define **loose muons** as the muons that satisfy $p_T > 5$, $|\eta| < 2.4$, $dxy < 0.5$ cm, $dz < 1$ cm, where dxy and dz are defined w.r.t. the PV and using the 'muonBestTrack'. Muons have to be reconstructed by either the Global Muon or Tracker Muon algorithm. Standalone Muon tracks that are only reconstructed in the muon system are rejected. Muons with `muonBestTrackType==2` (standalone) are discarded even if they are marked as global or tracker muons.

Loose muons with p_T below 200 GeVare considered identified muons if they also pass the PF muon ID (note that the naming convention used for these IDs differs from the muon POG naming scheme, in which the "tight ID" used here is called the "loose ID"). Loose muons with p_T above 200 GeVare considered identified muons if they pass the PF ID or the Tracker High-p_T ID, the definition of which is shown in Table 3.8. This relaxed definition is used to increase signal efficiency for the high-mass search. When a very heavy resonance decays to two bosons, both bosons will be very boosted. In the lab frame, the leptons coming from the decay of a highly boosted will be nearly collinear, and the PF ID loses efficiency for muons separated by approximately $\Delta R < 0.4$, which roughly corresponds to muons originating from bosons with $p_T > 500$ GeV.

TABLE 3.8: The requirements for a muon to pass the Tracker High-p_T ID. Note that these are equivalent to the Muon POG High-p_T ID with the global track requirements removed.

Plain-text description	Technical description
Muon station matching	Muon is matched to segments in at least two muon stations **NB: this implies the muon is an arbitrated tracker muon.**
Good p_T measurement	$\frac{p_T}{\sigma_{p_T}} < 0.3$
Vertex compatibility ($x - y$)	$d_{xy} < 2$ mm
Vertex compatibility (z)	$d_z < 5$ mm
Pixel hits	At least one pixel hit
Tracker hits	Hits in at least six tracker layers

An additional "ghost-cleaning" step is performed to deal with situations when a single muon can be incorrectly reconstructed as two or more muons:

- Tracker Muons that are not Global Muons are required to be arbitrated.
- If two muons are sharing 50% or more of their segments then the muon with lower quality is removed.

Muon Isolation

Particle-Flow based isolation is used for the muons. The so-called $\Delta\beta$ correction is applied in order to subtract the pileup contribution for the muons, whereby $\Delta\beta = \frac{1}{2}\sum_{\text{PU}}^{\text{charged had.}} p_T$ gives an estimate of the energy deposit of neutral particles (hadrons and photons) from pile-up vertices. The relative isolation for muons is then defined as:

$$\text{RelPFiso} = \frac{\sum^{\text{charged had.}} p_T + \max(\sum^{\text{neutral had.}} + \sum^{\text{photon}} - \Delta\beta, 0)}{p_T^{\text{lepton}}} \tag{3.10}$$

The isolation working point for muons was optimized in Ref. [**AN-15-277**] and the working point was chosen to be the same as electrons, namely RelPFiso($\Delta R = 0.3$) < 0.35.

Muon Impact Parameter Selection

We apply the same selection on the muon significance of impact parameter as for the electrons, as described in Sec. 3.3.1:

- $|\text{SIP}_{\text{3D}} = \frac{\text{IP}}{\sigma_{\text{IP}}}| < 4$

Muon Energy Calibrations

Similar to electrons the muon momentum scale is measured in data by fitting a Crystall-ball function to the di-muon mass spectrum around the Z peak in the $Z \to \mu\mu$ control region. At the moment no muon scale and resolution corrections are available but from Fig. 3.25 shows a very good agreement between data and simulation.

Muon Efficiency Measurements

Muon efficiencies are measured with the Tag and Probe (T&P) method performed on $\to \mu\mu$ and $\to \mu\mu$ events in bins of p_T and η. More details on the methodology can be found in Ref. [**CMS_AN_2015-277**]. Measurements are extracted using 2018 RunA,B,C,D data while the measurements corresponding to 2016 and 2017 datasets have already been reported in Ref. [**CMS_AN_2016-442**] and Ref. [**CMS_AN_2017-342**] respectively. The Z sample is used to measure the muon reconstruction and identification efficiency at high p_T, and the efficiency of the isolation and impact parameter requirements at all p_T. The sample is used to measure the reconstruction efficiency at low p_T, as it benefits from a better purity in that kinematic regime. In this case, events are collected using `HLT_Mu7p5_Track2_Jpsi_v*` when probing the reconstruction and identification efficiency in the muon system, and using the `HLT_Mu7p5_L2Mu2_Jpsi_v*` when probing the tracking efficiency.

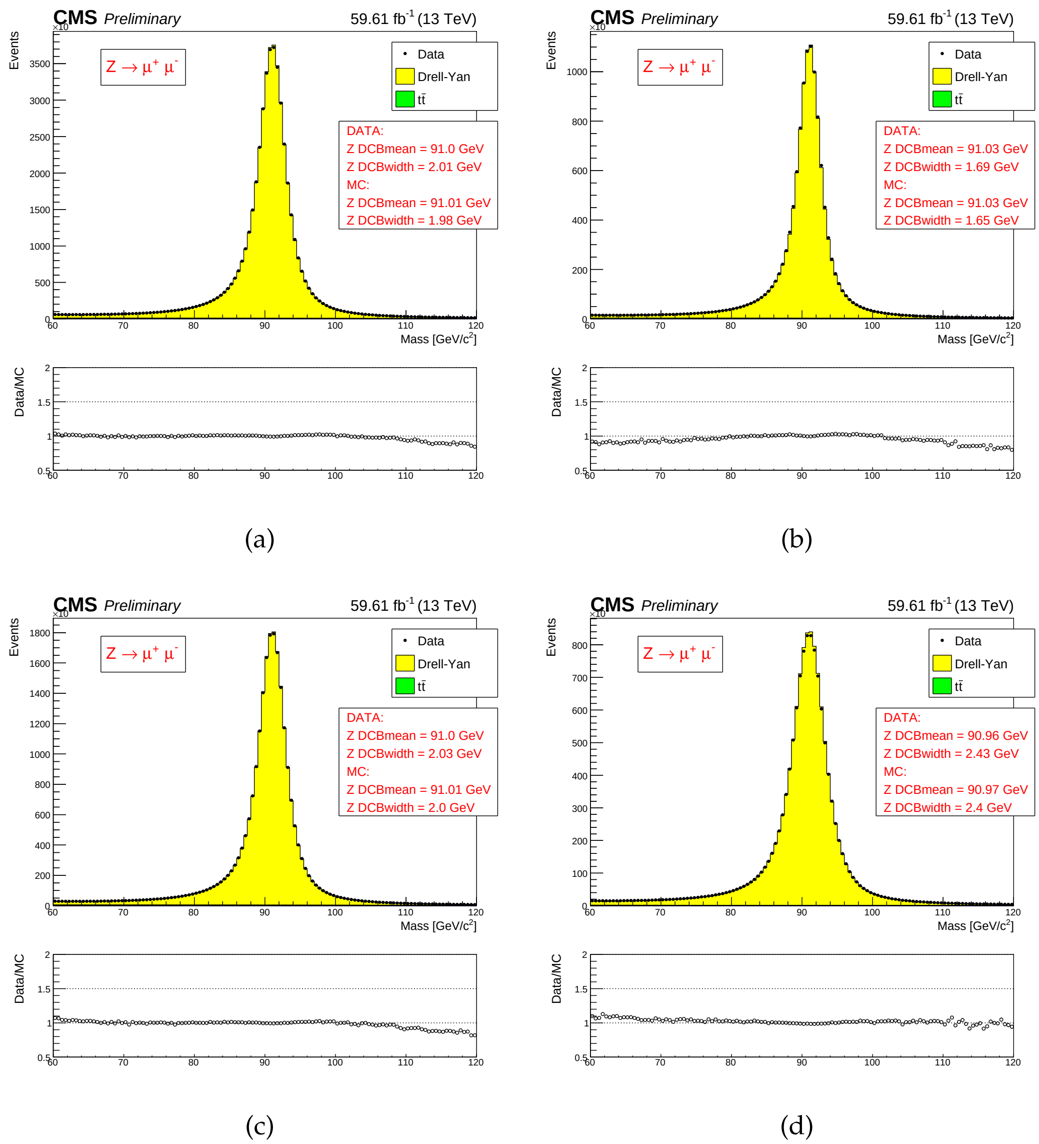


FIGURE 3.25: (a): muon energy scale measured in the $Z \to \mu\mu$ control region for all muons, for both muons in the barrel (b), for one muon in the barrel, one in the endcaps (c) and for both muons in the endcaps (d). The results of the Crystall-ball fit are reported in the figures.

Reconstruction and identification Results for the muon reconstruction and identification efficiency for $p_T > 20$ GeV have been derived by the Muon POG. The probe in this measurement are tracks reconstructed in the inner tracker, and the passing probes are those that are also reconstructed as a global or tracker muon and passing the Muon POG Loose muon identification. Results for low $p_T$muons were derived using events, with the same definitions of probe and passing probes. The systematic uncertainties are estimated by varying the analytical signal and background shape models used to fit the dimuon invariant mass. Details on the procedure can be found in Ref. [**AN-15-277**]. The efficiency and scale factors used for low p_T muons are the ones derived using single muon prompt-reco dataset.

The efficiency in data and simulation is shown in Fig. 3.26.

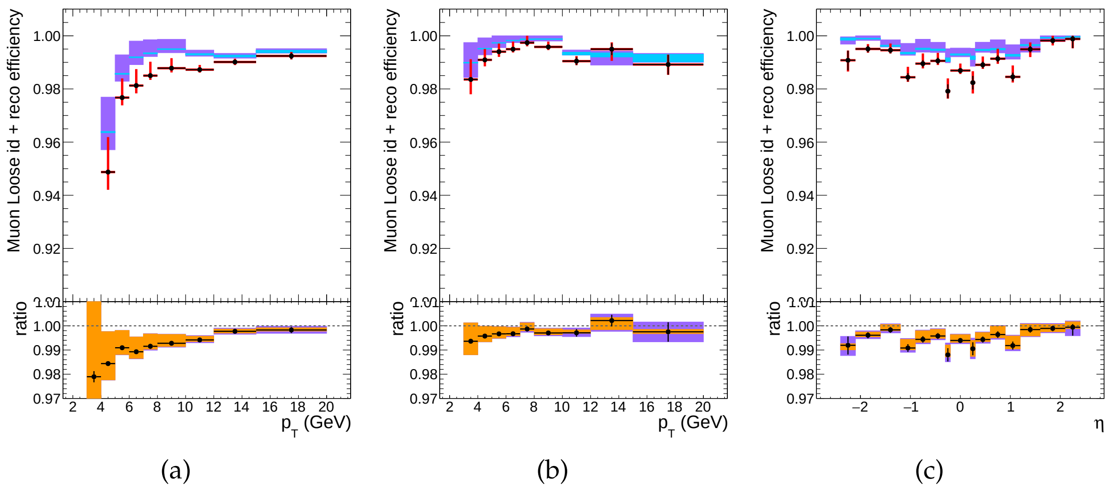


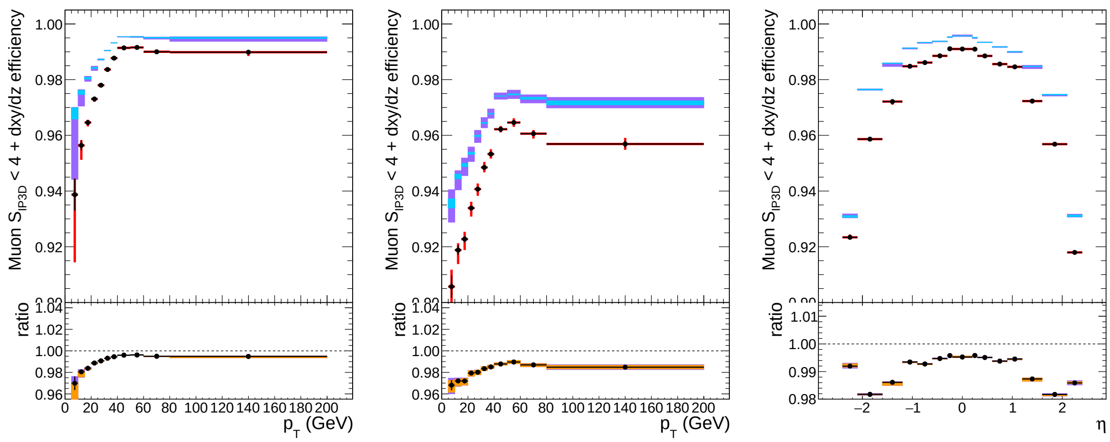
FIGURE 3.26: Muon reconstruction and identification efficiency at low p_T, measured with the tag&probe method on events, as function of $p_T$in the barrel (left) and endcaps (center), and as function of η for $p_T > 7$ GeV (right). In the upper panel, the larger error bars include also the systematical uncertainties, while the smaller ones are purely statistical. In the lower panel showing the ratio of the two efficiencies, the black error bars are for the statistical uncertainty, the orange rectangles for the systematical uncertainty and the violet rectangles include both uncertainties.

Impact parameter requirements The measurement is performed using Z events. Events are selected with `HLT_IsoMu27_v*` or `HLT_Mu50_v*` triggers. For this measurement, the probe is a muon passing the POG Loose identification criteria, and it is considered a passing probe if satisfies the SIP3D, dxy, dz cuts of this analysis. The results are shown in Fig. 3.27.

Isolation requirements The isolation efficiency is measured using events from the Z decay for any p_T. The events are selected with either of `HLT_IsoMu27_v*` or `HLT_Mu50_v*` triggers. The isolation of the muons are calculated after recovery of the FSR photons and subtracting their contribution to the isolation cone of the muons. More detailed description of the method can be found in Ref. [**AN-16-217**].

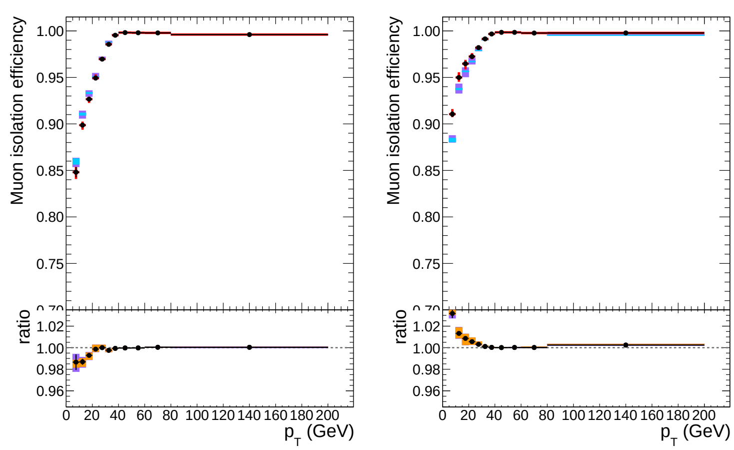
The results are shown in Fig. 3.28.

Tracking The efficiency to reconstruct a muon track in the inner detector is almost 100% and hence muon Physics Object Group no more recommends tracking scale factors to be applied to the MC in the measurement.



(a) (b) (c)

FIGURE 3.27: Efficiency of the muon impact parameter requirements, measured with the tag&probe method on Zevents, as function of $p_T$in the barrel (left) and endcaps (center), and as function of η for $p_T > 20$ GeV (right). In the upper panel, the larger error bars include also the systematical uncertainties, while the smaller ones are purely statistical. In the lower panel showing the ratio of the two efficiencies, the black error bars are for the statistical uncertainty, the orange rectangles for the systematical uncertainty and the violet rectangles include both uncertainties.



(a) (b)

FIGURE 3.28: Efficiency of the muon isolation requirement, measured with the tag&probe method on Zevents, as function of $p_T$in the barrel (left) and endcaps (right). In the upper panel, the larger error bars include also the systematical uncertainties, while the smaller ones are purely statistical. In the lower panel showing the ratio of the two efficiencies, the black error bars are for the statistical uncertainty, the orange rectangles for the systematical uncertainty and the violet rectangles include both uncertainties.

Overall results The product of all the data to simulation scale factors for muon tracking, reconstruction, identification, impact parameter and isolation requirements is shown in Fig. 3.29.

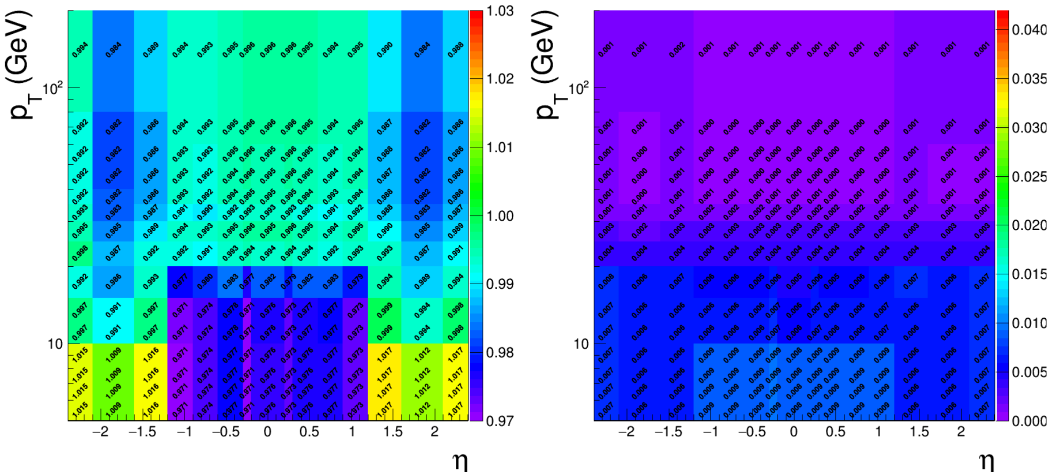


FIGURE 3.29: Left: Overall data to simulation scale factors for muons, as function of p_T and η. Right: Uncertainties on data to simulation scale factors for muons, as function of p_T and η.

3.3.3 Photons for FSR recovery

The FSR recovery algorithm was considerably simplified with respect to what was done in Run I, while maintaining similar performance. The selection of FSR photons is now only done per-lepton and no longer depends on any Z mass criterion, thus much simplifying the subsequent ZZ candidate building and selection. As regards the association of photons with leptons, the rectangular cuts on $\Delta R(\gamma, l)$ and $E_{T,\gamma}$ have been replaced by a cut on $\Delta R(\gamma, l)/E^2_{T,\gamma}$.

Starting from the collection of 'PF photons' provided by the particle-flow algorithm, the selection of photons and their association to a lepton proceeds as follows:

1. The preselection of PF photons is done by requiring $p_{T,\gamma} > 2$ GeV, $|\eta^\gamma| < 2.4$, and a relative Particle-flow isolation smaller than 1.8. The latter variable is computed using a cone of radius $R = 0.3$, a threshold of 0.2 GeV on charged hadrons with a veto cone of 0.0001, and 0.5 GeV on neutral hadrons and photons with a veto cone of 0.01, also including the contribution from pileup vertices (with the same radius and threshold as per charged isolation) .

2. Supercluster veto: we remove all PF photons that match with any electron passing both the loose ID and SIP cuts. The matching is peformed by directly associating the two PF candidates.

3. Photons are associated to the closest lepton in the event among all those pass both the loose ID and SIP cuts.

4. We discard photons that do not satisfy the cuts $\Delta R(\gamma, l)/E^2_{T,\gamma} < 0.012$, and $\Delta R(\gamma, l) < 0.5$.

5. If more than one photon is associated to the same lepton, the lowest-$\Delta R(\gamma, l)/E^2_{T,\gamma}$ is selected.

6. For each FSR photon that was selected, we exclude that photon from the isolation sum of all the leptons in the event that pass both the loose ID and SIP cuts. This concerns the photons that are in the isolation cone and outside the isolation veto of said leptons ($\Delta R < 0.4$ AND $\Delta R > 0.01$ for muons and $\Delta R < 0.4$ AND ($\eta^{\mathrm{SC}} < 1.479$ OR $\Delta R > 0.08$) for electrons).

More details on the optimization of the FSR photon selection can be found in Ref. [**AN-15-277**, **AN-16-217**].

3.3.4 Jets

Vector Boson Fusion (VBF) and other production mechanisms of Higgs Boson normally differ as regards the jet kinematics. In this analysis, jets are thus used for the event categorization, which will be introduced in Section **??**.

Jet Identification

Jets are reconstructed by using the anti-k_T clustering algorithm out of particle flow candidates, with a distance parameter $R = 0.4$, after rejecting the charged hadrons that are associated to a pileup primary vertex.

To reduce instrumental background, the tight working point jet ID suggested by the JetMET Physics Object Group is applied [**JetID2018**]. In addition, jets from Pile-Up are rejected using the PileUp jet ID criteria suggested by the JetMET POG [**JetPUID2017**]. In this analysis, the jets are required to be within $|\eta| < 4.7$ area and have a transverse momentum above 30 GeV. In addition, the jets are cleaned from any of the tight leptons (passing the SIP and isolation cut computed after FSR correction) and FSR photons by a separation criterion: $\Delta R(\text{jet,lepton/photon}) > 0.4$.

Jet Energy Corrections

The calorimeter response to particles is not linear and it is not straightforward to translate the measured jet energy to the true particle or parton energy, therefore we need Jet Energy Corrections. In this analysis, standard jet energy corrections are applied to the reconstructed jets, which consist of L1 Pileup, L2 Relative Jet Correction, L3 Absolute Jet Correction for both Monte Carlo samples and data, and also residual calibration for data [**JECMC2018**]. Jet Energy resolution corrections are also applied to MC.

B-tagging

For categorization purpose, we need to distinguish whether a jet is b-jet or not. The *DeepCSV* algorithm is used as our b-tagging algorithm. It combines the same information as the previous tagger *CSVv2*, impact parameter significance, secondary vertex and jet kinematics but uses information of more tracks. Also, the b-tag output discriminator is computed with a Deep Neural Network. In this analysis, a jet is considered to be b-tagged if it passes the *medium* working point, i.e. if its `|pfDeepCSVJetTags:probb+` pfDeepCSVJetTags:probbb| discriminator is greater than 0.4184 [**BTAG2018**].

Data to simulation scale factors for b-tagging efficiency are provided for this working point for the full dataset as a function of jet p_T, η and flavour. They are applied to simulated jets by downgrading (upgrading) the b-tagging status of a fraction of the b-tagged (untagged) jets that have a scale factor smaller (larger) than one.

Validation on data

Similarly to what was done with electrons and muons, we performed some validation studies to probe the agreement between data and simulated samples for jets. Events containing $\mathrm{Z} \to \mathrm{ee}$ or $\to \mu^+\mu^-$ were selecting, following the same trigger and lepton selection criteria of the main analysis, but increasing the p_T threshold on leptons to 30 GeV and asking the invariant mass of the two leptons to be within the 70-110 GeV range. Basic properties of additional jets in the event, with $p_T > 30$ GeV and $\eta < 4.7$, were then compared. Figure 3.30 shows the pseudo-rapidity of the leading jet in data and MC. Simulated samples of Drell-Yan and $^-$ are used. Uncertainties from the jet energy corrections are also displayed on the data/MC ratio plots.

As it has been shown, the Jet Energy Resolution corrections are creating the "‘horns "‘ one can see in the MC for η of about 2.8. Although not satisfactory, the disagreement we see between DATA and MC in this region is covered by the very large uncertainty. After discussion with jet experts in the JetMET POG, it has been decided that this is the configuration we should use for this preliminary result. We also verified that it was not inducing significant migration of events in the signal MC between the categories.

The disagreement between data and MC for the MET distribution prevents us at this stage to use this variable in the analysis, in particular to better target VH events where, for instance, a Z would decay into neutrinos.

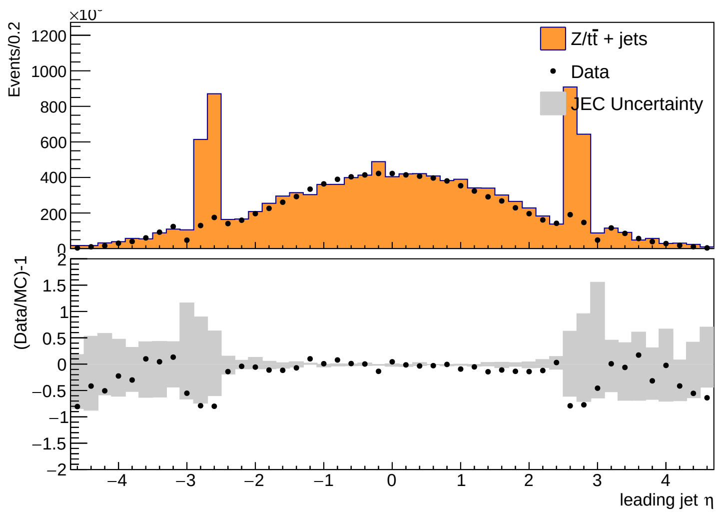


FIGURE 3.30: Comparison between data and MC for the leading jet η in Z$\rightarrow$ $\ell\ell$ +jets events. MC is normalized to data. Jet ID and Jet PUID are applied. MC samples include DY and ttbar. Data/MC ratio plots are shown in the bottom of each plots, together with the uncertainties (shaded histograms) from Jet Energy Corrections.

3.3.5 Object Summary

The requirements on all objects used for the analysis with 2018 data are summarized in the Table 3.9. In addition, a "ghost-cleaning" procedure is applied to the muons, as described in Sec. 3.3.2.

A lepton is declared **loose** if it passes the reconstruction, kinematics and dxy/dz cuts and declared **tight** if it passes in addition the identification, isolation and SIP3D cut.

TABLE 3.9: Summary of physics object selection for the analysis performed on the 2018 data.

Electrons
$\mathrm{p_T^e} > 7$ GeV $\quad \eta^e < 2.5$
$\mathrm{d_{xy}} < 0.5\,\mathrm{cm} \quad \mathrm{d_z} < 1\,\mathrm{cm}$
$\lvert\mathrm{SIP_{3D}}\rvert < 4$
BDT ID with isolation (Fall17v2 training) with cuts from Table 3.7
Muons
Global or Tracker Muon
Discard Standalone Muon tracks if reconstructed in muon system only
Discard muons with muonBestTrackType==2 even if they are globa or tracker muons
$\mathrm{p_T^\mu} > 5$ GeV $\quad \eta^\mu < 2.4$
$\mathrm{d_{xy}} < 0.5\,\mathrm{cm} \quad \mathrm{d_z} < 1\,\mathrm{cm}$
$\lvert\mathrm{SIP_{3D}}\rvert < 4$
PF muon ID if $p_T < 200$ GeV, PF muon ID or High-p_T muon ID (Table 3.8) if $p_T > 200$ GeV
$\mathcal{I}^\mu_{\mathrm{PF}} < 0.35$
FSR photons
$\mathrm{p_T^\gamma} > 2$ GeV $\quad \eta^\gamma < 2.4$
$\mathcal{I}^\gamma_{\mathrm{PF}} < 1.8$
$\Delta R(\ell,\gamma) < 0.5 \quad \frac{\Delta R(\ell,\gamma)}{(p_T^\gamma)^2} < 0.012\ \mathrm{GeV}^{-2}$
Jets
$\mathrm{p_T^{jet}} > 30$ GeV $\quad \eta^{\mathrm{jet}} < 4.7$
$\Delta R(\ell/\gamma, \mathrm{jet}) > 0.4$
Cut-based jet ID (tight WP)
Jet pileup ID (tight WP)
Deep CSV b-tagging (medium WP)

Chapter 4

Probing four lepton final-state for Higgs Cross section measurement

4.1 Introduction

Higgs particle has been the first fundamental scalar particle which was observed at LHC, CERN. Some years ago in 2012, both CMS and ATLAS collaborations announced its discovery [**Reference20**, **Reference21**] with its mass around 125 GeV. Later on, comprehensive properties of Higgs boson have been studied by these experiments [**Reference22**, **Reference21a**, **Reference21b**, **Reference21c**] which found to be compatible with SM Higgs boson.

In this chapter, measurements of the fiducial cross sections (inclusive and differential) for the Higgs boson production in the $H \to 4\ell$ decay channel ($\ell = e, \mu$) with the full Run 2 proton-proton collisions of CMS at $\sqrt{s} = 13TeV$ are presented which provide a direct test of perturbative QCD calculations of Standard Model (Higgs sector) at the TeV energy scale. Any deviation of the integrated or differential distributions from SM expectations would provide a direct hint to BSM contributions. These measurements with earlier or partial Run2 CMS data has also been performed Ref. [JHEP 04 (2016) 005, CMSPAS HIG-15-004, JHEP 11 (2017) 047, HIG-17-028] . Nearly similar measurements have already been reported by ATLAS experiment in $H \to \gamma\gamma[ATLAS-CONF-2019-029]$ [**ATLAS:Hgg_8TeV**, **ATLAS:Hgg_8TeV_13TeV**], $H \to 4\ell$ [ATLAS-CONF-2019-025], [**ATLAS:H4l_8TeV**, **ATLAS:H4l_8TeV_13TeV**] decay channels, combination of both channels [ATLAS-CONF-2019-032] [**ATLAS:H4l_Hgg_8TeV**, **ATLAS:H4l_Hgg_8TeV_13TeV**] and by CMS experiment in $H \to \gamma\gamma$ decay channel [**CMS:Hgg_8TeV**].

The integrated fiducial cross section measurements are performed using collisions data corresponding to integrated luminosity of 137.1 fb^{-1} at $\sqrt{s} = 13$ TeV, recorded with the CMS detector at LHC. To minimize the dependence of the results on the underlying models of the Higgs boson production mechanism and its properties, the cross sections are extracted in a fiducial phase space closely matching with the experimental acceptance as shown in Figure 4.1. The remaining model dependence is extracted by repeating the measurements for SM and other exotic models and reported as a distinct systematic uncertainty. All measurements includes the corrections for the finite detection efficiency and resolution effects. As

$\sigma \times BR$ strongly depends on the Higgs mass, the measurements have been extracted assuming $m_H = 125.09$ GeV, as measured using the $H \to 2\gamma$ and $H \to 4\ell$ decay channels by the CMS experiment [**Reference39**].

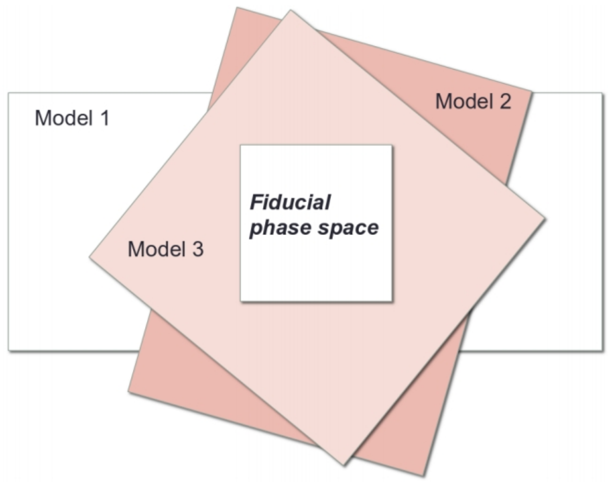


FIGURE 4.1: Fiducial phase definition among different models.

The differential fiducial cross sections are measured as a function of kinematic observables which are sensitive to the Higgs-boson production mechanisms: transverse momentum and the rapidity of the four-lepton system, transverse momentum of the leading jet and jet multiplicity.

The chapter is structured as follows. The datasets and simulated samples used for the analysis are introduced in Section 4.1.1. The analysis strategy is described in Section 4.2, where fiducial phase space definition, event selection and method for extracting integrated and differential fiducial cross sections are explained. In Section 4.3 details of systematic uncertainties affecting the measurement are explained. In Section 4.5 results are compared with SM theoretical predictions.

4.1.1 Datasets, Simulation Samples and Theoretical Predictions

4.1.2 Datasets

4.1.3 Data

Triggers and Datasets

This analysis uses a data sample recorded by the CMS experiment during 2016, 2017 and 2018, corresponding to $137.1\ \text{fb}^{-1}$ of data.

The datasets used are listed and documented in **??**. The analysis relies on four different primary datasets (PDs), *DoubleMuon*, *MuEG*, *EGamma*, and *SingleMuon*, each of which combines a certain collections of HLT paths. To avoid duplicate events from different primary datasets, events are taken:

- from EGamma if they pass the diEle or triEle or singleElectron triggers,
- from DoubleMuon if they pass the diMuon or triMuon triggers and fail the diEle and triEle triggers,
- from MuEG if they pass the MuEle or MuDiEle or DiMuEle triggers and fail the diEle, triEle, singleElectron, diMuon and triMuon triggers,
- from SingleMuon if they pass the singleMuon trigger and fail all the above triggers.

The HLT paths used for 2018 collision data is listed in **??**.

Trigger Efficiency

The efficiency in data of the combination of triggers used in the anlysis with respect to the offline reconstruction and selection is measured by considering 4ℓ events triggered by single lepton triggers. One of the four reconstructed leptons (the "tag") is geometrically matched to a trigger object passing the final filter of one of the single muon or single electron triggers. The other three leptons are used as "probes". In each 4ℓ event there are up to 4 possible tag-probe combinations, and all possible combinations are counted in the denominator of the efficiency. For each of the three probe leptons all matching trigger filter objects are collected. Then the matched trigger filter objects of the three probe leptons are combined in attempt to reconstruct any of the triggers used in the anlysis. If any of the analysis triggers can be formed using the probe leptons, the set of probes is also counted in the numerator of the efficiency.

This method does not have a perfect closure in MC events due to the fact that the presence of a fourth lepton increases the trigger efficiency, and this effect is not accounted for. Also, in the $2e2\mu$ final state, the three probe leptons cannot be combined to form all possible triggers which can collect events with two electrons and two muons (e.g. if the tag lepton is an electron, the three remaining leptons cannot pass a double electron trigger). Therefore the method is also exercised on MC and the difference between data and MC is used to determine the reliability of the simulation. The efficiency plotted as a function of the minimum p_{T} of the three probe leptons in data and MC using this method can be seen in Fig. 4.2 that shows the trigger efficiency in 2018 data for different final states. The MC efficiency describes well the data within the statistical uncertainties.

4.1.4 Simulation Samples and Theoretical Predictions

The SM Higgs boson production by gluon fusion ($\mathrm{gg} \to \mathrm{H}$) process descriptions (Figure) are acquired using the HRES **[deFlorian:2012mx, Grazzini:2013mca]**, POWHEG MINLO-HJ **[Hamilton:2012np]** and POWHEG V2 **[powheg, powhegvv]** generators.

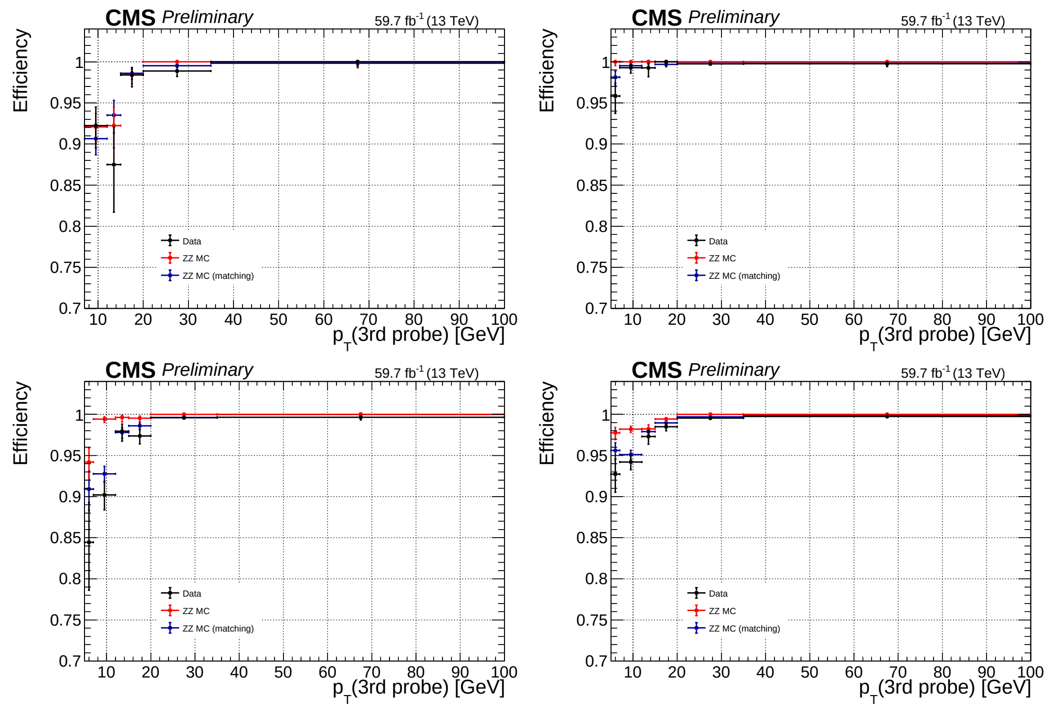


FIGURE 4.2: Trigger efficiency measured in 2018 data using 4ℓ events collected by single lepton triggers for the $4e$ (top left), 4μ (top right), $2e2\mu$ (bottom left) and 4ℓ (bottom right) final states.

The $\mathrm{gg} \to \mathrm{H}$ process description at next-to-next-to-leading order (NNLO) accuracy in perturbative QCD and next-to-next-to-leading logarithmic (NNLL) accuracy in resummation of soft-gluon effects at small transverse momenta is obtained by partonic level MC generator HRES. The inclusive jet associated activity predictions of fiducial cross section is attained by HRES because the resummation is inclusive over the QCD radiation recoiling against the Higgs boson. HRES predictions are obtained using the central values for the factorization and normalization scales to be $m_H = 125$ GeV. The partonic level matrix-element generator, the POWHEG, provides perturbative QCD calculations at next-to-leading (NLO) accuracy with an connection to parton-shower programs. The Higgs boson production via $\mathrm{gg} \to \mathrm{H}$ with zero associated jets can be described at NLO accuracy using the POWHEG. For this analysis, the p_T spectrum of the Higgs boson obtained by POWHEG has been reweighted to match closely with HRES generator in the full phase space. This factor shrinks additional jets emission in large p_T region, and raises the contribution in the region where p_T approaches to zero by the Sudakov form factor. The $\mathrm{gg} \to \mathrm{H}$ production in association with up to one additional jets is improved to next-to-leading logarithmic (NLL) accuracy by extending POWHEG generator to the POWHEG MINLO-HJ generator based on the MiNLO prescription **[Hamilton:2012np]**. The POWHEG MINLO-HJ generator provides NLO accuracy $\mathrm{gg} \to \mathrm{H}$ production in association with zero and one additional jets and

LO accuracy for two jets. Finite masses of top and bottom quarks has been considered for gg → H production for all generators. The SM Higgs boson production by

Signal Samples

Descriptions of the SM Higgs boson production are obtained using the POWHEG V2 [**Alioli:2008gx**, **Nason:2004rx**, **Frixione:2007vw**] generator for the five main production modes: gluon fusion (gg → H) including quark mass effects [**Bagnaschi:2011tu**], vector boson fusion (VBF) [**Nason:2009ai**], and associated production (WH, H and H [**Hartanto:2015uka**]). In the case of WH and H the MiNLO HVJ extension of POWHEG is used [**Luisoni:2013kna**]. The description of the decay of the Higgs boson to four leptons is obtained using the JHUGEN generator [**Gao:2010qx**]. In the case of WH, H and H, the Higgs boson is allowed to decay to H→→ $2\ell 2X$ such that 4-lepton events where two leptons originate from the decay of associated , W bosons or top quarks are also taken into account in the simulation. Showering of parton-level events is done using PYTHIA8.209, and in all cases matching is performed by allowing QCD emissions at all energies in the shower and vetoing them afterwards according to the POWHEG internal scale. All samples are generated with the NNPDF 3.1 NLO parton distribution functions (PDFs) [**Ball:2014uwa**]. The list of signal samples and their cross sections are shown in **??**. A subset of these samples are used to asses the model dependence of the measurement results are indicated in the Table **??**.

The MC samples generated by POWHEG and PYTHIA8.209 are used to extract fiducial phase space acceptance and reconstruction efficiencies for individual models separately as described in Section 4.2.4. These samples, along with the samples generated by using the HRES and the POWHEG MINLO-HJ with alternate representation of gg → H process, are utilized for the comparison of the measurement results with theoretical predictions based on SM calculation in Section 4.5.

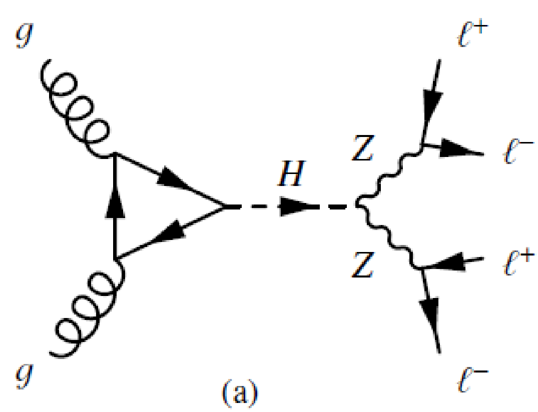


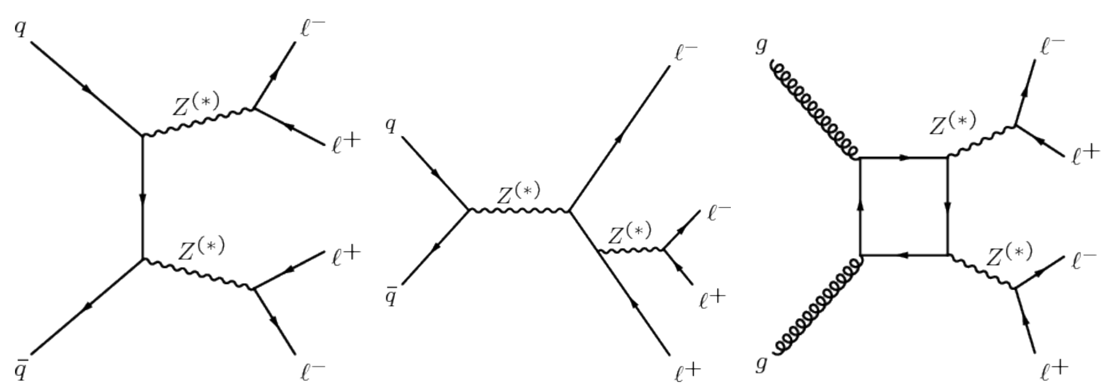
FIGURE 4.3: gg → H process fynman diagram.

All SM production modes have been produced individually to estimate the dependence of the measurements on underlying assumptions of Higgs boson production mechanism. We have also simulated a wide range of samples that provide descriptions for exotic Higgs-like resonances productions and their decay to the four-lepton final state to estimate the dependence of the measurements on the Higgs boson properties. These include spin-zero, spin-one and spin-two resonances with anomalous interactions with a pair of neutral gauge bosons (ZZ, $Z\gamma^*$, $\gamma^*\gamma^*$) described by higher-order operators, as discussed in details in Ref. [**Khachatryan:2014kca**]. The details of the method used to asses the model dependence will be described in Section 4.3. All these MC samples are produced using the POWHEG generator for the representation of NLO QCD effects in the production mechanisms, and JHUGEN [**Gao:2010qx**, **Bolognesi:2012mm**, **Anderson:2013afp**] to represent the decay of these exotic resonances to four-leptons containing all spin correlations.

Background Samples

For the estimations of the backgrounds using MC samples, the background process $\mathrm{q\bar{q}} \to \mathrm{ZZ} \to 4\ell$ was generated at NLO accuracy containing s-, t- and u-channel diagrams (Figure 4.4) using POWHEG V2 [**Nason:2013ydw**] and PYTHIA8, with the same settings as for the Higgs signal. As this simulation covers a large range of ZZ invariant masses, dynamical QCD factorization and renormalization scales have been chosen, equal to m_{ZZ}.

The $\mathrm{gg} \to ZZ$ process (Figure 4.4) is simulated at leading order (LO) with MCFM [**MCFM**, **Campbell:2013una**]. In order to match the $\mathrm{gg} \to \mathrm{H} \to ZZ$ transverse momentum spectra predicted by POWHEG at NLO, the showering for MCFM samples is performed with different PYTHIA8 settings, allowing only emissions up to the parton-level scale (“wimpy” shower).



(a) s-channel (b) t-channel (c) ggZZ-box

FIGURE 4.4: $\mathrm{q\bar{q}}to\mathrm{ZZ} \to 4\ell$ process s- and t- channel fynman diagrams (a) and (b). $\mathrm{gg} \to \mathrm{ZZ} \to 4\ell$ process fynman diagram (c).

The initial- and final-state radiation have been included for all events generators, which can guide to the existence of extra hard photons in the event. The

POWHEG and JHUGEN event generators also properly take into account interference between all contributing amplitudes of the $H \to 4\ell$ process, containing those affiliated to the reordering of identical leptons in the 4μ and $4e$ final states. The set of parton distribution functions (PDF) CTEQ6L [**cteq66**], CT10 [**CT10**] and MSTW2008 [**MSTW08**] are used for LO, NLO, and NNLO generators.

To include the fact of parton shower, underlying events and hadronization, all generated events were interfaced to PYTHIA 6.4 Tune $Z2^*$ [**Chatrchyan:2011id**]. The PYTHIA 6.4 $Z2^*$ tune is obtained from the Z1 tune [**Field:2010bc**], that takes the CTEQ5L parton distribution set, whereas $Z2^*$ follows CTEQ6L [**Pumplin:2002vw**]. HRES does not supply an interface with the programs that can include the effects of hadronization and multi-parton interaction. To correct HRES predictions for these effects, the POWHEG+JHUGEN events without hadronization and multi-parton interactions were reweighted to match closely the HRES events in a phase space minimally greater than fiducial phase space. After that, the hadronization and multi-parton interaction effects are included in reweighted POWHEG+JHUGEN events by an interface with PYTHIA. The procedure of reweighting events to match HRES to add non-perturbative effects to HRES partonic level predictions was repeated for all studied observables for integrated and differential cross section measurements.

All the generated events were processed via a full simulation of the CMS detector based on GEANT4 [**GEANT**] and were reconstructed with the same algorithms that are used for data analysis. Multiple interactions (pileup) are incorporated in simulations by reweighting the number of primary vertex distribution to match the distribution of the observed data. Energy deposits from pileup interactions and underlying events are subtracted when computing the energy of jets and isolation of reconstructed objects using the FASTJET technique [**Cacciari:2007fd**, **Cacciari:2008gn**, **Cacciari:2011ma**]. This technique is based on the calculation of the η-dependent average energy density evaluated individually for each event.

All the data-to-MC scale factors are determined using a "tag and probe" method [**CMS:2011lsa**], discussed in Sections 3.3.1 and 3.3.2, and are used to correct simulated samples to compensate the differences of the leptons selection efficiencies in data and simulation. The dependency of the selection efficiency on the number of reconstructed primary vertices in the event is found to be negligible for data samples. More details are presented and discussed in Ref. [**Chatrchyan:2013mxa**].

Pileup Reweighting

For each year, corresponding simulation samples are reweighted to match the pileup distribution in data. An example of reweighting procedure for 2018 data is shown in Fig. 4.5.

4.2 Analysis Strategy

The integrated and differential cross sections are measured in a fiducial region in order to reduce the effects of model dependent acceptances. The fiducial selection mimics the reconstruction level selection, which is optimized to detect a low

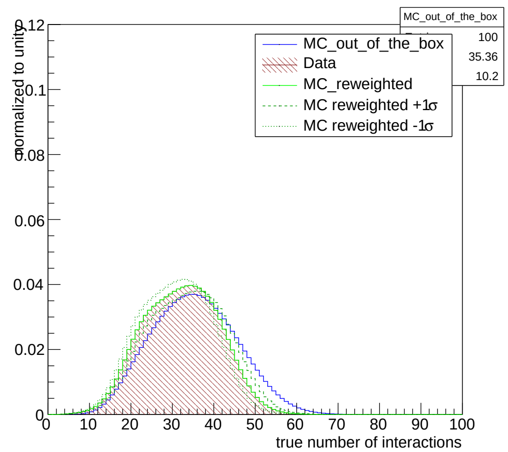


FIGURE 4.5: Distribution of pileup in 2018 MC and Data shown before and after the application of PU weights. Up and down variations of 5% in the minimum bias cross section when calculating the weighs are also shown.

mass (m$\sim$ 125 GeV) Higgs boson decaying to 4 leptons through two Z bosons. Since in the SM there are multiple production modes, the fiducial selection and measurement strategy are designed to be independent of how the Higgs boson is produced.

For this reason, the inclusion of isolation in the fiducial phase space definition is necessary in order to make the reconstruction efficiency independent with respect to the fiducial phase space independent of the number of jets.

Furthermore, in the sub-leading SM production modes there can be associated vector bosons which can decay to leptons, and therefore the reconstructed Higgs boson candidate may not be composed of the correct combination of the leptons. In order to retain the full information from an unbinned simultaneous fit of the signal and background pdfs to the four-lepton mass distribution, it is necessary to treat such "wrong combination" events as a background.

Further signal-induced background can be formed due to imperfect correspondence between quantities at the particle level and the corresponding quantities at the reconstruction level. Signal events which are from the fiducial phase space but fail the reconstruction are accounted in the fiducial efficiency (ϵ) and the expected contribution from events which pass the reconstruction selection but are not from within the fiducial phase space (f_{nonfid}) are referred to as "nonfiducial signal" events and are treated as a background, illustrated in Figure 4.6.

The effects of imperfect detector resolution can in general have a non-negligible impact on the shape of the distribution of the measured observables. For this reason, in case of the differential cross section measurements, a procedure to correct for the detection efficiencies and resolution effects is applied. Throughout this document, this procedure will be referred to as the unfolding procedure, and the unfolded differential distributions will be referred to as distributions at the fiducial

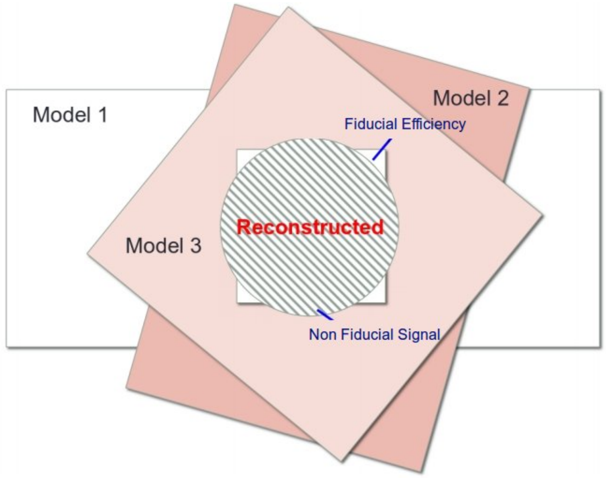


FIGURE 4.6: Fiducial efficiency(ϵ) and nonfiducial(f_{nonfid}) signal definition.

level. The reconstruction level selection, fiducial phase space definition, statistical procedure, and their model dependence will be described in the subsequent Sections.

4.2.1 Event Selection and Background Modeling

Event Selection

The event selection for these measurements are similar to the event selection used in previous Higgs boson properties measurements in the same final state [**HIG-18-001**, **HIG-16-041**, **Chatrchyan:2013mxa**, **Khachatryan:2014kca**]. The online event selection is based on having fired High Level Trigger paths as described in [**sec:trigpaths**]. Unlike in the Run I analysis, the trigger requirement does not depend on the selected final state: it is always the OR of all HLT paths. The reason is in Run II we will be targeting associated production modes that can come with additional leptons, thus improving trigger efficiency further. The events are required to have at least one good primary vertex (PV) fullfilling the following criteria: high number of degree of freedom ($N_{PV} > 4$), collisions restricted along the $z-$axis ($z_{PV} < 24$ cm) and small radius of the PV ($r_{PV} < 2$ cm).

The minimum p_T of 17 and 8 GeV is required for leading and subleading leptons for double leptons triggers, while 15, 8 and 5 GeV for triple electron trigger.

4.2.2 ZZ Candidate Selection

The four-lepton candidates are built from what we call **selected leptons**, which are the tight leptons (defined in sections 3.3.1 and 3.3.2) that pass the $\mathrm{SIP}_{3\mathrm{D}} < 4$ vertex constraint and the isolation cuts (defined in sections 3.3.1 and 3.3.2), where

FSR photons are subtracted as described in Section 3.3.3. A lepton cross cleaning is applied by discarding electrons which are within $\Delta R < 0.05$ of selected muons.

The construction and selection of four-lepton candidates proceeds according to the following sequence:

1. **Z candidates** are defined as pairs of selected leptons of opposite charge and matching flavour (e^+e^-, $\mu^+\mu^-$) that satisfy $12 < m_{\ell\ell(\gamma)} < 120$, where the Z candidate mass includes the selected FSR photons if any.

2. **ZZ candidates** are defined as pairs of non-overlapping Z candidates. The Z candidate with reconstructed mass $m_{\ell\ell}$ closest to the nominal Z boson mass is denoted as Z_1, and the second one is denoted as Z_2. ZZ candidates are required to satisfy the following list of requirements:

   - **Ghost removal** : $\Delta R(\eta, \phi) > 0.02$ between each of the four leptons.
   - **lepton** p_T: Two of the four selected leptons should pass $p_{T,i} > 20$ and $p_{T,j} > 10$.
   - **QCD suppression**: all four opposite-sign pairs that can be built with the four leptons (regardless of lepton flavor) must satisfy $m_{\ell\ell} > 4$. Here, selected FSR photons are not used in computing $m_{\ell\ell}$, since a QCD-induced low mass dilepton (eg. J/Ψ) may have photons nearby (e.g. from π_0).
   - Z_1 **mass**: $m_{\mathrm{Z1}} > 40$
   - **'smart cut'**: defining $\mathrm{Z_a}$ and $\mathrm{Z_b}$ as the mass-sorted alternative pairing Z candidates ($\mathrm{Z_a}$ being the one closest to the nominal Z boson mass), require NOT($|m_{\mathrm{Za}} - m_{\mathrm{Z}}| < |m_{\mathrm{Z1}} - m_{\mathrm{Z}}|$ AND $m_{\mathrm{Zb}} < 12$). Selected FSR photons are included in m_{Z}'s computations. This cut discards 4μ and $4e$ candidates where the alternative pairing looks like an on-shell Z + low-mass $\ell^+\ell^-$. (NB. In Run I, such a situation was avoided by choosing the best ZZ candidate before applying kinematic cuts to it, most precisely before the $m_{\mathrm{Z2}} > 12$ cut. The present smart cut allows to choose the best ZZ candidate after all kinematic cuts.)
   - **four-lepton invariant mass**: $m_{4\ell} > 70$

3. Events containing at least one selected ZZ candidate form the **signal region**.

4.2.3 Choice of the best ZZ Candidate

Unlike in the Run I analysis, the best ZZ candidate is now chosen after all kinematic cuts, a change that allows to test other selection strategies for this candidate choice. This is especially relevant for events with more than four selected leptons, having in mind the search for the Higgs boson production modes where associated particles can decay to leptons, such as VH and ttH.

For fiducial cross section measurement, Z_1 is chosen with mass closest to nominal Z boson mass and Z_2 is chosen from the Z candidates whose lepton pair gives higher p_T sum. For STXS measurement, we adopt a different approach: if more than one ZZ candidate survives the selection, we choose the one with the highest value of $\mathcal{D}^{\text{kin}}_{\text{bkg}}$ (defined in Section **??**, except when two candidates consist of the same four leptons in which case the candidate with Z_1 closest in mass to nominal Z boson mass is retained.

Reconstruction of the jets is incumbent to explore the differential cross section measurements with respect to jets related observables. Jets are reconstructed with $p_T \geq 30$ GeV and $|\eta| \leq 4.7$, discussed in Section **??**. To remove jets overlap with leptons or selected photons, it is required to have at least $\Delta R \equiv \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2} > 0.5$ [**Chatrchyan:2013mx**] from any of the selected leptons and FSR photons candidates.

Background Modeling

The dominant and sub-dominant irreducible backgrounds for $\mathrm{H} \to 4\ell$ process are ZZ production via $\mathrm{q\bar{q}}$ annihilation and gg fusion respectively. The reducible background is of the instrumental nature and arises from the process where partial jets are misidentified as leptons. The main processes which contributes in this background are: jets production in association with Z boson, jets production in association with WZ boson pair and the $\mathrm{t\bar{t}}$ pair production. From now on, this background will be denoted as Z+X.

The irreducible backgrounds $\mathrm{q\overline{q}} \to \mathrm{ZZ}$ and gg $\to \mathrm{ZZ}$ for $\mathrm{H} \to 4\ell$ integrated and differential measurements are estimated from Monte Carlo (MC) simulation as done earlier in Ref. [**Chatrchyan:2013mxa**]. In addition, following the study in Ref. [**Bonvini:2013**], in these measurements we have increased the expected yield for the gg $\to \mathrm{ZZ}$ background by assigning to the leading order (LO) background cross section a $m_{4\ell}$ dependent k-factor equal to the one used for the signal $\mathrm{H} \to 4\ell$ process. In the region $(105 < m_{4\ell} < 140$ GeV) this factor leads to an increase of the expected gg $\to$ ZZ background by a factor of ~ 2.66. After the inclusion of this k-factor, the gg $\to ZZ$ process is still a sub-dominant background in all $\mathrm{H} \to 4\ell$ measurements.

The reducible background (Z+X) is evaluated using a tight-to-loose "fake rate" method and the control regions in the data, following Ref. [**Chatrchyan:2013mxa**]. The yields of the Z+X background are determined by two methods (opposite sign (OS) or same sign (SS)) are found to be consistent with each within statistical uncertainties. The first control region is obtained by requiring that two leptons passing loose selections which does fulfill the criteria of Z_1 candidate (failing ID and isolation requirements), while the other two leptons in the event passes the criteria of Z_1 candidate. The sample is represented as "2 Prompt + 2 Fail" (2P2F) sample. This sample is populated with the events contains only two prompt leptons (mostly Drell-Yan with small fraction of Z/γ^* and $\mathrm{t\bar{t}}$ events). The second control region is obtained by requiring that three leptons pass the final requirements. The sample is represented as "3 Prompt + 1 Fail" (3P1F) sample.

The $m_{4\ell}$ shape of the reducible background is attained from OS method by performing a fit of $m_{4\ell}$ distributions of 3P1F and 2P2F events independently by empirical analytical functional constructed by Landau [**landau:1944**] in Ref. [**Chatrchyan:2013mxa**].

Prediction for the reducible background in all three channels together fitted using this empirical shape with indicated total uncertainty is shown in Figure **??**. In case of the differential $\mathrm{H} \to 4\ell$ measurements, the shape of the reducible background is obtained from the templates built from the control regions in data for each bin of the observable considered for the measurement, following a procedure similar to the one used in the spin-parity studies described in the Ref. [**Chatrchyan:2013mxa**]. More details on this procedure and template shapes for each of the differential $\mathrm{H} \to 4\ell$ measurements will be given in Section 4.2.7.

The number of estimated signal and background events, along with observed data events passing all selections in the mass range used for $\mathrm{H} \to 4\ell$ measurements ($m_{4\ell} > 70$ GeV) are listed in Table 4.1 for full Run 2 datasets.

TABLE 4.1: The number of observed candidates, expected background and signal events passing all the selections in the mass range of $m_{4\ell} > 70$ GeV corresponding to three different years. ZZ and signal events are estimated using MC simulations while Z+X is estimated by control regions in data.

Channel	4e	4μ	$2e2\mu$	4ℓ
2016 (13 TeV)				
$q\bar{q} \to ZZ$	$192.7^{+18.6}_{-20.1}$	$360.2^{+24.9}_{-27.3}$	$471.0^{+32.6}_{-35.7}$	$1023.9^{+68.9}_{-76.0}$
$gg \to ZZ$	$41.2^{+6.3}_{-6.1}$	$69.0^{+9.5}_{-9.0}$	$101.7^{+14.0}_{-13.3}$	$211.8^{+28.9}_{-27.5}$
Z + X	$21.1^{+8.5}_{-10.4}$	$34.4^{+14.5}_{-13.2}$	$59.9^{+27.1}_{-25.0}$	$115.4^{+31.9}_{-30.1}$
Sum of backgrounds	$255.0^{+23.9}_{-25.1}$	$463.5^{+31.9}_{-33.7}$	$632.6^{+44.2}_{-46.1}$	$1351.1^{+85.8}_{-91.2}$
Signal ($m_{\mathrm{H}} = 125$ GeV)	$12.1^{+1.3}_{-1.4}$	23.8 ± 2.1	30.3 ± 2.6	66.3 ± 5.6
Total expected	$267.1^{+24.9}_{-26.1}$	$487.4^{+33.1}_{-34.9}$	$662.9^{+45.7}_{-47.5}$	$1417.4^{+89.1}_{-94.3}$
Observed	293	505	681	1479
2017 (13 TeV)				
$q\bar{q} \to ZZ$	$233.86^{+y.y}_{-z.z}$	$441.08^{+y.y}_{-z.z}$	$569.51^{+y.y}_{-z.z}$	$1244.45^{+y.y}_{-z.z}$
$gg \to ZZ$	$49.11^{+y.y}_{-z.z}$	$81.50^{+y.y}_{-z.z}$	$121.05^{+y.y}_{-z.z}$	$251.66^{+y.y}_{-z.z}$
Z + X	$12.39^{+y.y}_{-z.z}$	$36.39^{+y.y}_{-z.z}$	$44.45^{+y.y}_{-z.z}$	$93.24^{+y.y}_{-z.z}$
Sum of backgrounds	$295.37^{+y.y}_{-z.z}$	$558.97^{+y.y}_{-z.z}$	$735.02^{+y.y}_{-z.z}$	$1589.35^{+y.y}_{-z.z}$
Signal ($m_{\mathrm{H}} = 125$ GeV)	$13.85^{+y.y}_{-z.z}$	$28.64^{+y.y}_{-z.z}$	$35.55^{+y.y}_{-z.z}$	$78.04^{+y.y}_{-z.z}$
Total expected	$309.22^{+y.y}_{-z.z}$	$587.60^{+y.y}_{-z.z}$	$770.57^{+y.y}_{-z.z}$	$1667.39^{+y.y}_{-z.z}$
Observed	307	602	797	1706
2018 (13 TeV)				
$q\bar{q} \to ZZ$	333^{+57}_{-53}	622^{+31}_{-44}	815 ± 73	1770^{+98}_{-101}
$gg \to ZZ$	$75.1^{+14.3}_{-13.5}$	$116.6^{+11.7}_{-12.8}$	176.9 ± 23.0	$368.5^{+29.5}_{-29.6}$
Z + X	19.3 ± 7.2	50.8 ± 15.2	64.6 ± 15.6	134.7 ± 22.9
Sum of backgrounds	$428^{+59.2}_{55.2}$	$790^{+36.4}_{-48.3}$	1057 ± 78.1	$2274^{+104.9}_{-107.7}$
Signal ($m_{\mathrm{H}} = 125$ GeV)	$19.6^{+3.3}_{-3.1}$	$40.8^{+2.5}_{-2.9}$	50.7 ± 5.6	$111.1^{+6.9}_{-7.0}$
Total expected	$447^{+59.3}_{55.2}$	$830^{+36.5}_{-48.4}$	1108 ± 78.3	$2385^{+105.1}_{-107.9}$
Observed	462	850	1130	2442

Figure 4.7 shows the reconstructed four-leptons invariant mass distribution for whole Run 2 datasets and MC samples in full mass range and high mass range.

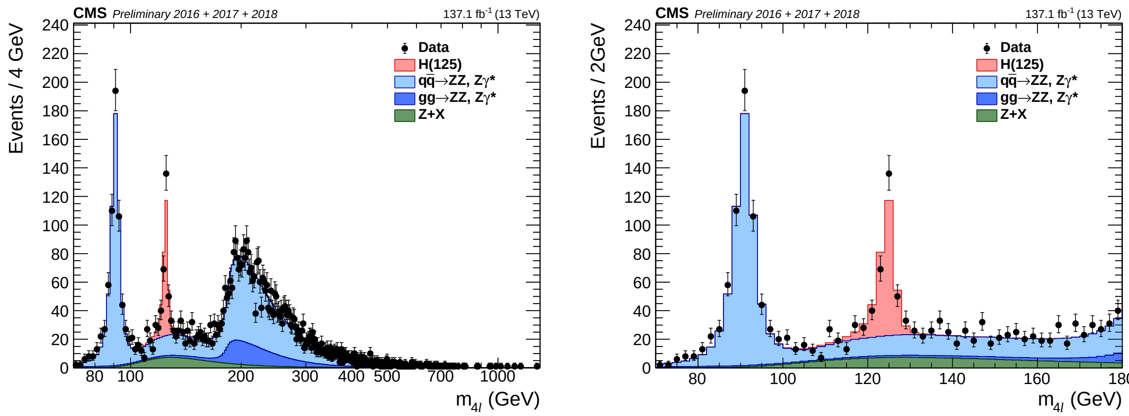


FIGURE 4.7: Distributions of invariant mass of four-leptons using full Run 2 datasets in the full mass range (left) and the low-mass range (right), along with predictions for Standard Model Higgs ($m_H = 125.0$ GeV), including $Z \to 4\ell$ resonance.

4.2.4 Definition of the Fiducial Volume

The fraction of all $\mathrm{H} \to 4\ell$ events that get selected following the event selection procedure can change significantly for various Higgs boson production mechanism and distinct exotic models of the Higgs boson. In order to minimize the dependence of the measurements on the underlying assumption on the models of the Higgs boson production and properties, the fiducial phase space for the $\mathrm{H} \to 4\ell$ cross section measurements is defined to match as closely as possible the experimental acceptance defined by the reconstruction level selection.

The fiducial phase space demands for at least four leptons having $p_T > 7(5)$ GeV for electrons (muons) with leading and sub-leading leptons having $p_T > 20$ GeV and $p_T > 10$ GeV. The four-momenta of each lepton is calculated at the hard scattering level (PYTHIA status code $= 3$), before any final state radiation occurs. All electrons (muons) must have pseudorapidity $|\eta| < 2.5(2.4)$. Each lepton isolation is calculated by adding p_T of all stable particles (PYTHIA status code $= 1$) within a cone of radius $\Delta R < 0.3$ around that lepton. All neutrinos and FSR photons are excluded from above summation. Then the relative isolation of the lepton is defined by the ratio of this sum and p_T of the considered lepton which is required to be less than 0.35, consistent with reconstruction level isolation requirement **[Chatrchyan:2013mxa]**. The requirement of isolation in fiducial phase space definition is crucial to control the difference of signal selection efficiency among different models. The studies based on simulation samples show that signal efficiency can change up to 45% among different models without requiring lepton isolation at fiducial level as shown in Table **??**, the maximum difference comes from $t\bar{t}\mathrm{H}$ and $\mathrm{gg} \to \mathrm{H}$ production modes. This behavior is expected as $t\bar{t}\mathrm{H}$ events has more hadronic activity around leptons as compared to $\mathrm{gg} \to \mathrm{H}$ events, so the cut at relative isolation kills more events in $t\bar{t}\mathrm{H}$ than $\mathrm{gg} \to \mathrm{H}$.

Furthermore, fiducial phase space is defined by several other topological and kinematics selections. There must be two SFOS lepton pairs, Z_1 is reconstructed with invariant mass closest to the world average Z boson mass by all possible combination of same flavor and opposite charge (SFOS) pairs of leptons within mass range of $40 \leq m(\mathrm{Z}_1) \leq 120$ GeV. Z_2 is reconstructed by remaining SFOS pair of leptons within inv. mass range of $12 \leq m(\mathrm{Z}_2) \leq 120$ GeV. If more than one Z_2 passes the criteria, the leptons pair with largest $\Sigma|p_T|$ is selected. To remove the overlapping between leptons, we ask for $\Delta R(\ell_i \ell_j) > 0.02$ for any $i \neq j$. To remove contamination from QCD low masses resonance, any opposite-charge pair of leptons is required to pass $m(\ell^+\ell'^-) \geq 4$ GeV. Finally, We require that all selected leptons must emerge from Higgs boson decay and invariant mass of four-leptons must fulfill 105 GeV $< \mathrm{m}(4\ell) < 140$ GeV. The demand for $\mathrm{m}(4\ell)$ helps to control the contamination from off-shell gg fusion production which is of considerable size, capable to increase the total cross sextion by a few percent [**Kauer:2013cga**, **Kauer:2012hd**]. The requirements that defines fiducial phase space are also gathered in Table 4.2.

TABLE 4.2: Overview of all requirements utilized to define the fiducial phase space for the $\mathrm{H} \to 4\ell$ cross section measurements.

Requirements for the $\mathrm{H} \to 4\ell$ fiducial phase space	
Lepton kinematics and isolation	
Leading lepton p_T	$p_T > 20$ GeV
Sub-leading lepton p_T	$p_T > 10$ GeV
Additional electrons (muons) p_T	$p_T > 7\ (5)$ GeV
Pseudorapidity of electrons (muons)	$\lvert\eta\rvert < 2.5\ (2.4)$
Sum of scalar p_T of all stable particles within $\Delta R < 0.3$ from lepton	$< 0.35 p_T$
Event topology	
Existence of at least two SFOS lepton pairs, where leptons satisfy criteria above	
Inv. mass of the Z_1 candidate	$40 < m(\mathrm{Z}_1) < 120$ GeV
Inv. mass of the Z_2 candidate	$12 < m(\mathrm{Z}_2) < 120$ GeV
Distance between selected four leptons	$\Delta R(\ell_i \ell_j) > 0.02$
Inv. mass of any opposite-sign lepton pair	$m(\ell_i^+ \ell_j^-) > 4$ GeV
Inv. mass of the selected four leptons	$105 < m_{4\ell} < 140$ GeV

It has been established based on simulation samples that with including isolation for the events passing fiducial phase space definition reconstruction efficiencies faintly depends on Higgs boson properties and production modes. The remaining model dependence will be quoted as a separate uncertainty as explained in Section 4.3. Part of the signal events passing fiducial phase space definition ($\mathcal{A}_{\text{fid}}$), and reconstruction efficiencies (ϵ) for SM production modes are registered in Table 4.3.

TABLE 4.3: Summary of different Standard Model signal models. The MC samples are from full Run 2 yearly productions (2016, 2017 and 2018) assuming $m_{\mathrm{H}} = 125.0$ GeV. The fiducial acceptance ($\mathcal{A}_{\mathrm{fid}}$) defined the fraction of events passing fiducial phase space selections, reconstruction efficiency (ϵ) represented by the ratio of reconstructed to accepted events. Also, f_{nonfid} which are reconstructed events but does not pass fiducial phase space definition at generator level. Values are obtained using 13 TeV signal models and the uncertainties shown are only statistical uncertainties.

Signal process	$\mathcal{A}_{\mathrm{fid}}$	ϵ	f_{nonfid}	$(1+f_{\mathrm{nonfid}})\epsilon$
Individual Higgs boson production modes				
gg→H (POWHEG)	0.402 ± 0.001	0.592 ± 0.002	0.053 ± 0.001	0.624 ± 0.002
VBF (POWHEG)	0.444 ± 0.002	0.605 ± 0.003	0.043 ± 0.001	0.631 ± 0.003
WH (POWHEG+MINLO)	0.325 ± 0.002	0.588 ± 0.003	0.075 ± 0.002	0.632 ± 0.004
ZH (POWHEG+MINLO)	0.340 ± 0.003	0.594 ± 0.005	0.081 ± 0.004	0.643 ± 0.006
ttH (POWHEG)	0.314 ± 0.003	0.585 ± 0.006	0.169 ± 0.006	0.684 ± 0.007
ggH(NNLOPS)	0.442 ± 0.001	0.595 ± 0.002	0.049 ± 0.001	0.624 ± 0.002

4.2.5 Signal Model and Statistical Procedure

The purpose is to extract integrated and differential fiducial cross sections after corrections of detector effects, resolution and systematic biases. To achieve this target, the corrections are extracted by using simulated samples and then encoded in the parametrization of $m_{4\ell}$ spectra at the reconstruction level. Then we directly extract the parameter of interest (σ_{fid}) by performing the maximum likelihood fit of background and signal parametrization to the observed $m_{4\ell}$ distribution. All the systematic uncertainties are encoded as nuisance parameters, which are profited properly in the fit procedure. To obtain results, we used asymptotic approach [**Cowan:2010js**] with the test statistics based on the profile likelihood ratio [**LHC-HCG**]. The maximization of the likelihood is performed by fitting simultaneously the signal cross section in each bin at the fiducial level with all reconstruction bins and all three final states. The procedure applied in case of the $\mathrm{H} \to 4\ell$ integrated and differential fiducial cross section measurements will be given in Sections 4.2.6 and 4.2.7. Finally, the unfolding procedure to extract the fiducial cross section in bins of kinematic observables at the fiducial level, corrected for the detection efficiency and resolution effects, is performed as discussed in Ref. [**Prosper:2011zz**] and similar to what is adopted for $H \to \gamma\gamma$ decay channel [**CMS:HIG-14-016**].

The shape of signal resonant contribution ($\mathcal{P}_{\mathrm{res}}(m_{4\ell})$) is denoted by double-sided Crystal Ball function [**Chatrchyan:2013mxa**] given below, with the yield proportional to the parameter of interest (σ_{fid}).

$$f_{dCB}(x;\alpha,n,s,\sigma) = N \left\{ \begin{array}{ll} e^{-\frac{(x-s)^2}{2\sigma^2}} & \text{if } -\alpha_L < \frac{x-s}{\sigma} < \alpha_R \\ A_L(B_L - \frac{x-s}{\sigma}^{-n_L}) & \text{if } \frac{x-s}{\sigma} \leq -\alpha_L \\ A_R(B_R + \frac{x-s}{\sigma}^{-n_R}) & \text{if } \frac{x-s}{\sigma} \geq \alpha_R \end{array} \right\} \tag{4.1}$$

$$A_L = (\tfrac{n_L}{\alpha})^{n_L} \times e^{-\frac{|\alpha_L|^2}{2}}$$

$$A_R = (\tfrac{n_R}{\alpha})^{n_R} \times e^{-\frac{|\alpha_R|^2}{2}}$$

$$B_L = \tfrac{n}{\alpha_L} - |\alpha_L|$$

$$B_R = \tfrac{n}{\alpha_R} - |\alpha_R|$$

N is the normalization constant.

In case of WH, ZH and $t\bar{t}$H production modes the event selection procedure will occasionally select 4ℓ candidates where at least one reconstructed lepton originates from associated vector bosons or jets, and not from the $\mathrm{H} \rightarrow 4\ell$ decays. Fraction of such signal events in the mass range $105\,\mathrm{GeV} < \mathrm{m}(4\ell) < 140\,\mathrm{GeV}$ amounts to about 4%, 18% and 22% for WH, ZH and $t\bar{t}$H production modes, respectively. These events have a broad invariant $m_{4\ell}$ mass distribution that depends on the exact production mechanism. The shape of the contribution by this background , $\mathcal{P}_{\mathrm{comb}}(m_{4\ell})$, is described a Landau distribution with parameters extracted from simulations.

The ratio of the reconstructed events which are from outside of the fiducial phase space to the events to reconstructed events which from within fiducial phase space(f_{nonfid}). The f_{nonfid} fraction describes the ratio of the nonfiducial and fiducial signal contribution in bin j at the reconstruction level.

$$f^{j}_{\mathrm{nonfid}} = \frac{N(\mathrm{j_{reco}}\ \&\&\ \mathrm{not\ fid.})}{N(\mathrm{j_{reco}}\ \&\&\ \mathrm{fid.})}. \tag{4.2}$$

These events arises due to imperfect detector resolution that can cause differences for the quantities calculated at generated and reconstructed levels such as lepton isolation and other quantities used for the event selection. This contribution is also treated as background and will be referred "non-fiducial signal“ throughout this document. It has been demonstrated with MC samples that these events can be described using same shape as resonant component. The values this fractions are extracted from simulation repeatedly for SM. This fraction changes from 4% for the $\mathrm{gg} \rightarrow \mathrm{H}$ production to 3% for the ttH production. The values for each model are listed in the Tables **??**, **??** and **??** for 2016, 2017 and 2018 respectively.

To make comparison with the theoretical predictions, measurements need to be corrected for detector resolution and efficiency effects. To unfold the reconstruction-level distribution to the distribution before all detector effects, the efficiency matrix (ϵ) is constructed using the signal MC samples. The efficiency matrix (ϵ) provides the probability of an event that was in i-th bin at the fiducial level and is reconstructed in j-th bin as:

$$\epsilon_{i,j} = \frac{N(\mathrm{j_{reco}}\ \&\&\ \mathrm{i_{gen}})}{N(\mathrm{i_{gen}})}\ , \tag{4.3}$$

where $N(i_{gen})$ is the number of the events in i-th bin at the fiducial level and $N(\mathrm{j_{reco}}\ \&\&\ \mathrm{i_{gen}})$ is the number of events that was in i-th bin at the fiducial level and reconstructed in j-th bin. For integrated measurements, where we have only one bin, fiducial efficiency is about 65% which can change upto 7% for other signal models. For differential measurements, it is described by detector response matrix for each observable. Figure **??** shows the efficiency matrix for $p_T(H)$ measurement using different SM production MC samples. The efficiency matrix is dominated by diagonal elements, but in general it can have non-negligible off-diagonal elements specially for jet related observables due to jet energy scale uncertainty. In case of differential variable, jet multiplicity, neighbour elements of diagonal varies from 3% to 21% as shown in Figures **??**-**??** in the Appendix **??**.

The number of estimated events in each final state f and in bin i of a studied observable are described as a function of $m_{4\ell}$ given by:

$$\begin{aligned} N^{\mathrm{f},i}_{\mathrm{obs}}(m_{4\ell}) &= N^{\mathrm{f},i}_{\mathrm{fid}}(m_{4\ell}) + N^{\mathrm{f},i}_{\mathrm{nonfid}}(m_{4\ell}) + N^{\mathrm{f},i}_{\mathrm{comb}}(m_{4\ell}) + N^{\mathrm{f},i}_{\mathrm{bkg}}(m_{4\ell}) \\ &= \sum_j \epsilon^{\mathrm{f}}_{i,j} \cdot \left(1 + f^{\mathrm{f},i}_{\mathrm{nonfid}}\right) \cdot \sigma^{\mathrm{f},j}_{\mathrm{fid}} \cdot \mathcal{L} \cdot \mathcal{P}_{\mathrm{res}}(m_{4\ell}) \\ &\quad + N^{\mathrm{f},i}_{\mathrm{comb}} \cdot \mathcal{P}_{\mathrm{comb}}(m_{4\ell}) + N^{\mathrm{f},i}_{\mathrm{bkg}} \cdot \mathcal{P}_{\mathrm{bkg}}(m_{4\ell}). \end{aligned} \tag{4.4}$$

Where $N^{\mathrm{f},i}_{\mathrm{fid}}(m_{4\ell})$, $N^{\mathrm{f},i}_{\mathrm{nonfid}}(m_{4\ell})$, $N^{\mathrm{f},i}_{\mathrm{comb}}(m_{4\ell})$ and $N^{\mathrm{f},i}_{\mathrm{bkg}}(m_{4\ell})$ denote the resonant fiducial signal, resonant nonfiducial signal, combinatorial contribution from fiducial signal, and background contributions in bin i as functions of $m_{4\ell}$, respectively. Similarly, the corresponding probability density functions for the resonant (fiducial and nonfiducial) signal, combinatorial signal, and background contributions are $\mathcal{P}_{\mathrm{res}}(m_{4\ell})$, $\mathcal{P}_{\mathrm{comb}}(m_{4\ell})$ and $\mathcal{P}_{\mathrm{bkg}}(m_{4\ell})$ respectively. $\epsilon^{\mathrm{f}}_{i,j}$ is the detector response matrix that connects the reconstructed level events in bin i with fiducial events in the bin j for studied differential observable. The f^{i}_{nonfid} fraction represents the ratio of the nonfiducial and fiducial signal contributions in bin i at the reconstruction level. The parameter $\sigma^{\mathrm{f},j}_{\mathrm{fid}}$ is the signal cross section for the final state f in bin j of the fiducial phase space.

4.2.6 Extraction of Integrated $\mathrm{H} \to 4\ell$ Fiducial Cross Section

Signal Model and Statistical Procedure

To measure the integrated cross sections within the fiducial phase space, we follow a general procedure for the measurements of the differential cross sections outlined in Section 4.2.5, treating it as a special case where the number of considered bins is one.

Table 4.4 shows the reconstruction efficiency (ϵ) for fiducial events, while Table 4.5 shows the ratio (f_{nonfid}) of reconstructed events which are from outside the fiducial phase space and reconstructed events which are from within the fiducial phase space , per final state for different Standard Model signal models. The primary source of the “nonfiducial signal” contributions (described with the factor f_{nonfid}) is the imperfect correspondence between the isolation at the particle level and isolation at the reconstruction level, as discussed at the beginning of Section 4.2. In a similar way, Table 4.6 shows fraction of signal events in the mass range 105.0–140.0 GeV where at least one lepton selected is not from the Higgs boson decay.

TABLE 4.4: Reconstruction efficiency (ϵ) for fiducial events per final state for different Standard Model signal models at $\sqrt{s} = 13$ TeV with 2018 data.

Sample	$4e$	4μ	$2e2\mu$	4ℓ
gg→H (POWHEG)	0.404 ± 0.002	0.799 ± 0.002	0.572 ± 0.002	0.593 ± 0.001
VBF (POWHEG)	0.431 ± 0.003	0.805 ± 0.002	0.586 ± 0.002	0.608 ± 0.002
WH (POWHEG+MINLO)	0.415 ± 0.004	0.780 ± 0.003	0.580 ± 0.003	0.593 ± 0.002
ZH (POWHEG+MINLO)	0.428 ± 0.006	0.780 ± 0.005	0.581 ± 0.005	0.597 ± 0.003
ttH (POWHEG)	0.429 ± 0.006	0.745 ± 0.005	0.573 ± 0.005	0.585 ± 0.003
ggH(NNLOPS)	0.425 ± 0.002	0.776 ± 0.002	0.578 ± 0.001	0.595 ± 0.001

TABLE 4.5: Ratio of reconstructed events which are from outside the fiducial phase space and reconstructed events which are from within the fiducial phase space (f_{nonfid}) per final state for different Standard Model signal models at $\sqrt{s} = 13$ TeV with 2018 data.

Sample	$4e$	4μ	$2e2\mu$	4ℓ
gg→H (POWHEG)	0.060 ± 0.002	0.046 ± 0.001	0.060 ± 0.001	0.055 ± 0.001
VBF (POWHEG)	0.050 ± 0.002	0.037 ± 0.001	0.046 ± 0.001	0.043 ± 0.001
WH (POWHEG+MINLO)	0.098 ± 0.004	0.058 ± 0.002	0.080 ± 0.002	0.075 ± 0.001
ZH (POWHEG+MINLO)	0.105 ± 0.007	0.071 ± 0.004	0.091 ± 0.004	0.087 ± 0.002
ttH (POWHEG)	0.278 ± 0.012	0.127 ± 0.005	0.193 ± 0.006	0.185 ± 0.004
ggH(NNLOPS)	0.051 ± 0.001	0.046 ± 0.001	0.051 ± 0.001	0.049 ± 0.001

TABLE 4.6: Fraction of signal events in the mass range 105.0–140.0 GeV where at least one lepton selected is not from the Higgs boson decay at $\sqrt{s} = 13$ TeV with 2018 data.

Sample	$4e$	4μ	$2e2\mu$	4ℓ
gg→H (POWHEG)	0.002	0.002	0.002	0.002
VBF (POWHEG)	0.002	0.003	0.003	0.003
WH (POWHEG+MINLO)	0.036	0.031	0.042	0.037
ZH (POWHEG+MINLO)	0.199	0.207	0.212	0.208
ttH (POWHEG)	0.175	0.156	0.164	0.163
ggH(NNLOPS)	0.003	0.003	0.004	0.003

4.2.7 Extraction of Differential $\mathrm{H} \to 4\ell$ Fiducial Cross Sections

Kinematic Observables

The differential cross section measurements are performed as a function of the kinematic properties of the four-lepton system and the jet multiplicity, and are

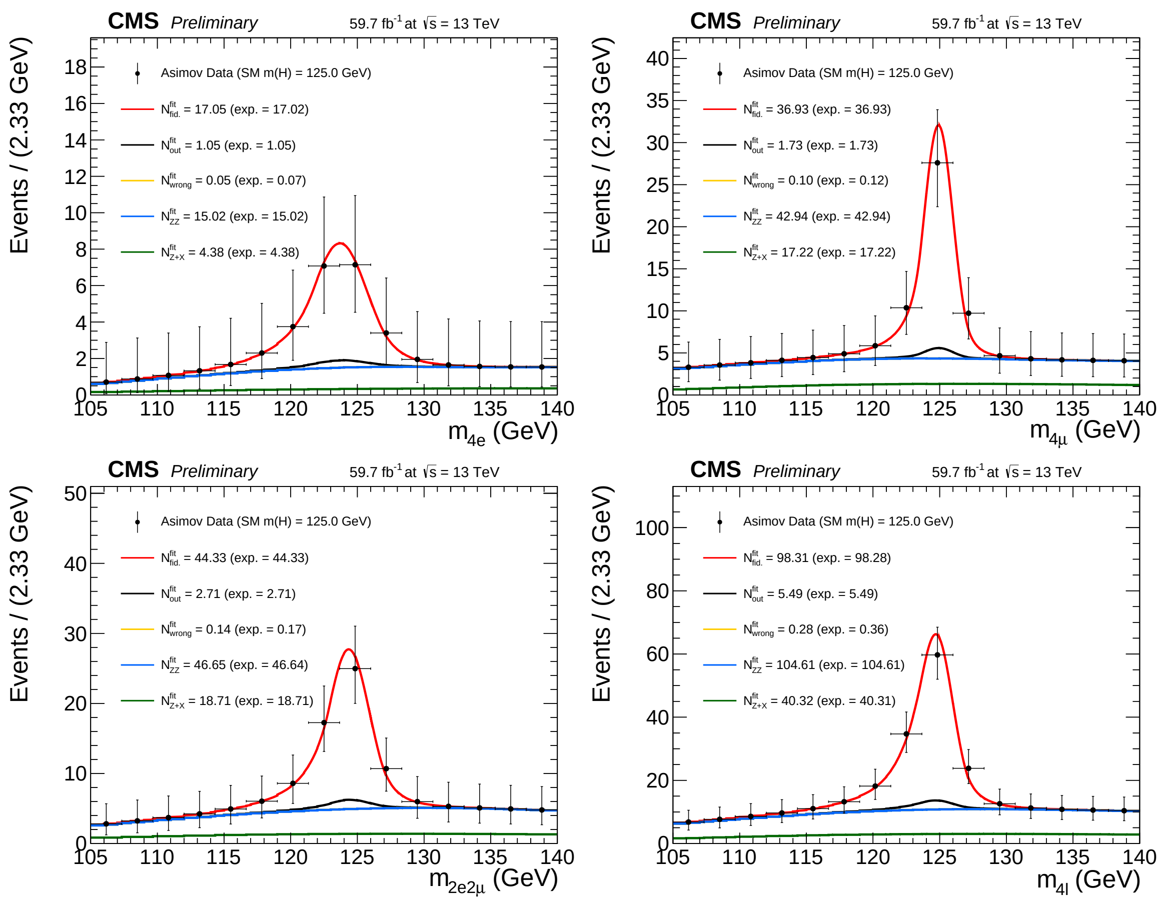


FIGURE 4.8: Inclusive four lepton mass distributions for an Asimov Dataset generated using SM cross section values and SM efficiencies where the gg→H prediction is from POWHEG+JHUGEN and resulting fitted values of the signal and background pdfs in different final states at $\sqrt{s} = 13$ TeV with 2018 data.

unfolded to the particle level. In this document, regarding the kinematic properties of the four-lepton system, the results are reported for the transverse momentum and rapidity of the four-lepton system (p_T(H) or $p_T(4\ell)$, $|y(\mathrm{H})|$ or $|y(4\ell)|$), N(jets), and $p_T(\mathrm{jet})$ which probe the production mechanism. The binning boundaries for each kinematic observable have been optimized in order to have a similar expected measured cross-section for every bin.

Background Model

In order to perform the differential measurements, the background is to be subtracted at the reconstruction level for each bin of the considered observable. In general, the background mass spectrum shape is not the same in all bins of the considered observable because the background does not have a resonance structure. The mass spectrum also can have non-negligible correlation with the observables.

In case of the irreducible backgrounds, the four-lepton mass spectra are modeled as the $m_{4\ell}$ template shapes using simulated events for each bin of the observable considered for the measurement. Similarly, in case of the reducible backgrounds the templates are built from the control regions in the data. This procedure is similar to the one described and used in the Ref. [**Chatrchyan:2013mxa**].

Fractions of background events in each bin of $p_T(4\ell)$ are presented in Table 4.7.

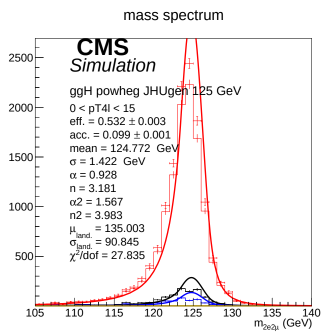
TABLE 4.7: Estimated fraction of background events in each bin of $p_T(4\ell)$, for each final state using 2018 samples.

Background	final state	$0-15$	$15-30$	$30-45$	$45-80$	$80-120$	$120-200$	$200-13000$
qqZZ	$2e2\mu$	0.483668	0.239196	0.130151	0.105276	0.0266332	0.011809	0.00326633
qqZZ	$4e$	0.459665	0.234399	0.136986	0.114916	0.0388128	0.00989346	0.00532725
qqZZ	4μ	0.492632	0.241203	0.110977	0.109173	0.0297744	0.0129323	0.00330827
ggZZ	$2e2\mu$	0.310446	0.309859	0.191901	0.168134	0.0187793	0.000880282	0.0
ggZZ	$4e$	0.3085	0.306332	0.19247	0.17433	0.0169994	0.00136908	0.0
ggZZ	4μ	0.321332	0.319311	0.194863	0.150987	0.0128703	0.000638196	0.0
Z+X (CR)	4l	0.0992521	0.211153	0.209307	0.305327	0.120303	0.0483797	0.00627828

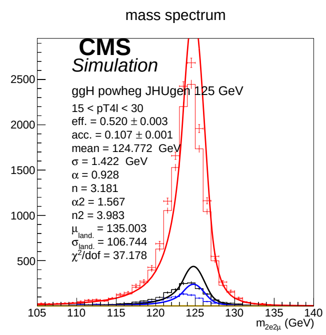
Signal Model and Statistical Procedure

In order to measure the differential cross sections within the fiducial phase space, we extract the cross sections in the bins of kinematic observables at the fiducial level following the procedure outlined in Section 4.2.5. To extract the signal yields, the signal mass spectrum is built in each cell of the response matrix using a double-sided Crytal Ball (CB) function, as was done in Ref. [**Chatrchyan:2013mxa**]. In practice, the same parameters of the CB function are used in each cell and the systematic uncertainties assigned on the energy scale and resolution are general enough to cover the discrepancy of the signal mass spectrum among each cell.

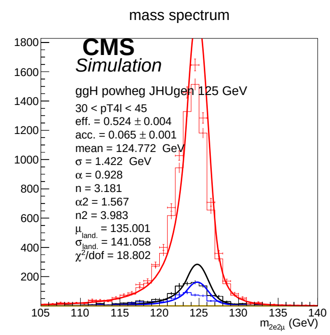
Examples of the fitted signal line shapes for differential bins of $p_T(\mathrm{H})$ from gg→H (POWHEG+JHUGEN 125 GeV 2018 sample) in the individual final states are shown in Figure 4.9, and examples of the efficiency matrices for $p_T(\mathrm{H})$ for the $2e2\mu$ final state are shown in Fig 4.10. Many more examples can be seen in Appendix **??**.



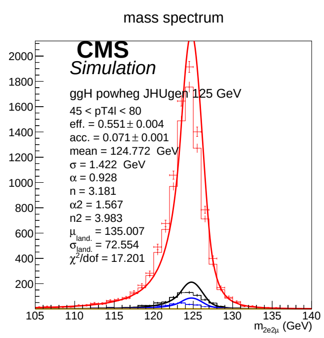
(a) $0.0 < p_T(\mathrm{H}) < 15.0$



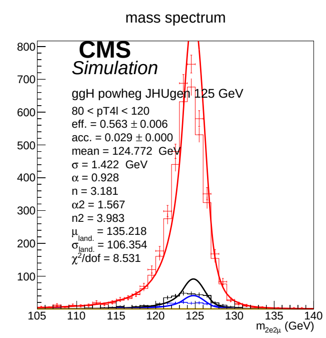
(b) $15.0 < p_T(\mathrm{H}) < 30.0$



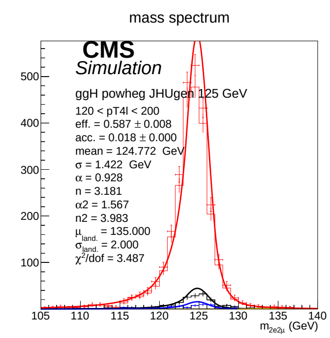
(c) $30.0 < p_T(\mathrm{H}) < 45.0$



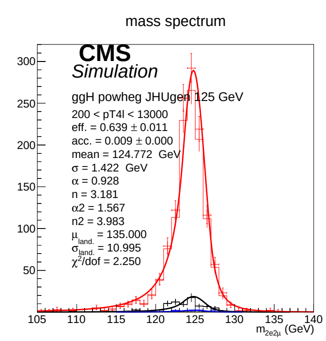
(d) $45.0 < p_T(\mathrm{H}) < 80.0$



(e) $80.0 < p_T(\mathrm{H}) < 120.0$

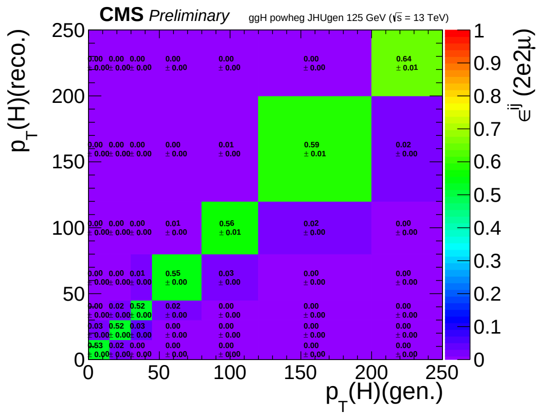


(f) $120.0 < p_T(\mathrm{H}) < 200.0$

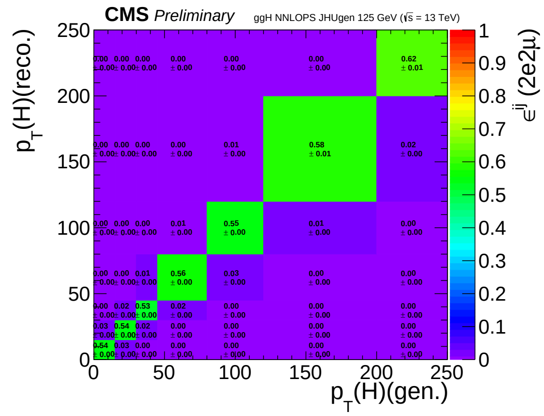


(g) $200.0 < p_T(\mathrm{H}) < 13000.0$

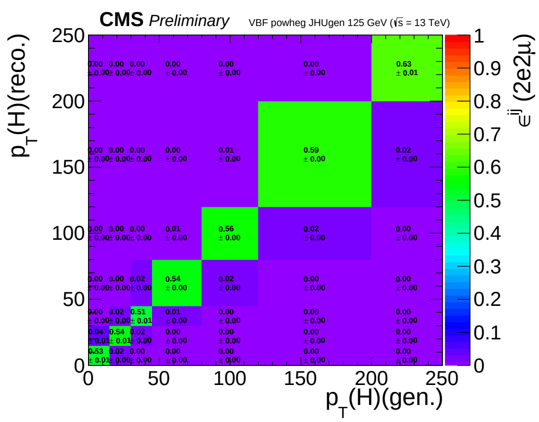
FIGURE 4.9: Distributions of m(4ℓ) in $2e2\mu$ final state for the $gg \to$ H production mode from POWHEG+JHUGEN in different bins of $p_T(\mathrm{H})$.



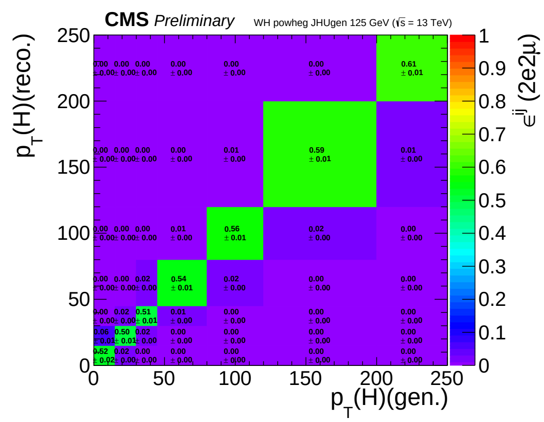
(a) $gg \to$H (POWHEG+JHUGEN)



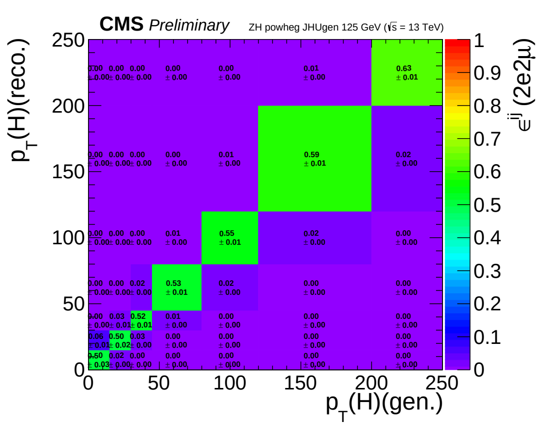
(b) $gg \to$H (MINLOHJJ)



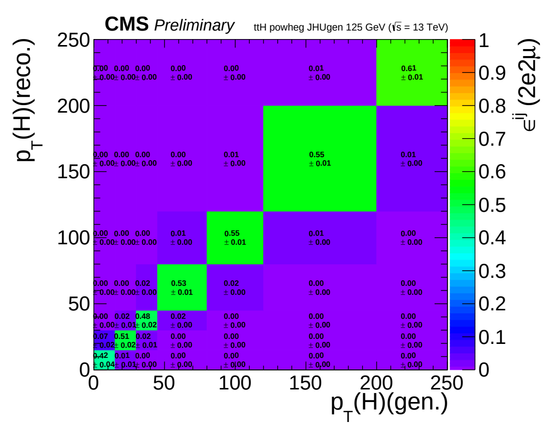
(c) VBF (POWHEG)



(d) WH (PYTHIA)



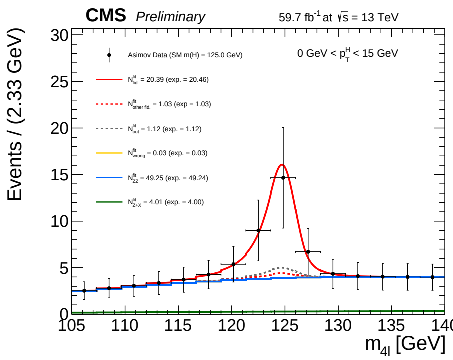
(e) ZH (PYTHIA)



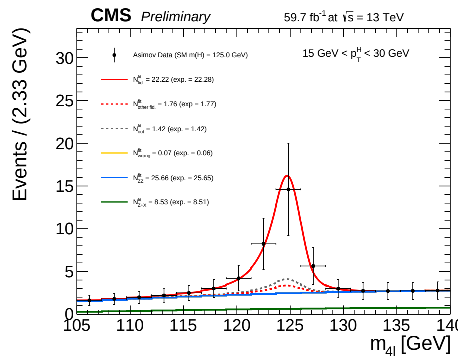
(f) ttH (PYTHIA)

FIGURE 4.10: Efficiency matrices for p_T(H) for different SM production modes in the $2e2\mu$ final state using 2018 samples.

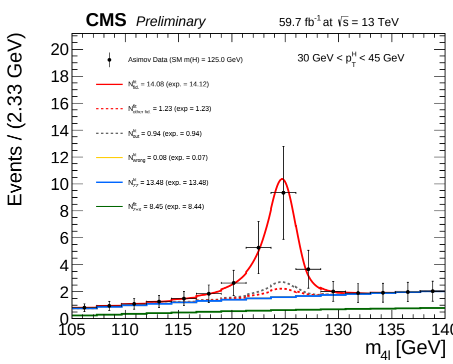
Figure 4.11 shows the four lepton mass distributions for an Asimov dataset generated using SM cross section values and efficiencies from the gg→H production mode from POWHEG+JHUGEN, and resulting fitted values of the signal and background pdfs in different bins of $p_T(\mathrm{H})$ (all final states combined).



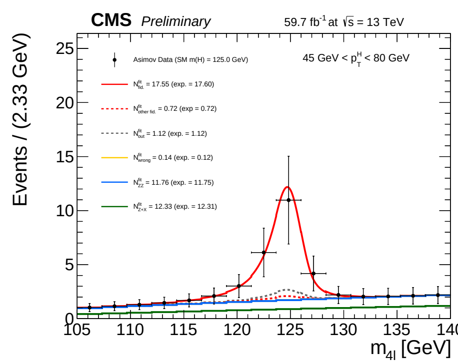
(a) 0.0 GeV $< p_T(\mathrm{H}) <$ 15.0 GeV



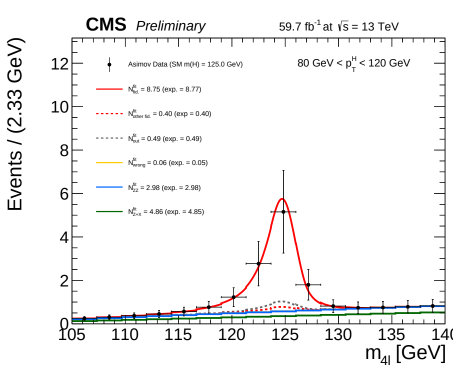
(b) 15.0 GeV $< p_T(\mathrm{H}) <$ 30.0 GeV



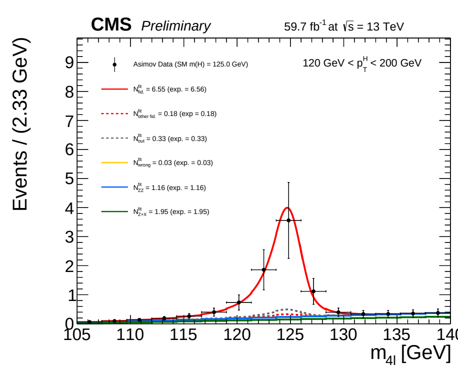
(c) 30.0 GeV $< p_T(\mathrm{H}) <$ 45.0 GeV



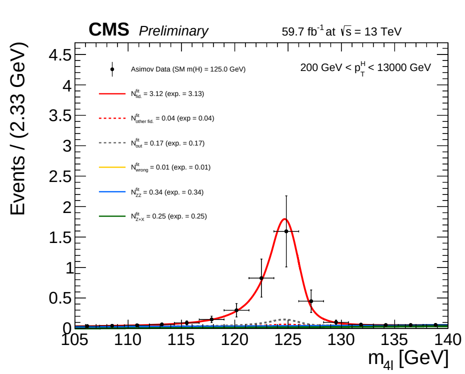
(d) 45.0 GeV $< p_T(\mathrm{H}) <$ 80.0 GeV



(e) 80.0 GeV $< p_T(\mathrm{H}) <$ 120.0 GeV



(f) 120.0 GeV $< p_T(\mathrm{H}) <$ 200.0 GeV



(g) 200.0 GeV $< p_T(\mathrm{H}) <$ 13000.0 GeV

FIGURE 4.11: Four lepton mass distributions for an Asimov dataset generated using SM cross section values and SM efficiencies where the gg→H prediction is from POWHEG+JHUGEN and resulting fitted values of the signal and background pdfs in different bins of $p_T(\mathrm{H})$ (all final states combined).

4.3 Systematic Uncertainties

4.4 Systematic uncertainties

The main experimental uncertainties which affect both signal and background are the uncertainty on the integrated luminosity (from 2.3% to 2.5%, depending on the year of data taking) and the uncertainty on the lepton identification and reconstruction efficiency (ranging from 2.5–16.1% on the overall event yield for the 4μ and $4e$ channels, respectively). Experimental uncertainties for the reducible background estimation, described in Section 4.2.3, originating from the background composition and misidentification rate uncertainty vary depending on the final state The mismatch in the composition of backgrounds between the samples where the fake rate is derived and where it is applied accounts between 24 and 43%.

The uncertainty of the lepton energy scale is assessed by propagating down to the four-leptons invariant mass the uncertainty associated the lepton momentum corrections on each indiviudal lepton. It is determined by considering the $Z \to \ell\ell$ mass distributions in data and simulation. Events are separated into categories based on the p_T and η of one of the two leptons, determined randomly, and integrating over the other. The dilepton mass distributions are then fit by a Breit-Wigner parameterization convolved with a double-sided Crystal Ball function. The uncertainty is treated as uncorrelated (all up or all down). The new four-leptons invariant mass obtained this way is then fitted with only the mean value floating. The difference between the new mean value and the default one gives an estimation of the impact of the lepton energy scale. The values are reported in the Table 4.8, together with the estimation of the impact of lepton momentum resolution, following a similar procedure.

TABLE 4.8: Difference (in %) between the nominal mean and width of the four-leptons Higgs mass distribution and those where the lepton momentum scale and resolution uncertainties have been propagated.

	4e	4μ	$2e2\mu$
Scale Difference (%)	0.087	0.049	0.069
Resolution Difference (%)	17	12	17

The results obtained this way suggest that the uncertainties from 2016 (20% on the resolution, 0.3% for the $4e$ channel for scale) are still valid.

Theoretical uncertainties which affect both the signal and background estimation include uncertainties from the renormalization and factorization scale and choice of PDF set. The uncertainty from the renormalization and factorization scale is determined by varying these scales between 0.5 and 2 times their nominal value while keeping their ratio between 0.5 and 2. In the case of ggF production mode, the QCD scale uncertainty is broke down to 9 different sources. The uncertainty from the PDF set is determined by taking the root mean square of the variation when using different replicas of the default NNPDF set. On the background, an additional uncertainty of 10% on the K factor used for the prediction is applied as described in Section 4.2.3. A systematic uncertainty of 2% on the branching ratio of only affects the signal yield. Additional uncertainties come from the imprecise knowledge of the jet energy scale and resolution (from 2% to 22%), b-tagging efficiency and light quarks (u, d, s, c) and gluon jet mistag rate. In the cross section measurement, the signal cross section uncertainties arise from theoretical sources are removed.

TABLE 4.9: Summary of the experimental systematic uncertainties in the $H \rightarrow 4\ell$ measurements of 2016 data.

Summary of relative systematic uncertainties	
Common experimental uncertainties	
Luminosity	2.6 %
Lepton identification/reconstruction efficiencies	2.5 – 9 %
Background related uncertainties	
Reducible background (Z+X)	36 – 43 %
Signal related uncertainties	
Lepton energy scale	0.04 – 0.3 %
Lepton energy resolution	20 %

TABLE 4.10: Summary of the experimental systematic uncertainties in the $H \rightarrow 4\ell$ measurements of 2017 data.

Summary of relative systematic uncertainties	
Common experimental uncertainties	
Luminosity	2.3 %
Lepton identification/reconstruction efficiencies	3. – 12.5 %
Background related uncertainties	
Reducible background fake rate variation (Z+X)	6 – 32 %
Signal related uncertainties	
Lepton energy scale	0.04 – 0.3 %
Lepton energy resolution	20 %

4.4.1 Systematic uncertainties treatment when combining the data sets

In the combination of the three data taking periods, the theoretical uncertainties as well as the experimental ones related to leptons or jets are treated as correlated while all other ones from experimental sources are taken as uncorrelated.

TABLE 4.11: TO BE UPDATED. COPY PASTE NUMBERS OF 2017 FOR NOW. Summary of the experimental systematic uncertainties in the $\mathrm{H} \to 4\ell$ measurements of 2018 data.

Summary of relative systematic uncertainties	
Common experimental uncertainties	
Luminosity	2.3 %
Lepton identification/reconstruction efficiencies	3. – 12.5 %
Background related uncertainties	
Reducible background fake rate variation (Z+X)	6 – 32 %
Signal related uncertainties	
Lepton energy scale	0.04 – 0.3 %
Lepton energy resolution	20 %

TABLE 4.12: Summary of the theory systematic uncertainties in the $\mathrm{H} \to 4\ell$ measurements for the inclusive analysis

Summary of inclusive theory uncertainties	
BR($H \to$ ZZ $\to 4\ell$)	2 %
QCD scale ($\mathrm{qq} \to$ ZZ)	+3.2/-4.2 % %
PDF set ($\mathrm{qq} \to$ ZZ)	+3.1/-3.4 %
Electroweak corrections ($\mathrm{qq} \to$ ZZ)	$\pm$ 0.1 %

4.5 Results

4.5.1 Measurement of Integrated Fiducial Cross Section

The resulting fits to the inclusive $\mathrm{m}(4\ell)$ distributions for the integrated $\mathrm{H} \to 4\ell$ fiducial cross section measurements for different final states, performed in the $m_{4\ell}$ range from 105 GeV to 140 GeV, are shown in Figure 4.13. The extracted inclusive cross sections as function of center of mass energy are shown in Figure 4.14(a) while the extracted integrated cross section results for different final states at 13 TeV corresponding to different data taking periods are shown in Figure 4.14(b). The red error bar represents the systematic uncertainty, while black error bar represents the combined statistical and systematic uncertainties, summed in quadrature. The sub-dominant component of the signal $\mathrm{XH} = \mathrm{VBF} + \mathrm{VH} + t\bar{t}\mathrm{H}$ is indicated in green separately. Above listed results are summarized in Tables 4.13 and 4.14.

The $\mathrm{H} \to 4\ell$ measurements are compared to the SM NNLO theoretical estimations where the acceptance of the dominant $\mathrm{gg} \to \mathrm{H}$ contribution is modelled using POWHEG at $\sqrt{s}$=13 TeVand HRES [**Grazzini:2013mca**, **deFlorian:2012mx**] at $\sqrt{s}$=7 and 8 TeVand the total gluon fusion cross section and uncertainty are taken from Ref. [**Anastasiou2016**]. The fiducial volume for $\sqrt{s}$=6–9 TeVuses the lepp ton

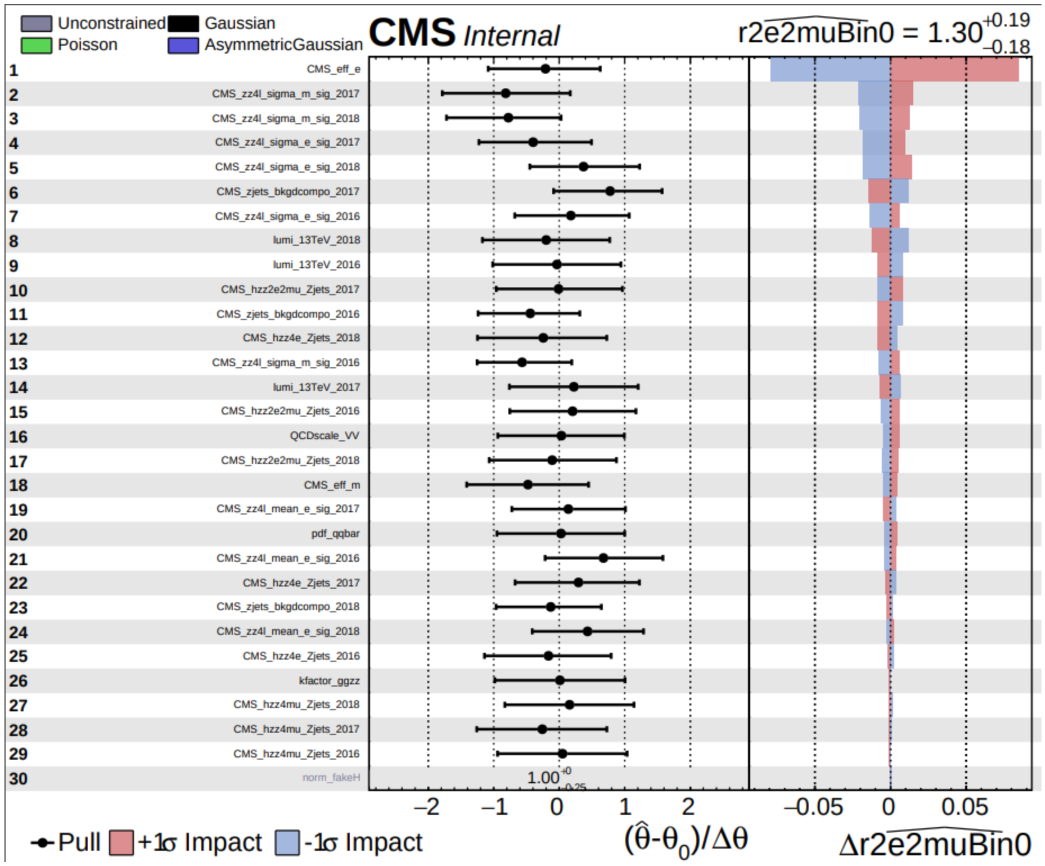


FIGURE 4.12: Pulls. Overall impact of different uncertainties sources on the inclusive measurement ($2e2\mu$ final state)

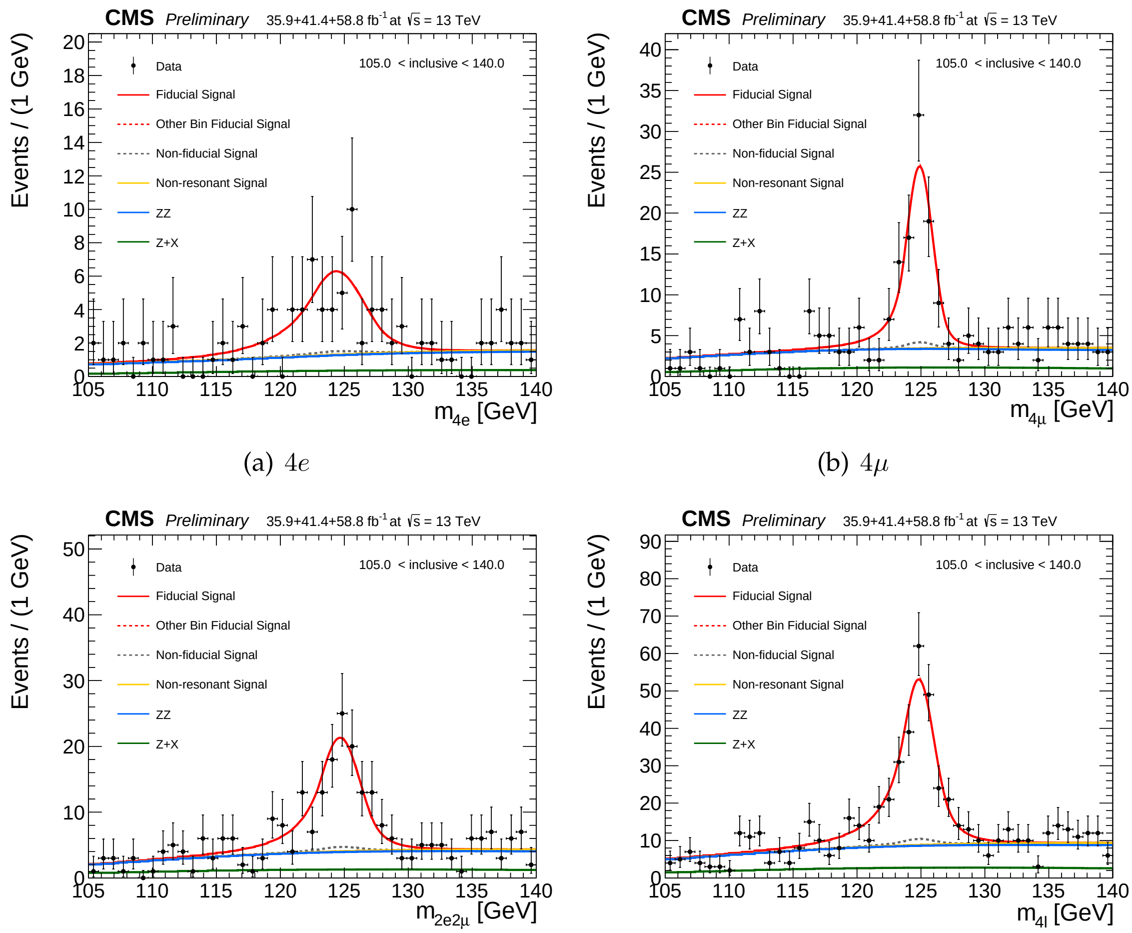
isolation definition from Ref. **[CMSH4lFiducial8TeV]**, while for $\sqrt{s}$=12–14 TeVthe definition descrr ibed in the text is used.

When extracting the total H$\to$ 4ℓ fiducial cross section, the parameter of interest is the total fiducial cross section and the fractions of $4e$, 4μ, and $2e2\mu$ is floated in the fit. When extracting the fiducial cross section for each final state, the three parameters of interest are the cross sections of three individual final states.

TABLE 4.13: Results of the $\mathrm{H} \to 4\ell$ integrated fiducial cross section measurements performed in the $m_{4\ell}$ range from 105 GeV to 140 GeV for proton collisions at 13 TeV.

Fiducial cross section $\mathrm{H} \to 4\ell$ **at 13 TeV**	
Measured	$2.73^{+0.30}_{-0.29} = 2.73^{+0.23}_{-0.22}(\mathrm{stat.})^{+0.24}_{-0.19}(\mathrm{syst.})$ fb (model)
gg→H(POWHEG) + XH	$0.93^{+0.10}_{-0.11}$fb
gg→H(NNLOPS) + XH	$0.94^{+0.10}_{-0.10}$fb

The measurement of the integrated $\mathrm{H} \to 4\ell$ fiducial cross section using 13 TeV



(a) $4e$ (b) 4μ

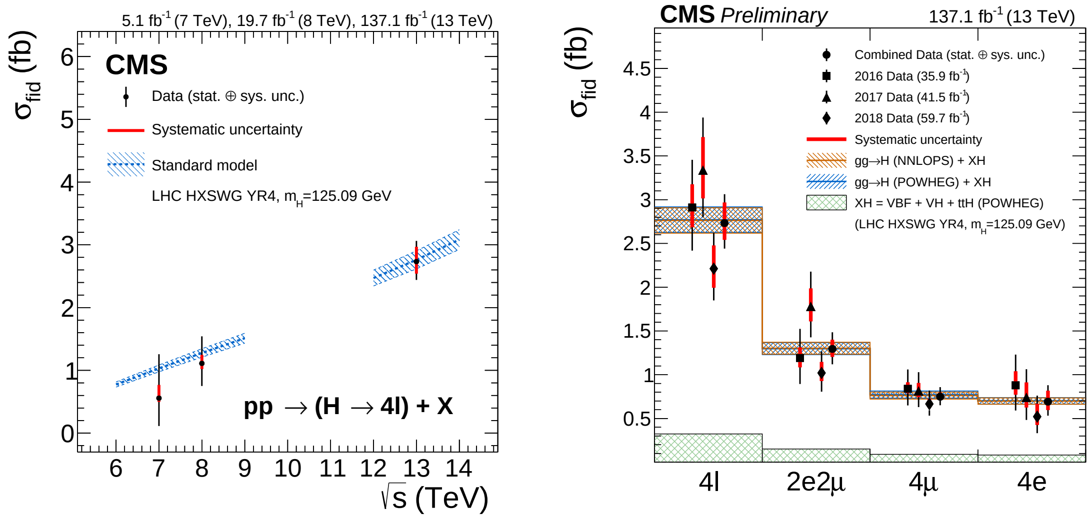
(c) $2e2\mu$ (d) 4ℓ

FIGURE 4.13: Inclusive four lepton mass distributions in data and the resulting fitted values of the signal and background pdfs in different final states.

data is found to be nice agreement with the theoretical predictions within the corresponding uncertainties. The dominant and sub-dominant uncertainties on the measurement are systematic and statistical uncertainties which are about 24% and 23% respectively. The theoretical uncertainty is about FIXME: 11%.

4.5.2 Measurement of Differential Fiducial Cross Sections

The observed data along with the prediction for a SM Higgs boson with m(H)=125.09 GeV and the expected background in each differential bin for the chosen observables are shown in Figures 4.15 for production related observables. The differential results are only produced using 13 TeV data. The extracted measurements of differential fiducial cross section for production related observables are shown in Figures 4.16. The results are produced for four-leptons transverse momentum and rapidity, transverse momentum of leading jet, jet multiplicity. The uncertainty in



(a) $\mathrm{H} \to 4\ell$ (7, 8 and 13 TeV) (b) $\mathrm{H} \to 4\ell$ (13 TeV)

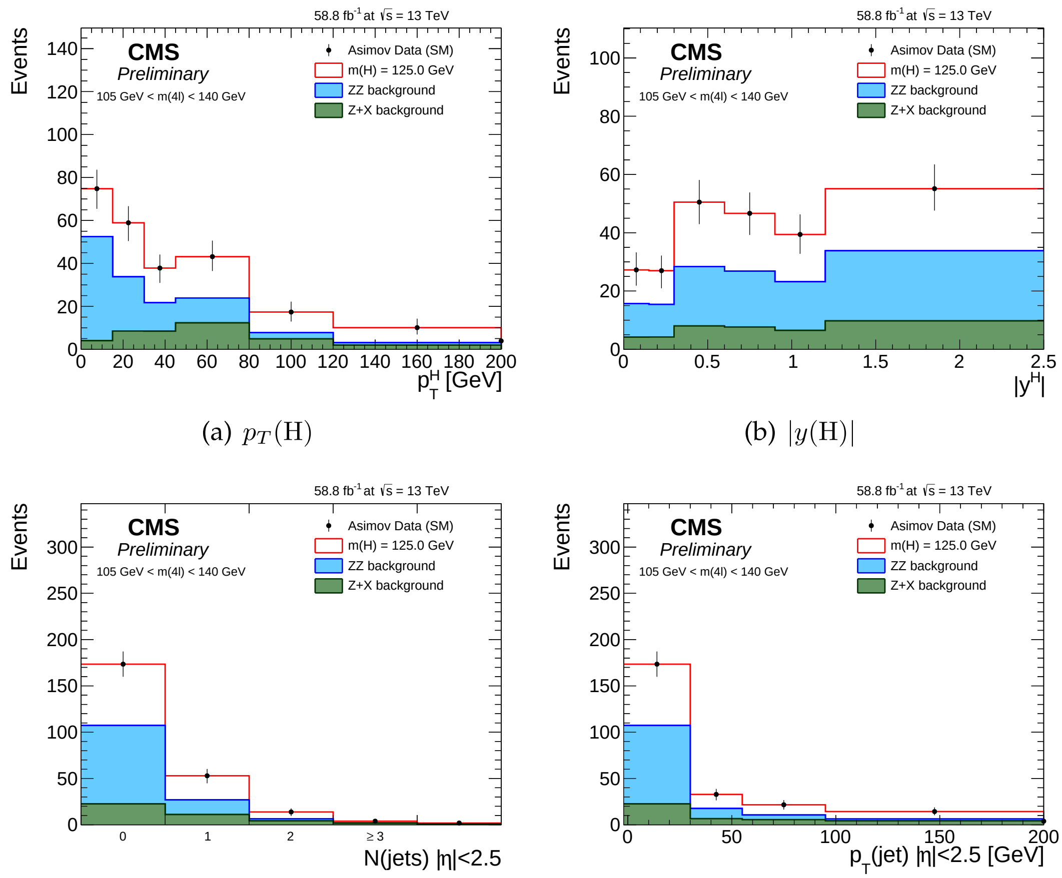
FIGURE 4.14: The measured fiducial cross section as a function of $\sqrt{s}$ (left). The measured inclusive fiducial cross section in different final states (right). The acceptance is calculated using at $\sqrt{s}$=13 TeVand HRES [**Grazzini:2013mca**, **deFlorian:2012mx**] at $\sqrt{s}$=7 and 8 TeVand the total gluon fusion cross section and uncertainty are taken from Ref. [**Anastasiou2016**]. The fiducial volume for $\sqrt{s}$=6–9 TeVuses the lepton isolation definition from Ref. [**CMSH4lFiducial8TeV**], while for $\sqrt{s}$=12–14 TeVthe definition described in the text is used.

TABLE 4.14: The results of the $\mathrm{H} \to 4\ell$ integrated fiducial cross section measurement performed in the $m_{4\ell}$ range from 105 GeV to 140 GeV.

Fiducial cross section H→ $4l$ (fb) at 13 TeV.				
	4l	$2e2\mu$	4μ	$4e$
Measured (stat. ⊕ sys.)(model)	$1.11^{+0.43+0.08}_{-0.36-0.02}$	$0.48^{+0.27+0.03}_{-0.22-0.03}$	$0.26^{+0.17+0.01}_{-0.13-0.01}$	$0.37^{+0.30+0.04}_{-0.22-0.02}$
gg→H(POHWHEG) + XH	$1.16^{+0.11}_{-0.11}$	$0.54^{+0.05}_{-0.05}$	$0.33^{+0.03}_{-0.03}$	$0.30^{+0.03}_{-0.03}$
gg→H(NNLOPS) + XH	$1.17^{+0.11}_{-0.11}$	$0.54^{+0.05}_{-0.05}$	$0.32^{+0.03}_{-0.03}$	$0.29^{+0.03}_{-0.03}$

the theoretical prediction for the dominant $\mathrm{gg} \to \mathrm{H}$ process is estimated in each bin of the studied observable with the generator used for the specific signal description POWHEG+JHUGEN. The uncertainty in the theoretical prediction for the associated production mechanisms are assumed to be constant among differential bins and are obtained from Ref. [**LHCHiggsCrossSectionWorkingGroup:2013tqa**]. The differential cross sections as a function of four-leptons transverse momentum and rapidity are sensitive to perturbative QCD calculations of the the dominant loop-mediated $\mathrm{gg} \to \mathrm{H}$ production mode, in which Higgs transverse momentum $p_T(H)$ is anticipated to be balanced with the emission of soft quarks and gluons. The differential distribution as a function rapidity of four-lepton system also probes the PDFs of the colliding protons and representation of the gluon fusion production modes. The differential observables, transverse momentum of leading jet and jet multiplicity N(jets) probe the hard quark and gluon radiation theoretical modelling in this process. The comparison of the measurements with theoretical predictions from POWHEG and NNLOPS for the gg→H process (shown in blue and brown colors, respectively) together with the predictions for XH = VBF (POWHEG) + VH + ttH(PYTHIA) (shown in green) is also shown. The error bars represents statistical and systematic uncertainty. The results are found to be compatible with SM based theoretical predictions within uncertainties.

When extracting the differential fiducial cross sections, the fractions of the $4e, 4\mu$ and $2e2\mu$ contributions to the fiducial cross sections in each bin is allowed to float. Results where the fractions are fixed to their SM expectations are included in Appendix **??**. Similarly, results where the model dependence uncertainty is assessed by taking into account the experimental constraints from the existing CMS measurements are included in Appendix **??**.



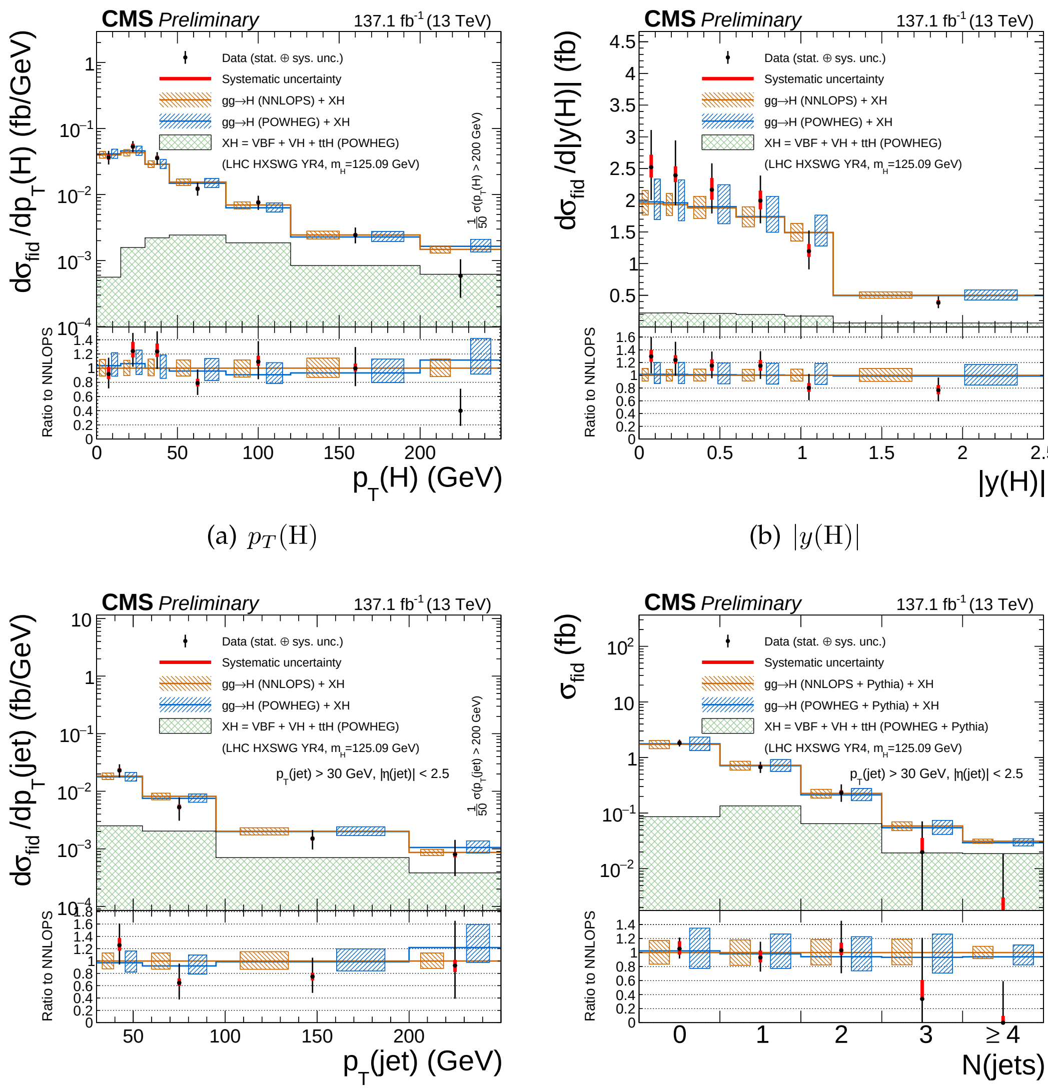
(a) $p_T(\mathrm{H})$

(b) $|y(\mathrm{H})|$

(c) N(jets), $|\eta| < 2.5$

(d) $p_T(\mathrm{jet}), |\eta| < 2.5$

FIGURE 4.15: Expected yield using 2018 Asimov data along with the SM signal (m(H)=125.09 GeV) and background expectations for different observables and for all final combined, in range $105 < m_{4\ell} < 140$ GeV used for the $\mathrm{H} \rightarrow 4\ell$ measurements. The extraction of the signal is performed by exploiting the differences in the expected $m_{4\ell}$ spectra of the signal and background contributions in each bin of an observable.



(a) p_T(H)

(b) $|y$(H)$|$

(c) p_T(leading jet), $|\eta| < 2.5$

(d) N(jets), $|\eta| < 2.5$

FIGURE 4.16: The differential fiducial cross section results for p_T(H) (top left), $|y$(H)$|$ (top right), p_T of the leading jet (bottom left) and N(jets) (bottom right) in H $\to 4\ell$ using full Run 2 CMS data and comparison with the theoretical predictions. Systematic uncertainty is indicated by red bars while black bars shows the total statistical and systematic uncertainties, combined in quadrature. The blue and brown colors show the theoretical predictions. The acceptance and theoretical uncertainties in the differential bins are are calculated using . The sub-dominant production contributions XH = VBF + VH + $t\bar{t}$H are shown in green separately. The total inclusive cross section is normalized to the SM estimate computed at NNLL+NNLO accuracy for all theoretical predictions. The systematic uncertainties correspond to the generators accuracy are also taken in to account for differential predictions. The fraction of $4e, 4\mu$ and $2e2\mu$ in each bin is allowed to float in the fit.